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Nonlinear Dynamics and Control

Analysis and control of a tumor-growth model

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Chapter 1

Context and model description

1.1 Biological context

The model analyzed in this report is a Three-dimensional cancer model focused on the interaction between tumor cells and the cell-mediated immune response. This model is an extension of the seminal work by Kuznetsov et al. [2], who first introduced the intuition of a nonlinear recruitment process of immune effectors from a pool of precursors. The 3D representation explicitly accounts for the transition between **resting predator cells** (precursors) and **hunting predator cells**.

1.2 Model assumptions

The mathematical structure is based on specific biological assumptions regarding the prey-predator relationship between the tumor and the immune system [1]:

1. Predators actively attack and destroy tumor cells.
2. The predator population exists in two distinct states: hunting and resting.
3. Macrophages release cytokines upon tumor interaction to activate resting cells.
4. Resting cells are stimulated by tumor antigens to convert into hunting cells; they do not possess direct killing capabilities.

5. The conversion process involves the consumption of the resting cell pool as they mature into effectors.

6. Tumor destruction follows the law of mass action. Once activated, hunting cells do not return to the resting state and face a constant probability of death.

7. Tumor cells possess a significant proliferating advantage over normal tissue cells.

1.3 Mathematical formulation

The interaction is initially described by a system of three ordinary differential equations for tumor cells (T), hunting cells (H), and resting cells (R):

$$\frac{dT}{dt} = r_1 T \left(1 - \frac{T}{k_1}\right) - a_{12} TH - a_{13} TR \tag{1.1}$$

$$\frac{dH}{dt} = r_2 H \left(1 - \frac{H}{k_2}\right) - a_{21} TH \tag{1.2}$$

$$\frac{dR}{dt} = \left(\frac{r_3 TR}{T + k_3}\right) - a_{31} TR - d_3 R \tag{1.3}$$

where r_1 defines the tumor growth potential and the a_{ij} coefficients represent the interaction and inactivation rates.

1.4 Nondimensionalization

To facilitate the dynamical analysis and reduce the number of independent parameters, the dimensional model is converted into a dimensionless form. This process follows the scaling logic established by Lodhi et al., ensuring that the state variables and parameters are normalized relative to the system's biological limits, such as carrying capacities and growth rates.

1.4.1 Variable transformation

The transition from physical populations — measured in number of cells — to dimensionless variables (x_1, x_2, x_3) is achieved by scaling

each population by a characteristic reference value. We define:

$$x_1 = \frac{T}{k_1}, \quad x_2 = \frac{H}{k_2}, \quad x_3 = \frac{R}{k_3} \tag{1.4}$$

The tumor, hunting and resting cells have been normalized with reference to their relative carrying capacity.

This transformation allows for a generalized interpretation of the model: regardless of the absolute number of cells (which could vary significantly between patients), the variables x_i always provide the **percentage of saturation** or the relative strength of each component.

1.4.2 Time scaling and the role of r_1

The introduction of dimensionless time τ is obtained by scaling the physical time t by the tumor growth rate r_1:

$$\tau = r_1 t \implies t = \frac{\tau}{r_1} \tag{1.5}$$

By applying the chain rule to the differential equations, the parameter r_1 is effectively normalized to unity:

$$\frac{dT}{dt} = \frac{d(k_1 x_1)}{d\tau}\frac{d\tau}{dt} = k_1 r_1 \frac{dx_1}{d\tau} \tag{1.6}$$

Substituting this into the original growth term we obtain the dimensionless term $x_1(1 - x_1)$.

1.4.3 Interpretation of simulation results

The combined scaling of states and time provides a clear framework for interpreting the simulation graphs presented in the following chapters:

1. **Magnitude of Response:** If a plot shows $x_1(\tau)$ dropping from 0.8 to 0.01, it signifies that the control action has reduced the tumor from 80% of its carrying capacity to a nearly eradicated state (1%).

2. **Temporal Window:** Based on the biological parameters validated by de Pillis et al. [5], the tumor growth rate is estimated

as $r_1 \approx 0.514\, day^{-1}$. Under these conditions, each unit on the x-axis of our plots ($\Delta\tau = 1$) corresponds to:

$$t = \frac{1}{r_1} \approx \frac{1}{0.514} \approx 1.95\,\text{days} \tag{1.7}$$

3. **Clinical Mapping:** A simulation showing stabilization within 50 time units corresponds to a real-world therapeutic window of approximately 100 days. This allows the dimensionless results to be translated into realistic medical protocols by simply adjusting the r_1 factor.

1.4.4 Interpretation and mapping of the control input

The control variable $u(\tau)$ represents the normalized rate of external immune recruitment.

A value of $u = 1$ signifies a recruitment rate that matches the maximum potential growth rate of the tumor mass. To ensure clinical feasibility, the control input must be constrained within the interval $u \in [0, 1]$.

1.4.5 The dimensionless equations

The original dimensional model is transformed into three first-order dimensionless ordinary differential equations:

$$\dot{x}_1 = x_1(1 - x_1) - A_{12}x_1x_2 - A_{13}x_1x_3 \tag{1.8}$$

$$\dot{x}_2 = R_2x_2(1 - x_2) - A_{21}x_1x_2 \tag{1.9}$$

$$\dot{x}_3 = \frac{R_3x_1x_3}{x_1 + k_3} - A_{31}x_1x_3 - D_3x_3 + u \tag{1.10}$$

The parameters shown in these equations are obtained via the following positions:

$$A_{12} = \frac{a_{12}k_2}{r1} \tag{1.11}$$

$$A_{21} = \frac{a_{21}k_1}{r1} \tag{1.12}$$

$$A_{13} = \frac{a_{13}k_3}{r1} \tag{1.13}$$

$$A_{31} = \frac{a_{31}k_1}{r1} \tag{1.14}$$

1.5 Physical meaning of terms and parameters

The equations incorporate key biological mechanisms. The term $x_1(1 - x_1)$ represents **logistic growth**. The immune recruitment in Eq. (1.10) follows a **Michaelis-Menten function**, which is a standard mathematical structure used to describe the saturable activation of the immune system in response to tumor antigens [3].

The system behavior is governed by eight parameters:

- A_{12}, A_{13}: Killing efficiency of active and resting cells against the tumor.
- R_2: Intrinsic growth rate of hunting cells.
- A_{21}: Inactivation rate of hunting cells by the tumor.
- R_3: Immune recruitment rate.
- k_3: Tumor density at which recruitment reaches half of its maximum.
- A_{31}: Suppression of resting cells by the tumor.
- D_3: Natural death rate of resting cells.
- u: External control input.

1.6 System nominal values

The nominal parameter values used for the numerical analysis are summarized in Table 1.1.

Parameter	**Value**	**Parameter**	**Value**
R_2	0.6	R_3	4.5
D_3	0.5	A_{12}	1.0
A_{21}	1.5	A_{13}	2.5
A_{31}	0.2	k_3	1.0

Table 1.1: Nominal dimensionless parameter values

Chapter 2

Open-loop analysis

2.1 Equilibrium points and stability

The equilibrium points are found by setting the state derivatives to zero, assuming $u = 0$. The system to solve is:

$$x_1(1 - x_1 - A_{12}x_2 - A_{13}x_3) = 0 \tag{2.1}$$

$$x_2(R_2(1 - x_2) - A_{21}x_1) = 0 \tag{2.2}$$

$$x_3\left(\frac{R_3x_1}{x_1 + k_3} - A_{31}x_1 - D_3\right) = 0 \tag{2.3}$$

Considering the nominal parameters in table 1.1, we identify all possible combinations of solutions:

1. **Origin (P_0):** $(x_1, x_2, x_3) = (0, 0, 0)$. This represents the trivial state where no cell populations are present.

2. **Healthy State (P_H):** For $x_1 = 0$, Eq. (2.2) admits $x_2 = 1$. Since Eq. (2.3) implies $x_3 = 0$, we obtain $P_H = (0, 1, 0)$. This represents a tumor-free state with a maximum population of hunting cells.

3. **Maximum Tumor (P_1):** For $x_2 = 0$ and $x_3 = 0$, Eq. (2.1) gives $x_1 = 1$. Thus, $P_1 = (1, 0, 0)$.

4. **Dormancy / Partial Equilibrium (P_{13}):** Setting $x_2 = 0$ but $x_3 \neq 0$, x_1 must satisfy the bracket in Eq. (2.3):

$$A_{31}x_1^2 - (R_3 - A_{31}k_3 - D_3)x_1 + D_3k_3 = 0 \tag{2.4}$$

The solutions are $x_{1a} \approx 0.1325$ and $x_{1b} \approx 18.86$, but only x_{1a} is biologically feasible. Substituting x_{1a} into Eq. (2.1) with $x_2 = 0$, we find $x_3 = \frac{1-0.1325}{2.5} \approx 0.347$. Thus, $P_{13} = (0.1325, 0, 0.347)$.

5. **Coexistence Equilibrium (P_3):** Setting $x_2 \neq 0$ and $x_3 \neq 0$, x_1 is still determined by Eq. (2.4), giving $x_1 = 0.1325$. From Eq. (2.2), we find

$$x_2 = 1 - \frac{A_{21}}{R_2} x_1 \approx 0.66875. \tag{2.5}$$

Finally, from Eq. (2.1), we get

$$x_3 = \frac{1 - x_1 - A_{12}x_2}{A_{13}} \approx 0.0795. \tag{2.6}$$

Thus, $P_3 = (0.1325, 0.66875, 0.0795)$.

2.2 Stability analysis via linearization

To characterize the behavior of the system around the equilibrium points, we employ the method of linearization. In order to do so, first we have to derive the Jacobian matrix of the system, which is obtained as:

$$J(x) = \begin{bmatrix} \frac{\partial \dot{x}_1}{\partial x_1} & \frac{\partial \dot{x}_1}{\partial x_2} & \frac{\partial \dot{x}_1}{\partial x_3} \\ \frac{\partial \dot{x}_2}{\partial x_1} & \frac{\partial \dot{x}_2}{\partial x_2} & \frac{\partial \dot{x}_2}{\partial x_3} \\ \frac{\partial \dot{x}_3}{\partial x_1} & \frac{\partial \dot{x}_3}{\partial x_2} & \frac{\partial \dot{x}_3}{\partial x_3} \end{bmatrix} \tag{2.7}$$

2.2.1 Numerical validation and Julia implementation

The equilibria and their stability properties were validated using the **Julia** programming language, leveraging the DynamicalSystems.jl ecosystem. Specifically, we utilized the fixedpoints function from the ChaosTools.jl library.

This numerical approach is particularly robust as it employs **Interval Root-Finding**. Unlike local optimization methods, this algorithm scans the entire defined biological domain $x_i \in [0, 1]$ to guarantee that all equilibrium points are identified. For each point, the software automatically computes the Jacobian and classifies the

equilibrium based on the eigenvalues of the Jacobian evaluated at the point.

The following table summarizes the unique equilibrium points identified. The results perfectly match the analytical derivations:

Point	**Coordinates** (x_1, x_2, x_3)	**Eigenvalues** (λ_i)	**Nature**
P_0	$(0, 0, 0)$	$1.0, 0.6, -0.5$	Saddle point-Unstable
P_H	$(0, 1, 0)$	$-0.6, -0.5, 0.0$	Non-hyperbolic
P_1	$(1, 0, 0)$	$-1.0, -0.9, 1.55$	Saddle Point-Unstable
P_{13}	$(0.1325, 0, 0.3470)$	$-0.066 \pm 0.613j, 0.401$	Saddle-Focus-Unstable
P_3	$(0.1325, 0.6688, 0.0795)$	$-0.614, 0.040 \pm 0.235j$	**Saddle-Focus-Unstable**

Table 2.1: Classification of unique equilibrium points for $R_3 = 4.5$ with Julia

2.2.2 The Julia script

To reproduce these results, the following Julia script was developed:

```
using DynamicalSystems
using LinearAlgebra
using StaticArrays
using IntervalArithmetic

function kuznetsov_rule(u, p, t)
    x1, x2, x3 = u
    R2, R3, D3, A12, A21, A13, A31, k3 = p

    dx1 = x1 * (1 - x1) - A12 * x1 * x2 - A13 * x1
       * x3
    dx2 = R2 * x2 * (1 - x2) - A21 * x1 * x2
    dx3 = (R3 * x1 * x3) / (x1 + k3) - A31 * x1 *
       x3 - D3 * x3

    return SVector(dx1, dx2, dx3)
end

p_vals = [0.6, 2.71, 0.5, 1.0, 1.5, 2.5, 0.2, 1.0]
ds = CoupledODEs(kuznetsov_rule, [0.1, 0.1, 0.1],
   p_vals)



search_box = IntervalBox(0..1, 0..1, 0..1)
```

```julia
fps_raw, eigs_raw, stabilities_raw = fixedpoints(ds, search_box)


seen = Set{Vector{Float64}}()
for i in 1:length(fps_raw)
    rounded = round.(Vector(fps_raw[i]), digits=4)
    if !(rounded in seen)
        push!(seen, rounded)
        type_str = get_point_type(eigs_raw[i])
        println("Point: $(rounded)")
        println("  Nature: $type_str")
        println("  Eigenvalues: $(round.(eigs_raw[i], digits=3))")
        println("-"^60)
    end
end
```

Listing 2.1: Julia script for global equilibrium analysis

2.3 Phase space analysis

To visually investigate the global dynamics of the system, a three-dimensional phase portrait of the system was generated, which allows us to observe how trajectories evolve from various initial conditions:

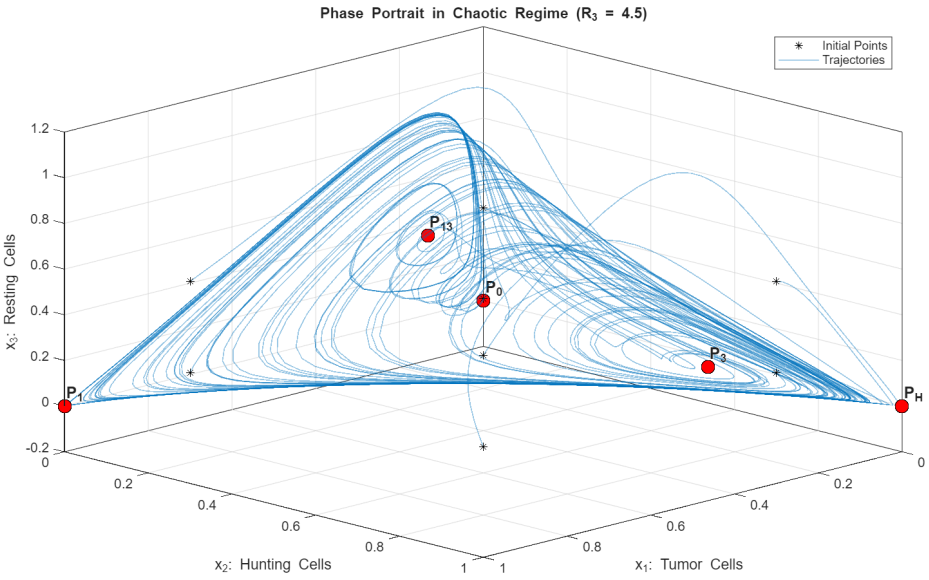


Figure 2.1: Phase portrait of the system for $R_3 = 4.5$

2.3.1 Phase portrait interpretation

As shown in Figure [2.1], all identified equilibrium points are marked in red, indicating their instability. The trajectories, initialized from a grid of points, do not converge to any steady state. Instead, they are repelled by the the equilibria and remain confined within a bounded region of the state space.

The motion around the coexistence equilibrium P_3 is of particular interest: trajectories exhibit the characteristic spiraling behavior predicted by the complex eigenvalues of the **Saddle-Focus**. This deterministic yet non-periodic evolution is the visual signature of the strange attractor that characterizes the tumor-immune interaction in this regime.

2.4 Simulink implementation

In order to analyze the temporal evolution of the system state variables, the model was implemented within the Simulink environment. The initial condition of the system was chosen to be $[0.1; 0.1; 0.1]^T$, the same selected in [1]. The following sections present the block diagram and the simulation results, illustrating the time-domain behavior of the state variables:

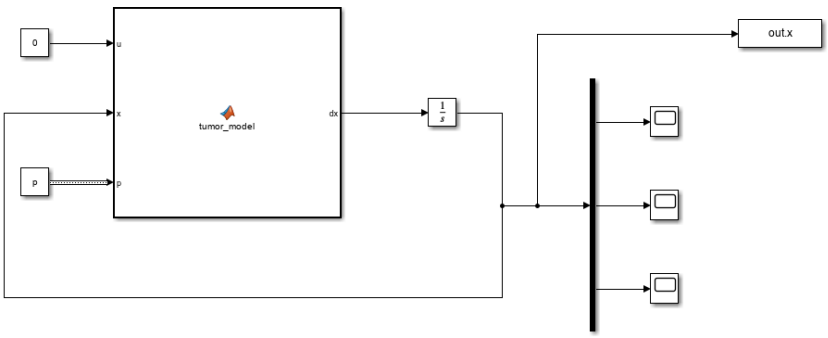


Figure 2.2: Simulink block scheme of open loop scheme in chaotic regime

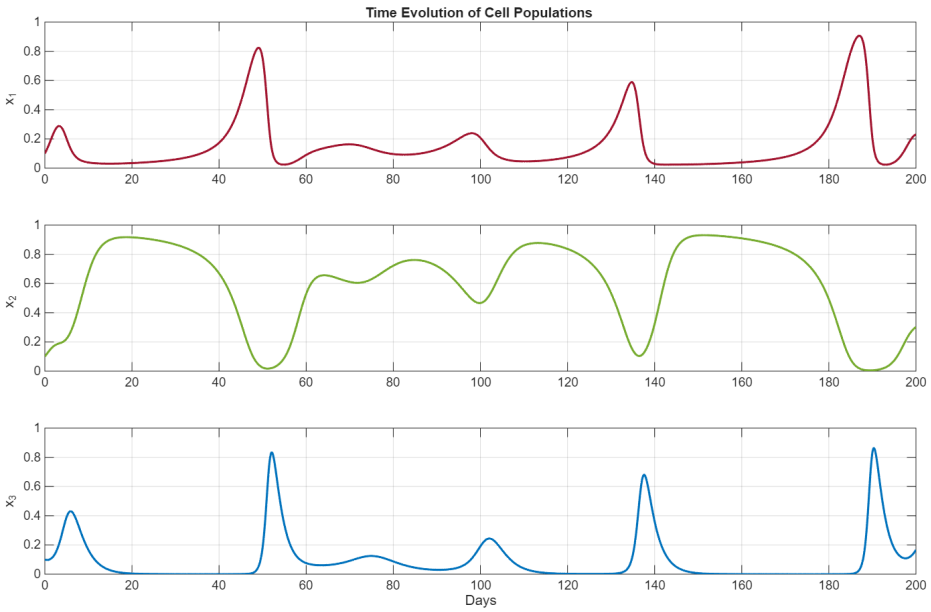


Figure 2.3: Time evolution of state variables

With the external control u set to zero, the system's state variables temporal evolution reflects the aperiodic nature of the chaotic regime. As illustrated in Figure [2.3], the populations do not reach a steady state:

- **Tumor Cells population :** Exhibits high-amplitude and irregular burstsm with peak values varying significantly.
- **Immune Response:** Hunting and resting cell populations fluctuate in response to tumor growth, failing to suppress cancer effectively.

2.5 Bifurcation analysis

The chaotic regime observed is the result of a precise sequence of changes in the system's dynamics. To understand how it transitions toward chaos, we treat the recruitment rate R_3 as the bifurcation parameter.

2.5.1 Analysis for low recruitment

At $R_3 = 1.0$, immune recruitment is too weak to allow the existence of any coexistence equilibria. To verify this, we recall the analysis of the internal equilibrium points ($x_2 = 0, x_3 \neq 0$) performed in Section 2.1.

As derived in Eq. (2.4), the coordinate x_1 of such equilibria must satisfy the quadratic equation:

$$A_{31}x_1^2 + (A_{31}k_3 + D_3 - R_3)x_1 + D_3k_3 = 0 \tag{2.8}$$

Substituting the nominal parameters and $R_3 = 1.0$, the equation becomes:

$$0.2x_1^2 + (0.7 - 1.0)x_1 + 0.5 = 0 \implies 0.2x_1^2 - 0.3x_1 + 0.5 = 0 \tag{2.9}$$

Calculating the discriminant:

$$\Delta = (-0.3)^2 - 4(0.2)(0.5) = 0.09 - 0.4 = -0.31 < 0 \tag{2.10}$$

Since $\Delta < 0$, no real solutions exist for x_1. Consequently, the system dynamics are entirely dominated by the structural equilibria P_0, P_H and P_1.

2.5.2 Stability analysis of structural points

To understand the evolution of the system, we analyze the Jacobian matrix $J(x)$ evaluated at the points that do not depend on R_3:

$$J(x^*) = \begin{bmatrix} 1 - 2x_1 - x_2 - 2.5x_3 & -x_1 & -2.5x_1 \\ -1.5x_2 & 0.6(1 - 2x_2) - 1.5x_1 & 0 \\ \mathcal{D}_{31} & 0 & \mathcal{D}_{33} \end{bmatrix}_{x=x^*} \tag{2.11}$$

Where:

- $\mathcal{D}_{31} = \frac{\partial \dot{x}_3}{\partial x_1} = \frac{R_3 x_3}{(x_1+1)^2} - 0.2x_3$
- $\mathcal{D}_{33} = \frac{\partial \dot{x}_3}{\partial x_3} = \frac{R_3 x_1}{x_1+1} - 0.2x_1 - 0.5$

The equilibria of interest are:

- **Origin:** Evaluated at P_0, $J = diag(1, 0.6, -0.5)$. The presence of positive eigenvalues confirms that the origin is a saddle, regardless of R_3.
- **Tumor-Free State (P_H):** At $(0, 1, 0)$, J yields eigenvalues $\lambda = \{0, -0.6, -0.5\}$. The zero eigenvalue indicates a non-hyperbolic equilibrium.

Using the Julia implementation described in Listing 2.1, we validated the existence and stability of these points, as summarized in table [2.2].

Point	**Coordinates** (x_1, x_2, x_3)	**Eigenvalues** (λ_i)	**Nature**
P_0	$(0, 0, 0)$	$1.0, 0.6, -0.5$	Saddle
P_H	$(0, 1, 0)$	$0.0, -0.6, -0.5$	Non-hyperbolic
P_1	$(1, 0, 0)$	$-1.0, -0.9, -0.2$	**Stable Sink**

Table 2.2: Numerical classification of unique equilibrium points for $R_3 = 1.0$.

At this stage, the phase portrait of the system looks like this:

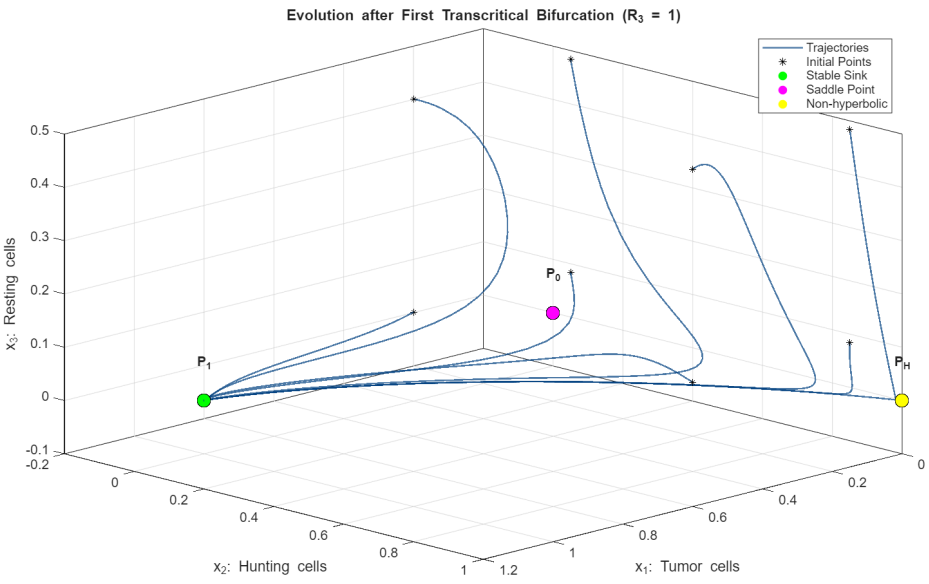


Figure 2.4: Phase portrait of the system with R3=1

2.5.3 First transcritical bifurcation

The stability of the tumor-saturated state P_1 depends on R_3. At $(1, 0, 0)$, the eigenvalues are $\lambda = \{-1, -0.9, \frac{R_3}{2} - 0.7\}$. When $R_3 = 1.4, \lambda_3 = 0$ and P_1 loses its stability. For $R_3 > 1.4$, the tumor can no longer escape immune detection completely, and the partial dormancy state P_{13} enters the biological domain. The phase portrait of the system changes accordingly:

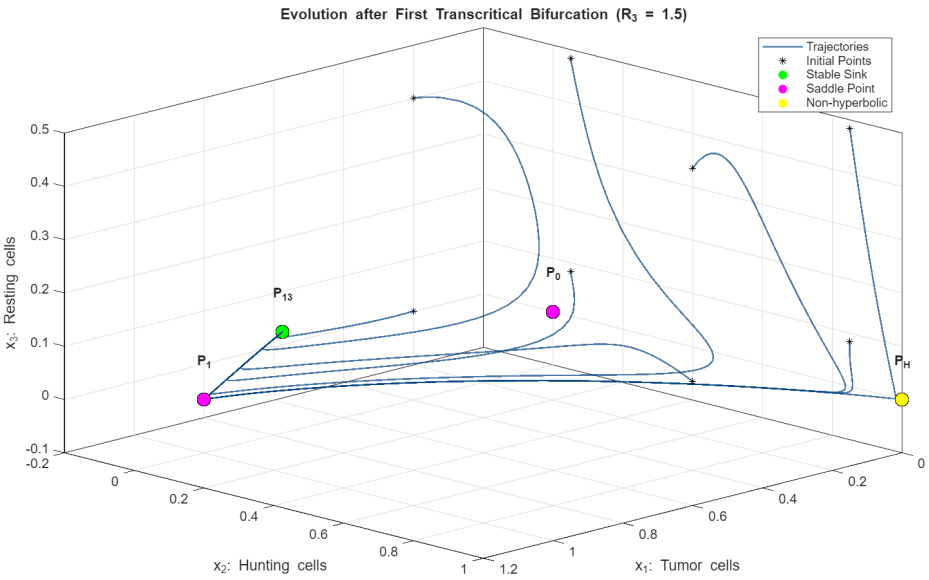


Figure 2.5: Phase portrait of the system with R3=1.5

The Julia script previously used, now updated with the new $R3$ value, confirms these conclusions as summerized in table [2.3]:

Point	**Coordinates** (x_1, x_2, x_3)	**Eigenvalues** (λ_i)	**Nature**
P_0	$(0, 0, 0)$	$1.0, 0.6, -0.5$	Saddle Point
P_H	$(0, 1, 0)$	$-0.6, -0.5, 0.0$	Non-hyperbolic
P_1	$(1, 0, 0)$	$-1.0, -0.9, 0.05$	Saddle Point
P_{13}	$(0.7753, 0, 0.0899)$	$-0.707, -0.563, -0.068$	**Stable Sink**

Table 2.3: Classification of unique equilibrium points for $R_3 = 1.5$.

2.5.4 Evolution of dormancy and the second transcritical bifurcation

As the recruitment rate R_3 increases beyond 1.4, the newfound equilibrium moves along the $x_2 = 0$ plane.

The coordinates of P_{13} depend strictly on R_3 through the quadratic relationship (2.8). As R_3 grows, the tumor coordinate x_1^* decreases, while the resting cell population x_3^* increases.

The critical question regarding the tumor-immune dynamics is whether the dormancy state $P_{13} = (x_1^*, 0, x_3^*)$, is robust against the re-emergence of the specific immune response.

To understand what will happen to the hunting cells population, we can reorder the rows and columuns of the Jacobian matrix so that it assumes a block upper-triangular form:

$$J_{ordered} = \left[\begin{array}{cc|c} J_{11} & J_{13} & J_{12} \\ J_{31} & J_{33} & J_{32} \\ \hline 0 & 0 & J_{22} \end{array}\right] \tag{2.12}$$

We can observe that, since we exchanged the second row with the third, and the second column with the third, the sign of the determinant remains unchanged. Consequently, the stability of the hunting cell population decouples from the rest of the system and is governed solely by the isolated element in the bottom-right block, known as the **transverse eigenvalue**:

$$\lambda_{x2} = J_{22}(P_{13}) = R_2(1 - 2x_2) - A_{21}x_1\Big|_{x_2=0} = 0.6 - 1.5x_1^* \tag{2.13}$$

For P_{13} to remain a stable sink, this eigenvalue must be negative. This leads to the critical condition $x_1^* > 0.4$.

$$\lambda_{x2} = 0.6 - 1.5x_1^* \tag{2.14}$$

The second **Transcritical bifurcation** occurs when P_{13} loses its stability, exactly when $x_1^* = 0.4$. By substituting this value into the equilibrium equation, we can derive the critical recruitment rate $R_{3,T2}$:

$$0.2(0.4)^2 + (0.7 - R_3)(0.4) + 0.5 = 0 \tag{2.15}$$

Solving for R_3, we obtain the threshold:

$$0.4R_3 = 0.812 \implies R_{3,T2} \approx 2.03 \tag{2.16}$$

For $R_3 > 2.03$, a new stable equilibrium, P_3, enters the biological

domain, with the phase portrait changing again:

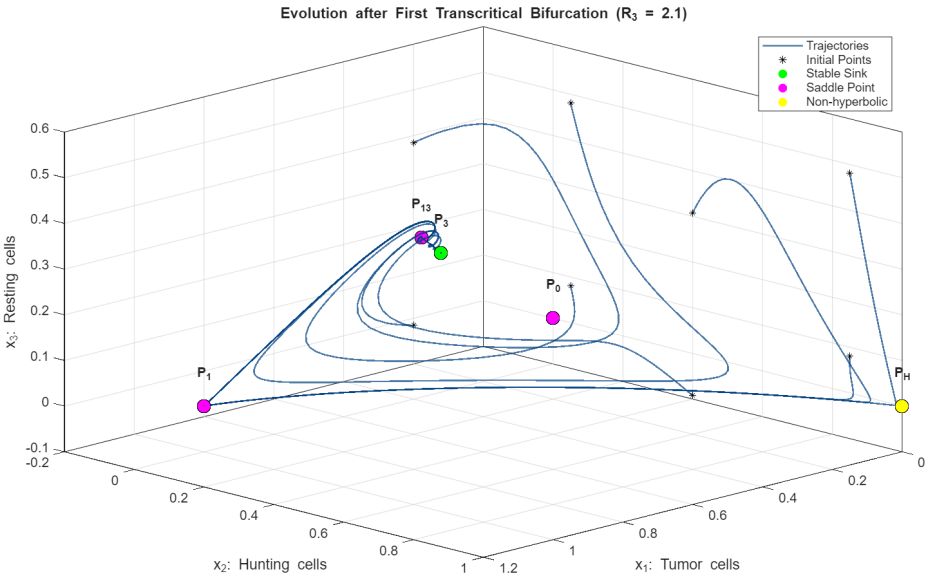


Figure 2.6: Phase portrait of the system with R3=2.1

2.6 Numerical bifurcation tracking

While the previous sections relied on analytical derivations to identify critical thresholds, a more robust validation is required to ensure no bifurcations phenomena are missed. To this end, we utilize the `BifurcationKit.jl` library in Julia to perform numerical continuation.

2.6.1 Conceptual framework: numerical continuation

The numerical continuation method is a powerful tool for tracing the solution manifold of the system equation $F(\mathbf{x}, R_3) = 0$ as the parameter R_3 varies. Unlike brute-force simulation, this algorithm employs a **Pseudo-Arclength Continuation (PALC)** scheme. This method parameterizes the curve by its arc length s, treating the parameter R_3 as a variable rather than a fixed constant.

The tracking proceeds through a **Predictor-Corrector** iterative process:

- **Predictor step:** Given a known point on the equilibrium curve, the algorithm estimates the next point by moving along the tangent vector of the continuation curve.

- **Corrector step:** The prediction is refined using a **Newton-type** solver to project the point orthogonally back onto the equilibrium manifold $F(\mathbf{x}, R_3) = 0$.

2.6.2 Implementation and validation

To automate the detection of singular points and verify the analytical thresholds derived in Section 2.5, a dedicated algorithm was implemented using the Julia language and the `BifurcationKit.jl` library:

```
1
2  using BifurcationKit, Plots, LinearAlgebra,
       StaticArrays
3
4  using BifurcationKit, Plots, LinearAlgebra,
       StaticArrays
5
6  function Kuznetsov!(f, x, p)
7      x1, x2, x3 = x
8      R2, R3, D3, A12, A21, A13, A31, k3, u = p
9
10     f[1] = x1 * (1 - x1) - A12 * x1 * x2 - A13 * x1
           * x3
11     f[2] = R2 * x2 * (1 - x2) - A21 * x1 * x2
12     f[3] = (R3 * x1 * x3) / (x1 + k3) - A31 * x1 *
          x3 - D3 * x3 + u
13     return f
14 end
15
16 function Jac_Kuznetsov(x, p)
17     x1, x2, x3 = x
18     R2, R3, D3, A12, A21, A13, A31, k3, u = p
19     J = [
20         (1 - 2*x1 - A12*x2 - A13*x3)   (-A12 * x1)
                                  (-A13 * x1);
21         (-A21 * x2)                    (R2 * (1 -
              2*x2) - A21 * x1)     (0.0);
22         ((R3*x3*k3)/(x1 + k3)^2 - A31*x3) (0.0)
                                    ((R3*x1)/(x1 + k3
              ) - A31*x1 - D3)
23     ]
```

```
    return J
end

record_fields(x, p; kwargs...) = (x1 = x[1], x2 = x[2], x3 = x[3], param = p)

p_valore = [0.6, 1.0, 0.5, 1.0, 1.5, 2.5, 0.2, 1.0, 0.0]
x0 = [1.0, 0.0, 0.0]

prob = BifurcationProblem(Kuznetsov!, x0, p_valore, 2;
    J = Jac_Kuznetsov,
    record_from_solution = record_fields)

opts_br = ContinuationPar(
    p_min = 0.5,
    p_max = 5.0,
    ds = 0.01,
    dsmax = 0.05,
    detect_bifurcation = 3,
    n_inversion = 10,
    max_steps = 1000
)


br1 = BifurcationKit.continuation(prob, PALC(), opts_br; plot = false)

br2 = BifurcationKit.continuation(br1, 1, opts_br; plot = false)

br3 = BifurcationKit.continuation(br2, 2, opts_br; plot = false)

hopf_points = filter(x -> x.type == :hopf, br3.specialpoint)

if !isempty(hopf_points)
    for (i, h) in enumerate(hopf_points)
        println("Hopf bifurcation #$i found at R3 =
```

```
            ", round(h.param, digits=4))
    end
else
    println("No Hopf found.")
end
```

Listing 2.2: Julia script for bifurcation identification

The script employs a *Branch Switching* strategy to reconstruct the complete history of the system's equilibria.

Detection of the Hopf bifurcation

Beyond validating the transcritical thresholds, the numerical continuation reveals a critical dynamic phenomenon along the third branch. The algorithm identifies a **Hopf Bifurcation**, where a pair of complex conjugate eigenvalues crosses the imaginary axis.

The script output localizes this singularity at the value:

$$R_{3,H} \approx 2.686 \tag{2.17}$$

This result indicates that for $R_3 > R_{3,H}$, the coexistence equilibrium P_3 loses its stability not to another fixed point, but by generating a stable **Limit Cycle**:

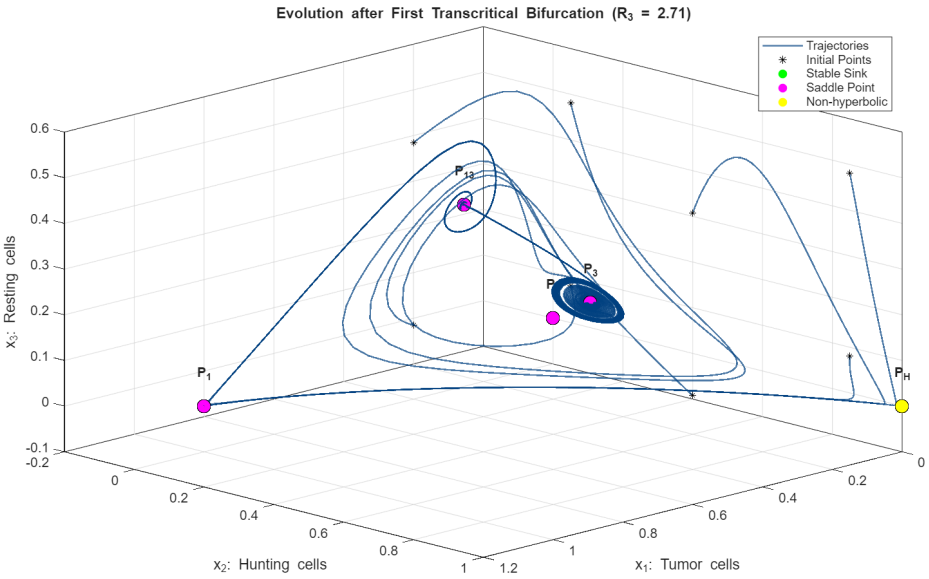


Figure 2.7: Phase portrait of the system with R3=2.71

The plots of the state variables are the following:

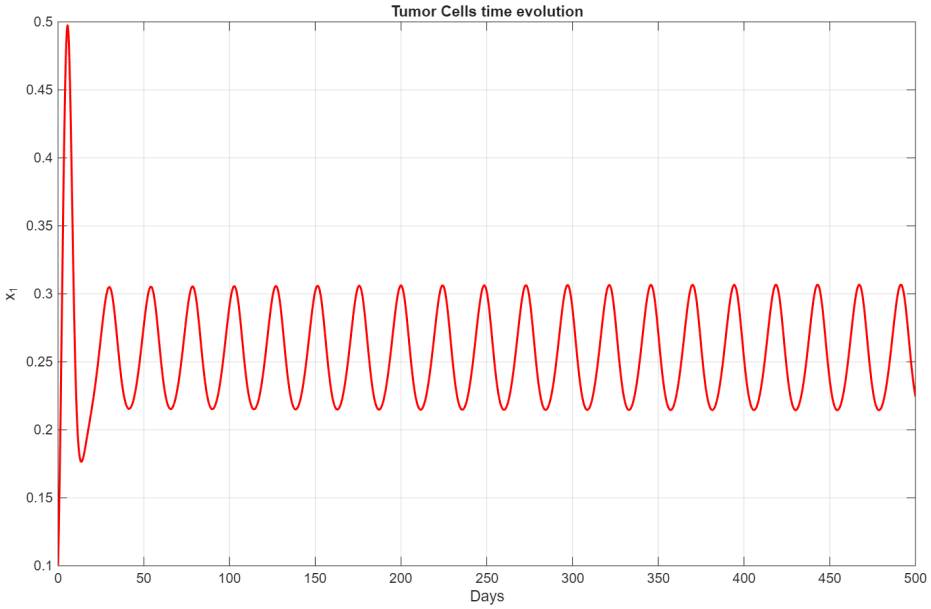


Figure 2.8: Time evolution of tumor cells with $R_3 = 2.71$

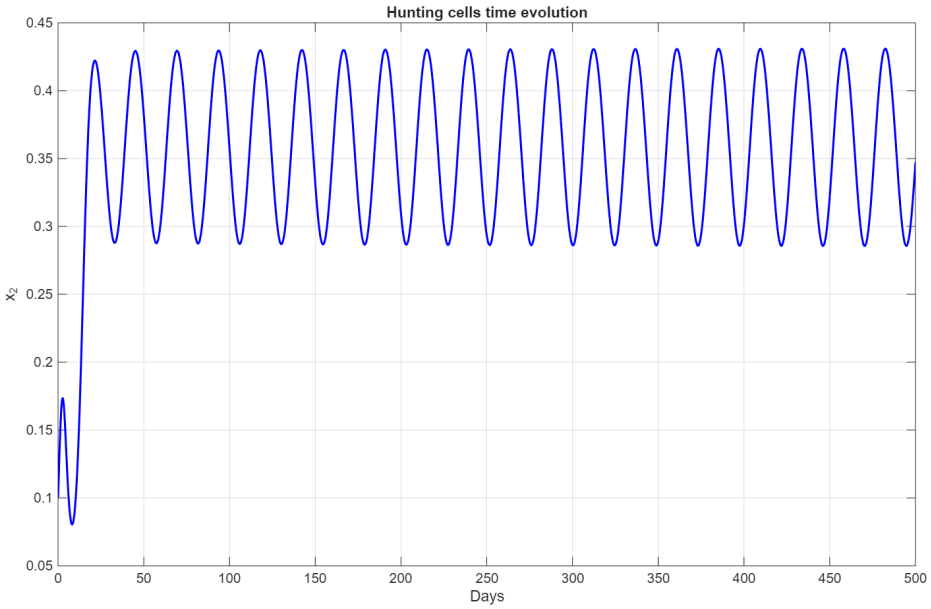


Figure 2.9: Time evolution of hunting cells with $R_3 = 2.71$

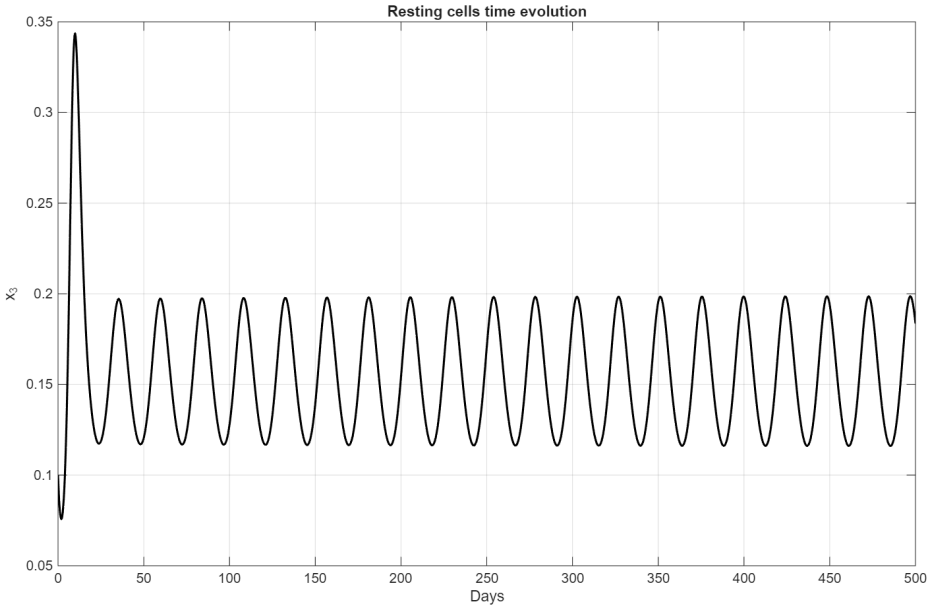


Figure 2.10: Time evolution of resting cells with $R_3 = 2.71$

2.7 The period-doubling cascade

The detection of the Hopf bifurcation at $R_{3,H} \approx 2.686$ marks the onset of oscillatory behavior. However, numerical simulations at the nominal value $R_3 = 4.5$ revealed complex, aperiodic dynamics. To understand the global mechanism connecting the limit cycle to the chaotic regime, we performed an analysis using a **Poincaré Map approach**.

2.7.1 Methodology: the Poincaré section of maxima

We defined a Poincaré section Σ corresponding to the local maxima of the tumor population:

$$\Sigma = \left\{ \mathbf{x} \in \mathbb{R}^3 \mid \dot{x}_1 = 0, \ddot{x}_1 < 0 \right\} \tag{2.18}$$

This discretization technique, based on the sequence of successive local maxima, follows the methodological approach originally introduced by Lorenz [9] to analyze the deterministic nature of aperiodic flows.

By sampling the trajectory only when it crosses Σ, the continuous flow is transformed into a discrete map $x_{k+1} = P(x_k, R_3)$. This

allows us to distinguish between different dynamic regimes:

- **Period-1 limit cycle:** Appears as a single fixed point on the map.
- **Period-N limit cycle:** Appears as N distinct points.
- **Chaos:** Appears as a dense set of points.

2.7.2 Numerical reconstruction of the Feigenbaum tree

Using a numerical algorithm implemented in MATLAB, we constructed the bifurcation tree by varying the recruitment rate R_3 in the range $[2.5, 4.5]$. The resulting diagram Figure [2.11] illustrates the complete route to chaos:

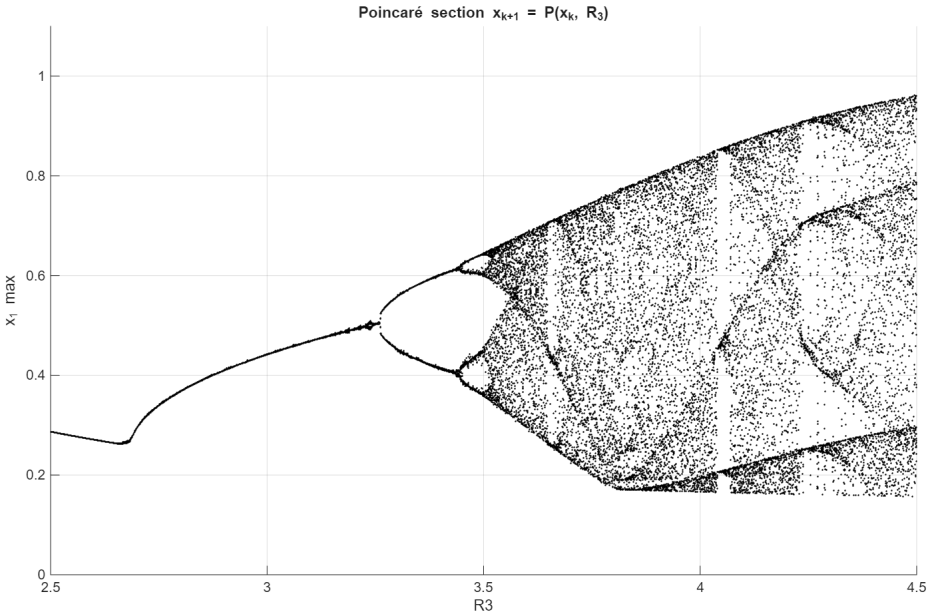


Figure 2.11: Period Doubling cascade

The analysis identifies the following sequence of events:

1. **Stable oscillation** ($2.686 < R_3 < 3.29$)**:** Following the Hopf bifurcation, the system exhibits a stable Period-1 limit cycle. The tumor population oscillates with a constant amplitude.
2. **First flip bifurcation** ($R_3 \approx 3.29$)**:** The branch splits into two. The Period-1 cycle loses stability, and a Period-2 limit cycle emerges.

3. **The cascade ($R_3 > 3.44$):** Further increases in R_3 trigger a rapid succession of Flip bifurcations ($P_2 \to P_4 \to P_8 \dots$), occurring in geometrically shrinking intervals.

2.8 Conclusion of the open-loop analysis

The analysis conducted in this chapter demonstrates that the system is structurally unstable for high values of immune recruitment (R_3). This behavior justifies the need for the **Control synthesis** presented in the next chapter.

Chapter 3

Control synthesis

This chapter focuses on the synthesis of a control strategy to suppress the chaotic behavior exhibited by the system and drive it toward a healthy state.

3.1 Control problem formulation

The primary objective of the control synthesis is the complete eradication of the tumor population, which corresponds to the stabilization of the system at the tumor-free equilibrium point:

$$P_H = [x_1^*, x_2^*, x_3^*]^T = [0, 1, 0]^T \tag{3.1}$$

The performance specifications and constraints for the controller design are as follows:

- **Asymptotic stability:** The control law must ensure that the state trajectories converge asymptotically to the healthy equilibrium:

$$\lim_{t\to\infty} \mathbf{x}(t) = P_H \tag{3.2}$$

- **Control effort constraints:** The control input $u(t)$ is subject to strict physiological constraints. Following the normalization standards established in literature for dimensionless tumor models, we set:

$$0 \leq u(t) \leq u_{max}, \quad u_{max} = 1 \tag{3.3}$$

  This threshold ensures that the therapeutic intervention remains within a range that is biologically sustainable, avoid-

ing the systemic damage often associated with high-dose immunotherapy [4].

.

3.2 Linear control strategy: LQR design

The first approach aimed at achieving tumor eradication involves the design of a Linear Quadratic Regulator. This technique requires a linear state-space representation of the system dynamics, which is obtained by approximating the nonlinear model in the neighborhood of a specific equilibrium point.

3.2.1 Linearization and deviation variables

To apply linear control theory to the nonlinear model, we describe the system's behavior in terms of deviations from a desired equilibrium state. Let $\mathbf{x} = [x_1, x_2, x_3]^T$ be the state vector and u the control input. We define the operating point as (P_ϵ, u^*), where P_ϵ is the chosen equilibrium and u^* is the nominal control input required to maintain that equilibrium (in our case, $u^* = 0$ for the healthy state).

We introduce the **deviation variables** as:

$$\delta\mathbf{x}(t) = \mathbf{x}(t) - P_\epsilon, \quad \delta u(t) = u(t) - u^* \tag{3.4}$$

Under the assumption that the system remains in a sufficiently small neighborhood of P_ϵ, the nonlinear vector field $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, u)$ can be approximated using a first-order Taylor series expansion:

$$\dot{\mathbf{x}} = \mathbf{f}(P_\epsilon, u^*) + \left.\frac{\partial \mathbf{f}}{\partial \mathbf{x}}\right|_{P_\epsilon, u^*} \delta\mathbf{x} + \left.\frac{\partial \mathbf{f}}{\partial u}\right|_{P_\epsilon, u^*} \delta u + \underbrace{\text{HOT}}_{\text{Higher order terms}} \tag{3.5}$$

Since P_ϵ is an equilibrium point, $\mathbf{f}(P_\epsilon, u^*) = 0$. Neglecting the higher-order terms, we obtain the standard linear state-space form:

$$\dot{\delta\mathbf{x}} = A\delta\mathbf{x} + B\delta u \tag{3.6}$$

where:

- $A \in \mathbb{R}^{3\times 3}$ is the **Jacobian matrix** evaluated at the linearization point.

- $B \in \mathbb{R}^{3\times 1}$ is the **input matrix**, which defines how the external control u affects each state variable.

3.2.2 Choice of the linearization point P_ϵ

A stability analysis through linearization at the tumor-free equilibrium P_H reveals a critical structural issue. Evaluating the Jacobian matrix $A = J|_{P_H}$, the first row becomes $[1 - A_{12}, 0, 0]$. Since the control $u(t)$ enters through the third equation (x_3), the input matrix is $B = [0, 0, 1]^T$.

In this state, the **loss of controllability** is evident: the tumor variable x_1 is decoupled from the control action in the linear approximation because the term $-A_{13}x_1x_3$ has a null partial derivative with respect to x_3 when $x_1 = 0$. Mathematically, the controllability matrix $Co = [B, AB, A^2B]$ loses full rank.

To circumvent this limitation, we linearize the system around a **perturbed tumor-free point** P_ϵ:

$$P_\epsilon = [\epsilon, 1, 0]^T, \quad \epsilon \ll 1 \tag{3.7}$$

Choosing $\epsilon = 0.01$ ensures that the coupling between x_3 and x_1 is captured by the linear model, restoring the rank of the controllability matrix.

3.2.3 LQR synthesis

The LQR controller minimizes the quadratic cost function:

$$J = \int_0^\infty (\delta\mathbf{x}^T Q \delta\mathbf{x} + R\delta u^2)dt \tag{3.8}$$

where $Q = \text{diag}(q_1, q_2, q_3)$ and R are selected to prioritize tumor eradication ($q_1 \gg q_2, q_3$). The optimal feedback gain K is then computed such that the control law $\delta u = -K\delta\mathbf{x}$ stabilizes the system.

3.3 Simulation results

In this section, we present the numerical results obtained from the Simulink implementation of the controlled system. The primary focus is to validate the effectiveness of the LQR controller in driving the state trajectories toward the healthy equilibrium point.

3.3.1 Local validation in the neighborhood of P_ϵ

The first batch of simulations assesses the controller's performance when the system is initialized near the linearization point $P_\epsilon = [0.01, 1, 0]^T$.

The Simulink control scheme which was implemented to evaluate the controller performances is the following:

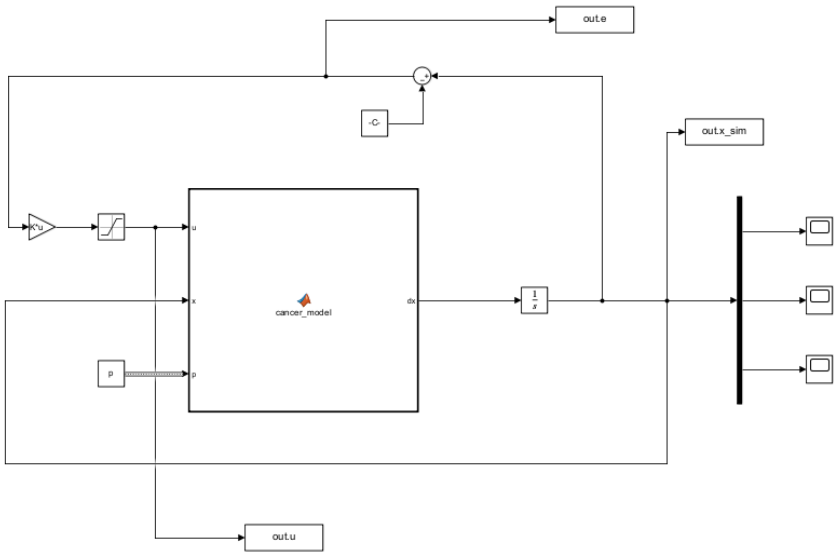


Figure 3.1: Simulink block scheme for LQR control strategy

Simulation results are now presented, showing the performances of the controller in terms of state variables time evolution, error time evolution and control effort time evolution:

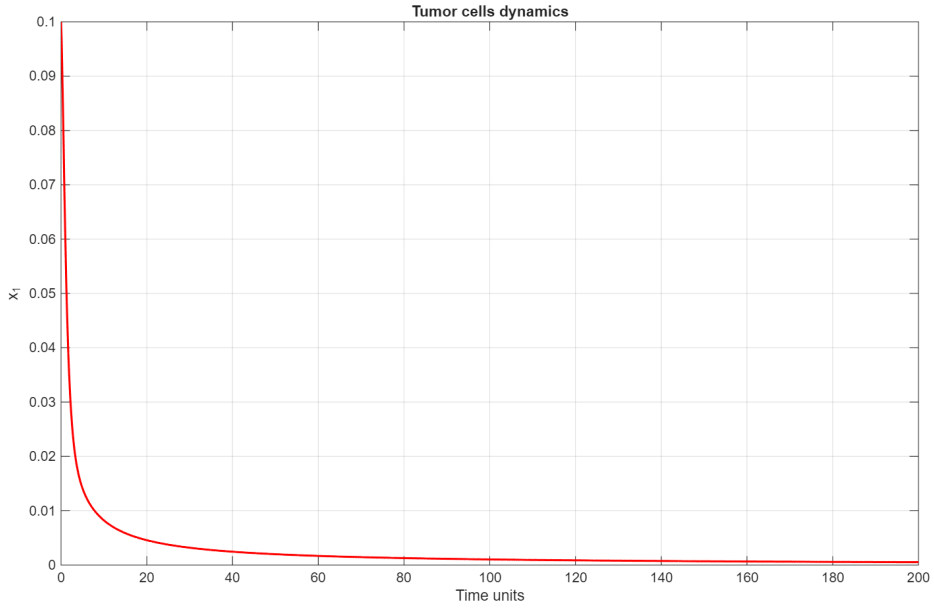


Figure 3.2: Tumor cells time evolution

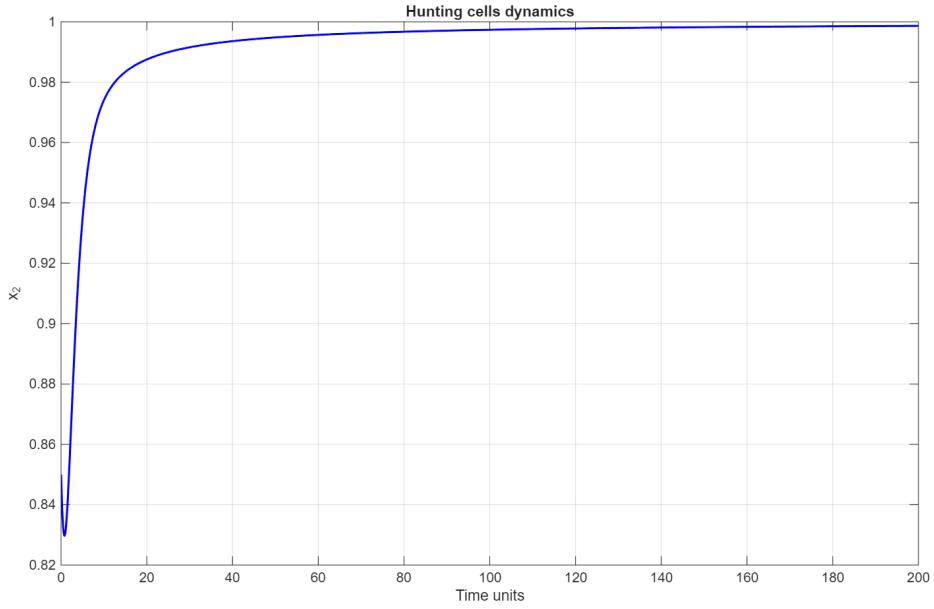


Figure 3.3: Hunting cells time evolution

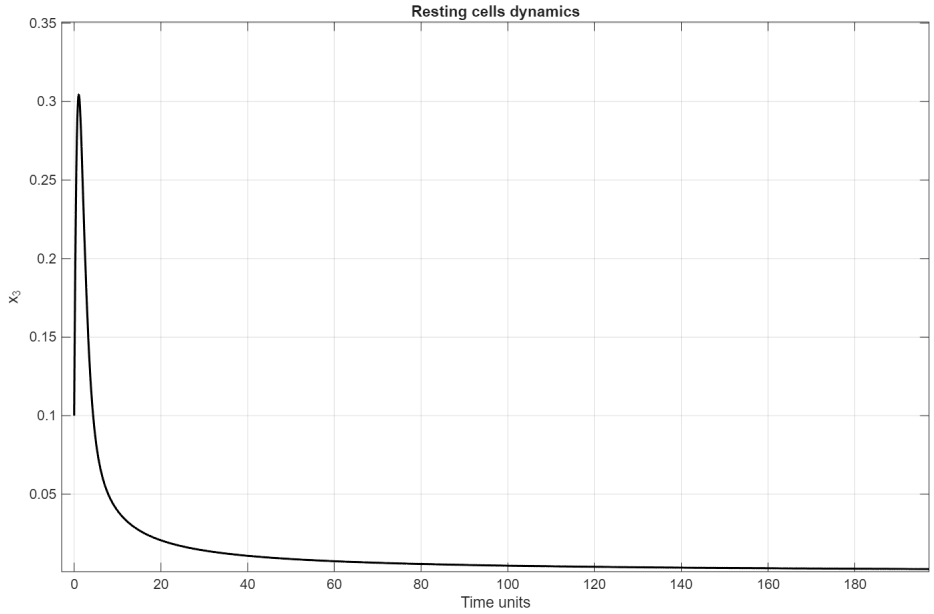


Figure 3.4: Resting cells time evolution

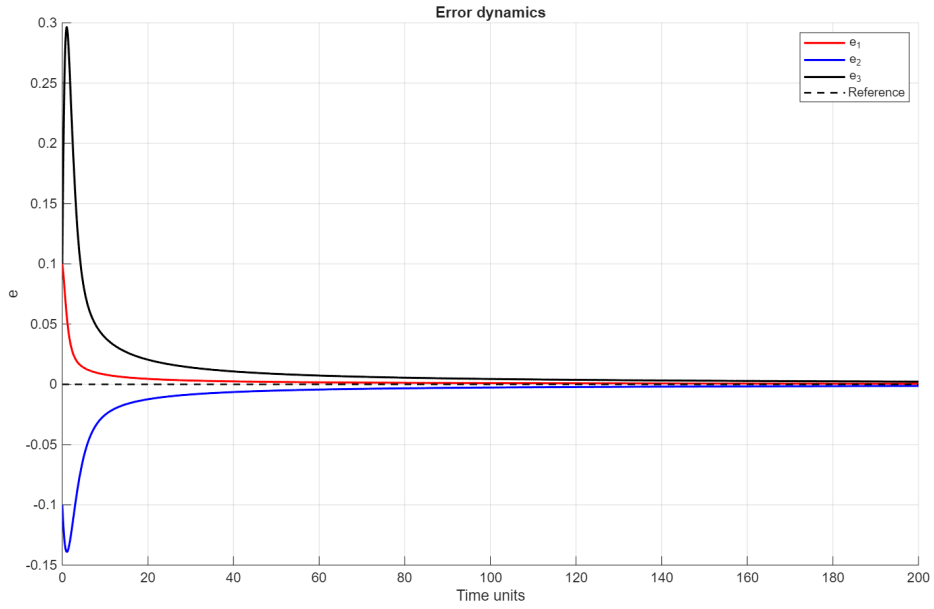


Figure 3.5: Error dynamics

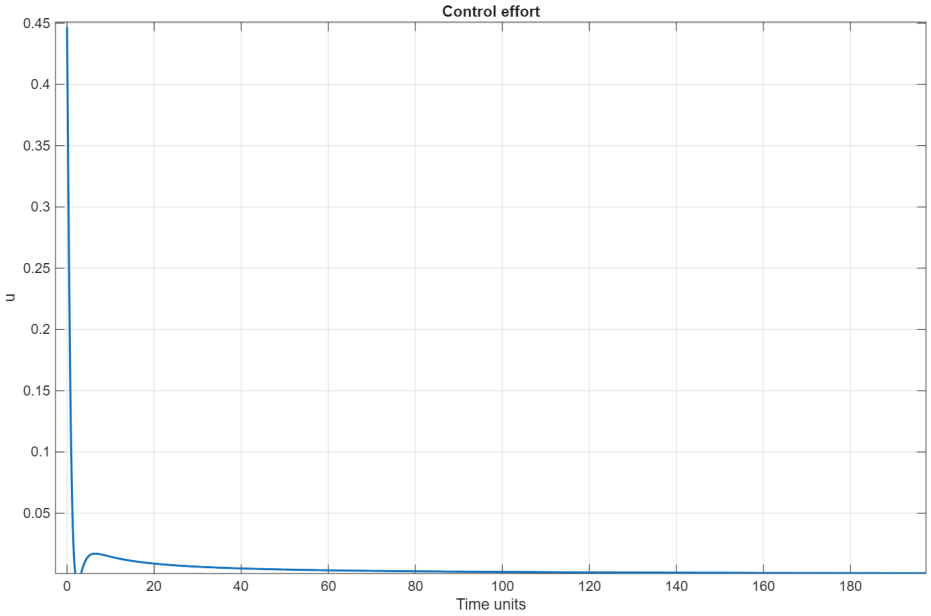


Figure 3.6: Control effort

The simulation results demonstrate the following behaviors:

- **Tumor eradication:** The tumor population x_1 exhibits a rapid descrease.
- **Immune system recovery:** The state variables x_2 and x_3 converge asymptotically to their target values.
- **Control effort:** The control signal $u(t)$ displays an initial peak before rapidly converging toward the origin.

While the theoretical synthesis allows for unconstrained control inputs to be allowed, the practical implementation considers the physical non-negativity of the treatment, as can be seen by the saturation introduced in the control scheme shown in [3.1].

3.3.2 Robustness analysis against parametric uncertainty

To provide a rigorous validation of the LQR controller, a stress test was performed under significant parametric uncertainty. We define a **Global perturbation** where every parameter is varied by + 20% from its nominal value, while the stabilizing gain computed by the LQR is based on the nominal values of the parameters:

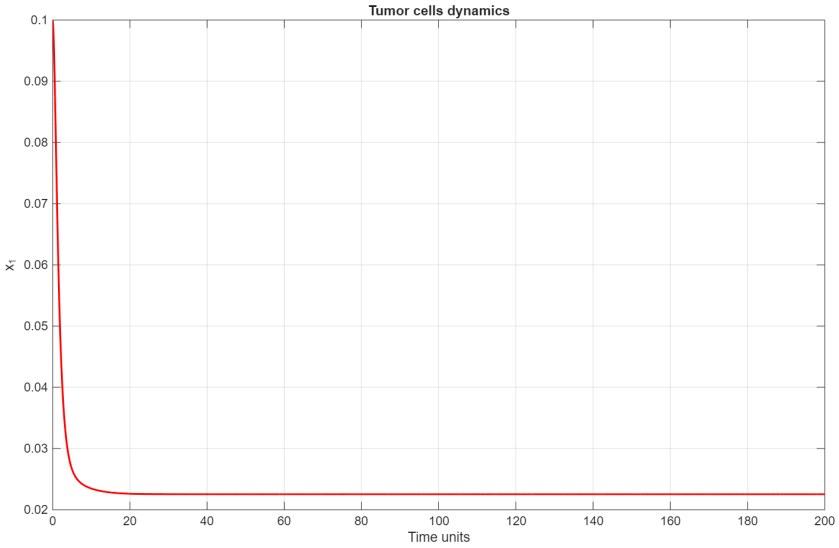


Figure 3.7: Tumor cells dynamics with parametric uncertainty

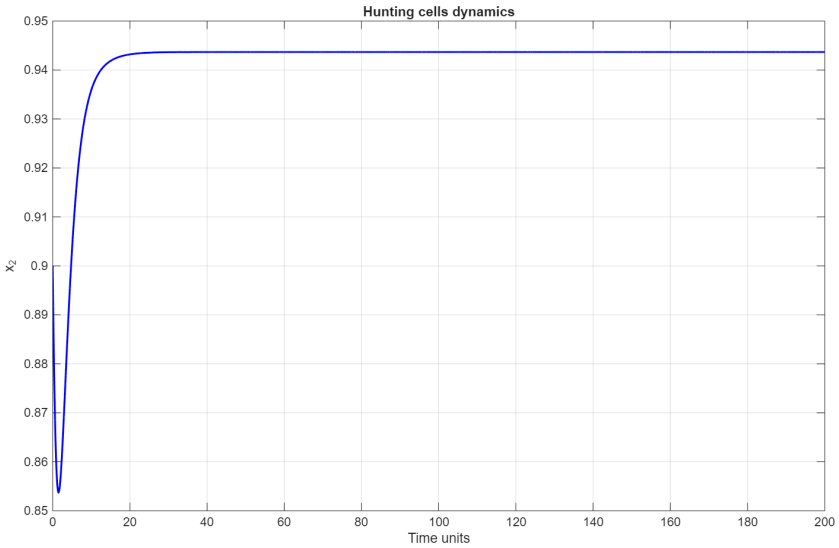


Figure 3.8: Hunting cells dynamics with parametric uncertainty

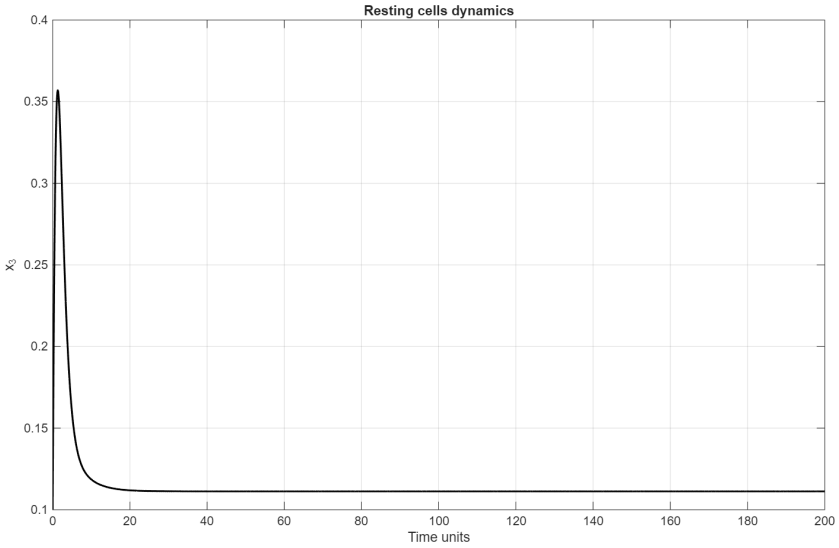


Figure 3.9: Resting cells dynamics with parametric uncertainty

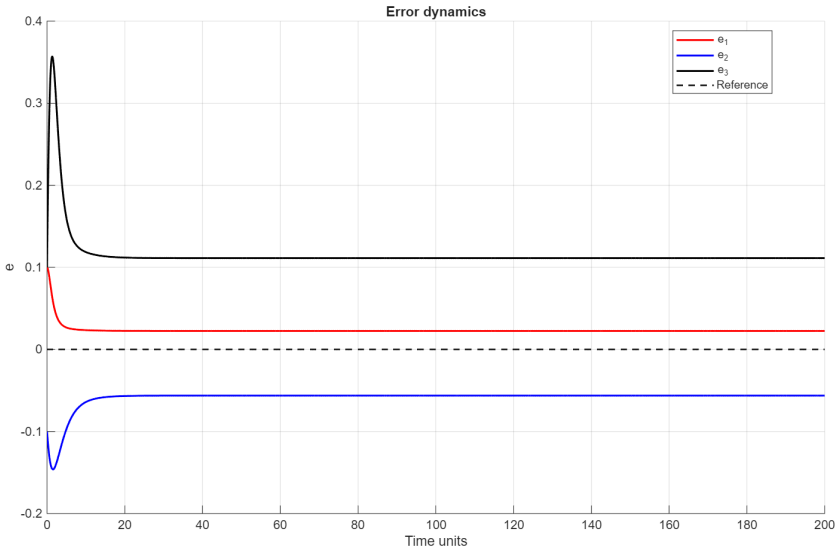


Figure 3.10: Error dynamics with parametric uncertainty

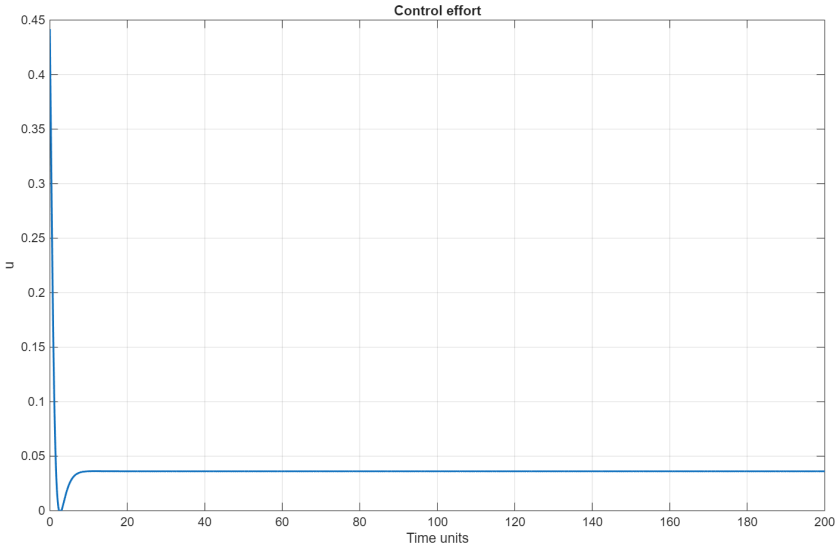


Figure 3.11: Control effort with parametric uncertainty

A detailed observation of the simulation results reveals that, although the tumor is successfully eradicated, the state variables do not converge exactly to the ideal healthy equilibrium $P_H = [0, 1, 0]^T$, thus showing a steady-state error. Nevertheless, even in a stress scenario like the one presented, the controller has shown good performances.

3.3.3 Disturbance rejection analysis

To further assess the robustness of the LQR controller, we analyzed its ability to reject external additive disturbances affecting the tumor population dynamics . This test simulates the outbreak of an unpredictable biological event. The system was modified by adding a time-limited step disturbance to the tumor growth as shown in the following scheme:

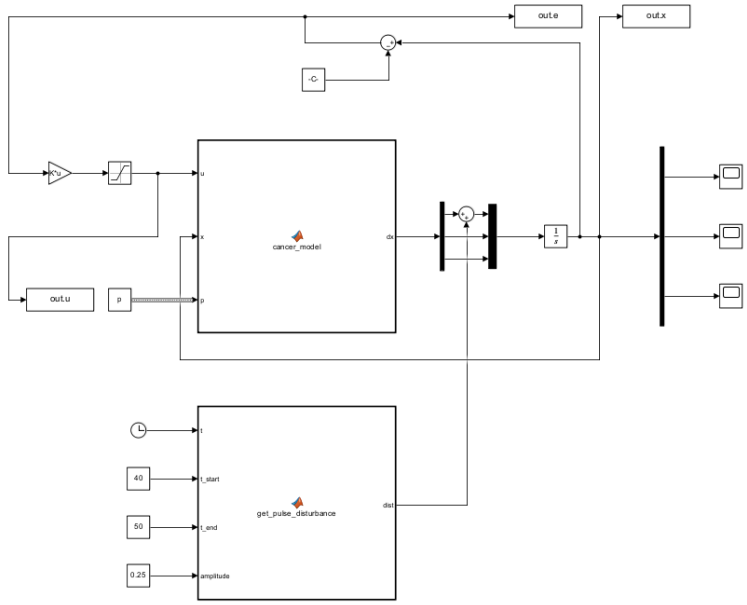


Figure 3.12: Error dynamics with parametric uncertainty

The performances of the controller under the presence of the disturbance are shown below:

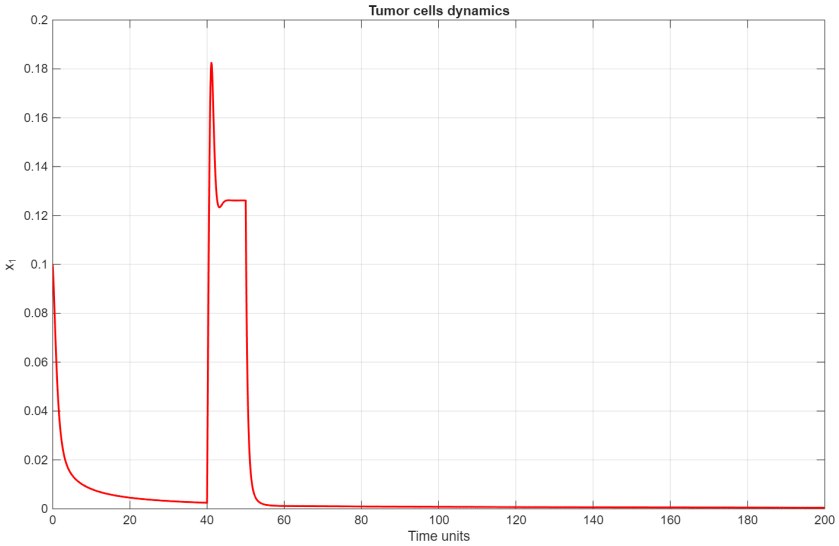


Figure 3.13: Tumor cells time evolution with disturbance

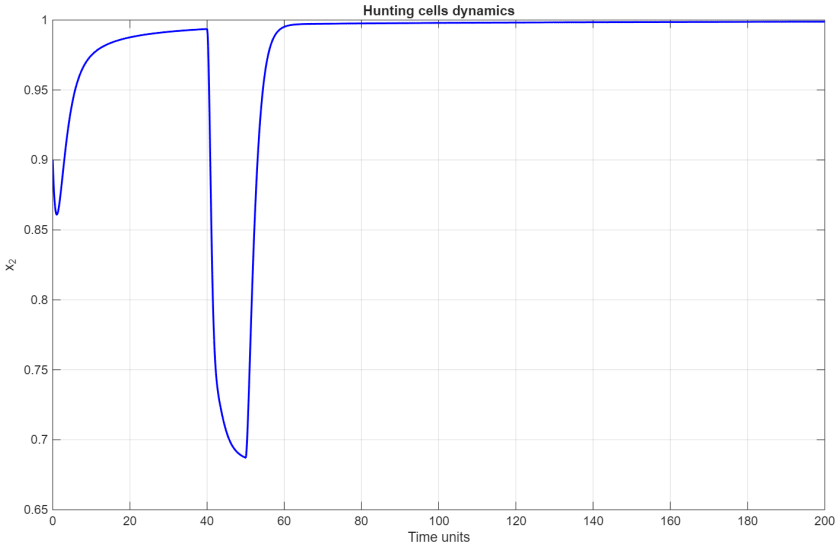


Figure 3.14: Hunting cells time evolution with disturbance

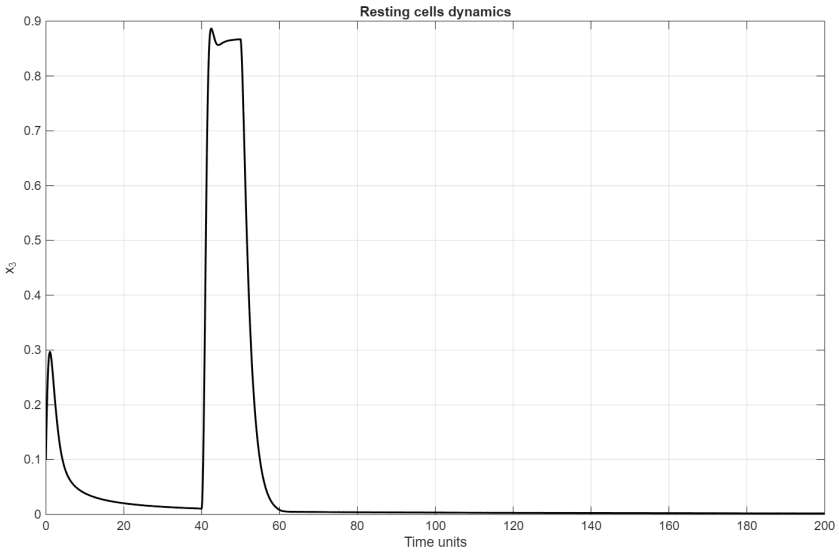


Figure 3.15: Resting cells time evolution with disturbance

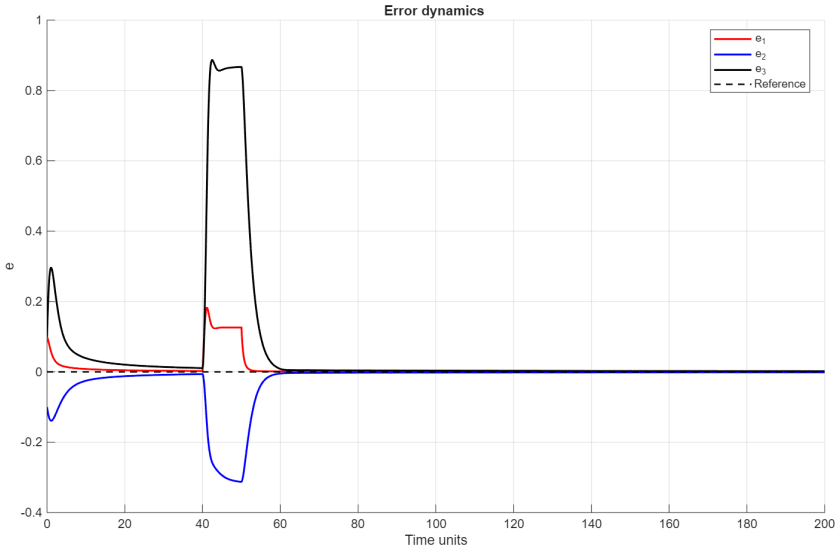


Figure 3.16: Error dynamics with disturbance

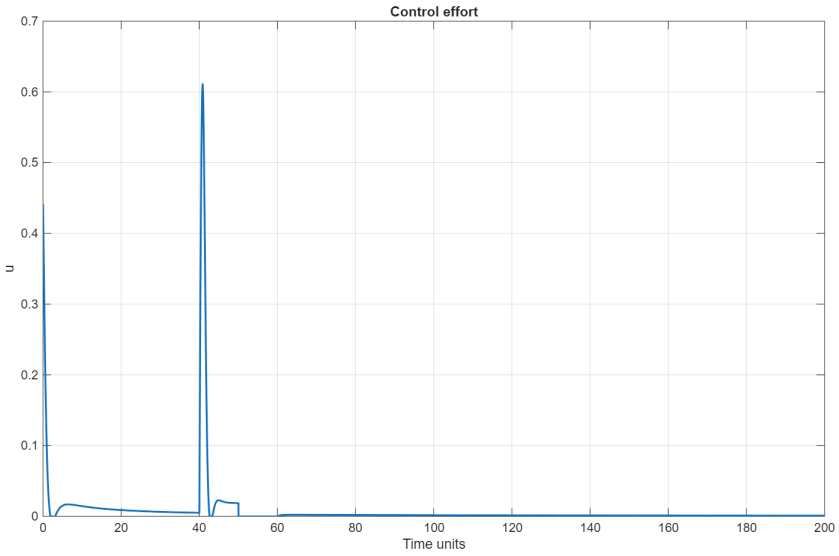


Figure 3.17: Control effort with disturbance

The simulation results confirm that the LQR controller successfully handles the disturbance applied to the system.

3.4 Nonlinear control synthesis

The LQR controller is inherently limited by its local validity. To overcome this restriction, nonlinear control strategies are required.

3.4.1 Input-output feedback linearization

According to the geometric control framework, feedback linearization is a control technique aimed at transforming a nonlinear system into a linear one through a suitable nonlinear control law u. Unlike jacobian linearization, this approach is intended to be valid throughout a region of interest in the state space and not just locally.

Mathematical foundation: Lie derivatives

The fundamental tool for this synthesis is the **Lie derivative**. Given a smooth scalar function $h(x)$ and a smooth vector field $f(x)$, the Lie derivative of h with respect to f is defined as:

$$L_f h = \nabla h \cdot f \tag{3.9}$$

This operation can be nested to define higher-order derivatives, $L_f^r h = L_f(L_f^{r-1} h)$, with $L_f^0 h = h$.

Relative degree and control law computation

The core idea of I/O FBL is to differentiate the output equation $y = h(x)$ repeatedly until the control input u appears explicitly. The number of required differentiations r is defined as the **relative degree** of the system. For an affine SISO system in the form$\dot{x} = f(x) + g(x)u$, the first derivative is:

$$\dot{y} = L_f h(x) + L_g h(x) u \tag{3.10}$$

If $L_g h(x) \neq 0$, the relative degree is $r = 1$. In this case, the linearizing controller is chosen as :

$$u = \frac{1}{L_g h(x)} \left[-L_f h(x) + v \right] \tag{3.11}$$

resulting in the closed-loop linear dynamics $\dot{y} = v$, where v is an auxiliary input which aims at stabilizing the now linearized system.

3.4.2 Control output selection

Before proceeding with the control design, a critical analysis of the output selection is required. A naive approach would be to choose

the tumor population itself as the controlled output, i.e., $y = h_1(x) = x_1$, and apply I/O FBL to drive it to zero. However, we demonstrate here that this choice leads to a singularity in the control law exactly at the target equilibrium.

Let us consider the system in the control-affine form:

$$\dot{\mathbf{x}} = \underbrace{\begin{bmatrix} x_1(1-x_1) - A_{12}x_1x_2 - A_{13}x_1x_3 \\ R_2x_2(1-x_2) - A_{21}x_1x_2 \\ \frac{R_3x_1x_3}{x_1+k_3} - A_{31}x_1x_3 - D_3x_3 \end{bmatrix}}_{f(\mathbf{x})} + \underbrace{\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}}_{g(\mathbf{x})} u \tag{3.12}$$

Case A: Direct tumor control

If we select $y = x_1$, the first Lie derivative of the output along g is:

$$L_g h_1(x) = \nabla h_1 \cdot g = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = 0 \tag{3.13}$$

Since $L_g h_1(x) = 0$, the relative degree is strictly greater than 1, we proceed to derive the output again:

$$\dot{y} = L_f h_1(x) = \dot{x}_1 = \; x_1(1-x_1) - A_{12}x_1x_2 - A_{13}x_1x_3 \tag{3.14}$$

$$\ddot{y} = L_f^2 h_1(x) + L_g(L_f h_1(x))u \tag{3.15}$$

The decoupling term $L_g L_f h_1(x)$ is calculated as:

$$L_g L_f h_1(x) = \frac{\partial \dot{x}_1}{\partial x_3} \cdot 1 = -A_{13}x_1 \tag{3.16}$$

The feedback linearization control law u would take the form:

$$u = \frac{1}{L_g L_f h_1(x)}(v - L_f^2 h_1(x)) = \frac{1}{-A_{13}x_1}(\dots) \tag{3.17}$$

The control law contains the term x_1 in the denominator. Since the control objective is tumor eradication, the required control effort diverges as the system approaches the target. This makes the direct control of x_1 physically unrealizable and mathematically ill-posed for the eradication task.

Case B: Indirect control via resting cells

Conversely, selecting the the output as $y = h_2(x) = x_3$, yields:

$$L_g h_2(x) = \nabla h_2 \cdot g = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = 1 \neq 0 \tag{3.18}$$

The system has a relative degree $r = 1$ everywhere in the state space. Therefore, we adopt the strategy of controlling x_3 to a reference value sufficient to induce tumor regression through the stable internal dynamics, as detailed in the following sections.

Following this approach, the nonlinear feedback linearizing control law is defined as:

$$u = -\left(\frac{R_3 x_1 x_3}{x_1 + k_3} - A_{31} x_1 x_3 - D_3 x_3 \right) + v \tag{3.19}$$

This controller effectively cancels the nonlinearities in the third state equation, transforming the input-output dynamics into the linear integrator:

$$\dot{y} = \dot{x}_3 = v. \tag{3.20}$$

3.4.3 Analysis of the zero-dynamics

Before considering the controlled system, we investigate the stability of the healthy state P_H by analyzing the *zero-dynamics*. This corresponds to the case where the third population is absent, $x_3 \equiv 0$, which implies $\dot{x}_3 = 0$.

In error coordinates relative to $P_{H,ZD} = (x_1, x_2) = (0, 1)$, we define:

$$e_1 = x_1, \quad e_2 = x_2 - 1 \tag{3.21}$$

The system dynamics on this invariant manifold are given by:

$$\begin{cases} \dot{e}_1 = -e_1(e_1 + e_2) \\ \dot{e}_2 = -(1 + e_2)(R_2 e_2 + A_{21} e_1) \end{cases} \tag{3.22}$$

The biological domain in the error coordinates is: $\mathcal{D} = \{(e_1, e_2) \in \mathbb{R}^2 \mid 0 \leq e_1 \leq 1, -1 \leq e_2 \leq 0\}$

Since the Jacobian matrix at the origin has a zero eigenvalue

($\lambda_1 = 1 - A_{12} = 0$ for $A_{12} = 1$), the stability of the equilibrium $\mathbf{e} = 0$ cannot be determined via linearization.

Using the Matlab app **Phase plane and Slope field**, it was possible to derive the phase portrait of the 2D-system in error coordinates, showing that the origin of the system is unstable:

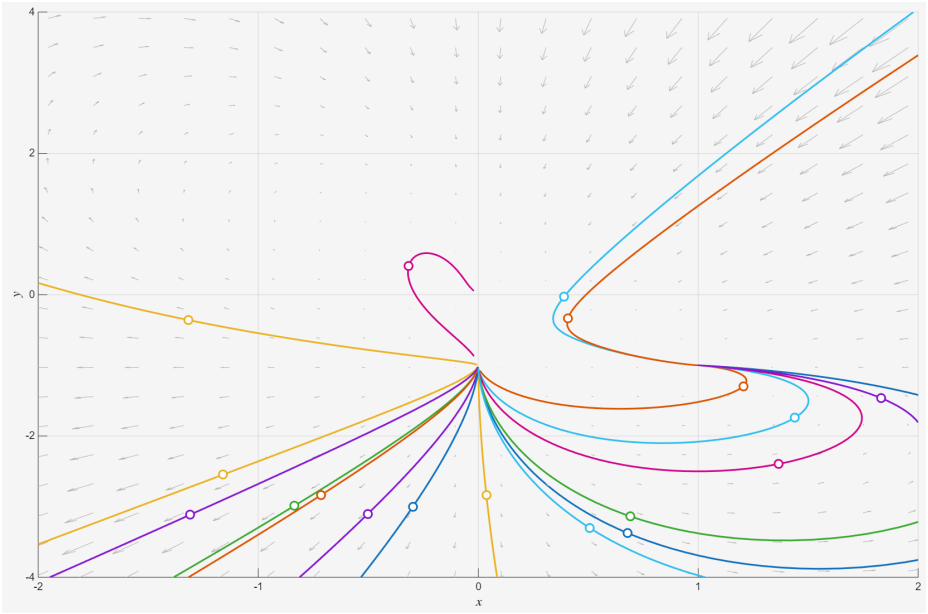


Figure 3.18: Phase portrait of the error system

This proves that this type of analysis is not sufficient to prove the overall stability of the system controlled via I/O FBL.

Computation of the threshold value for x_3

Through the I/O FBL, the non-linearities affecting x_3 are canceled. To ensure the state x_3 tracks a constant reference $x_{3,d}$, we define the auxiliary control law as a proportional controller:

$$v = -k(x_3 - x_{3,d}) \tag{3.23}$$

Substituting equation (3.23) into (3.20) yields the error dynamics for $e_3 = x_3 - x_{3,d}$:

$$\dot{e}_3 = -ke_3 \implies e_3(t) = e_3(0)e^{-kt} \tag{3.24}$$

The selection of $k > 0$ ensures that the error on x_3 asymptotically converges to zero with exponential speed. This allows us to perform

the following analysis by treating $x_3 = x_{3,d}$ as a constant rather than a variable.

To guarantee that the tumor population e_1 is eradicated regardless of the initial state, we apply Lyapunov analysis to the e_1-subsystem. Let us consider the candidate function $V_1(e_1) = e_1$.

The requirement for asymptotic stability of the tumor-free state is $\dot{V}_1 = \dot{e}_1 < 0$. Substituting the dynamics under the I/O FBL control (assuming x_3 has reached $x_{3,d}$):

$$\dot{V}_1 = e_1(1 - e_1 - x_2 - A_{13}x_{3,d}) \tag{3.25}$$

Since $e_1 > 0$, the stability condition $\dot{V}_1 < 0$ is satisfied if and only if:

$$1 - e_1 - x_2 - A_{13}x_{3,d} < 0 \implies A_{13}x_{3,d} > 1 - e_1 - x_2 \tag{3.26}$$

To find a value of $x_{3,d}$ that guarantees that $\dot{V}_1 < 0$, we must satisfy this inequality in the **worst-case scenario**. The right-hand side is maximized when e_1 and x_2 are at their minimum values ($e_1 \to 0, x_2 \to 0$)

Using the nominal parameter $A_{13} = 2.5$, the stabilization threshold is:

$$x_{3,d} > \frac{1}{2.5} = 0.4 \tag{3.27}$$

Setting $x_{3,d} = 0.5$ provides a safety margin.

3.4.6 Global asymptotic stability via composite Lyapunov

To prove the Global asymptotic stability of the healthy equilibrium P_H under the treatment condition $x_{3,d} = 0.5$, we follow the methodology for interconnected systems described in Khalil [7].

We first map our error dynamics into the formal structure:

$$\dot{e}_i = h_i(e_i) + g_i(e), \quad i = 1, 2 \tag{3.28}$$

By expanding the system around the equilibrium $(e_1, e_2) = (0, 0)$ and substituting $x_{3,d} = 0.5$ we obtain the following decomposition:

1. **Tumor Subsystem:**

$$\dot{e}_1 = \underbrace{e_1(-1.25 - e_1)}_{h_1(e_1)} + \underbrace{(-e_1e_2)}_{g_1(e)} \tag{3.29}$$

2. **Healthy Cell Subsystem:**

$$\dot{e}_2 = \underbrace{-e_2(1+e_2)}_{h_2(e_2)} + \underbrace{(-1.5e_1(1+e_2))}_{g_2(e)} \tag{3.30}$$

To prove global stability, we define the composite Lyapunov candidate $V = V_1(e_1) + \kappa V_2(e_2)$. Following the entropy-like functions for population persistence proposed by Goh [6], we define:

$$V(e) = e_1 + \kappa[e_2 - \ln(1+e_2)] \tag{3.31}$$

where $\kappa > 0$ is the weighting constant. The derivative with respect to time of V is:

$$\dot{V} = \dot{e}_1 + \kappa\left(\frac{e_2}{1+e_2}\right)\dot{e}_2 \tag{3.32}$$

Substituting the dynamics and expanding the terms:

$$\dot{V} = -e_1^2 - \kappa e_2^2 - 1.25e_1 - e_1e_2 - 1.5\kappa e_1e_2 \tag{3.33}$$

Regrouping the terms we get:

$$\dot{V} = -e_1^2 - \kappa e_2^2 - e_1[1.25 + e_2(1+1.5\kappa)] \tag{3.34}$$

In the biological domain $\mathcal{D}$, the healthy cell error e_2 is bounded in $[-1, 0]$. To ensure $\dot{V} < 0$ globally, the term multiplying e_1 must remain strictly positive. Evaluating the worst-case scenario at $e_2 = -1$:

$$1.25 - 1(1+1.5\kappa) > 0 \implies 0.25 > 1.5\kappa \implies \kappa < 0.166 \tag{3.35}$$

By selecting $\kappa = 0.1$, the term becomes $[1.25+1.15e_2]$, which is ≥ 0.1 for all $e_2 \in \mathcal{D}$. Thus, $\dot{V}$ is strictly negative definite, proving that the healthy state is **Globally asymptotically stable**.

3.5 Simulation results and analysis

In this section, we evaluate the performance of the control system. We compare the results of the ideal I/O FBL against a realistic implementation that employs a sigmoid activation function for drug delivery.

3.5.1 Control system performances with no saturation implemented

This section presents the results of the simulations with the unconstrained control input. The block scheme implementing the I/O FBL controller is the following:

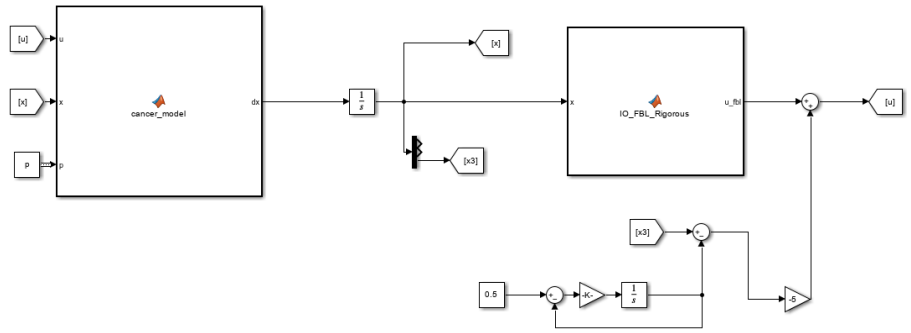


Figure 3.19: Simulink block scheme implementing I/O FBL controller

One thing that can be noticed is the pre-filtering logic applied to the reference signal, which proved to be useful in order to avoid unnatural reactions from the system. The value chosen for the stabilizing gain k is 5, which proved to be sufficiently high to allow x_3 to reach the desired value. The performances attained by the control system are the following:

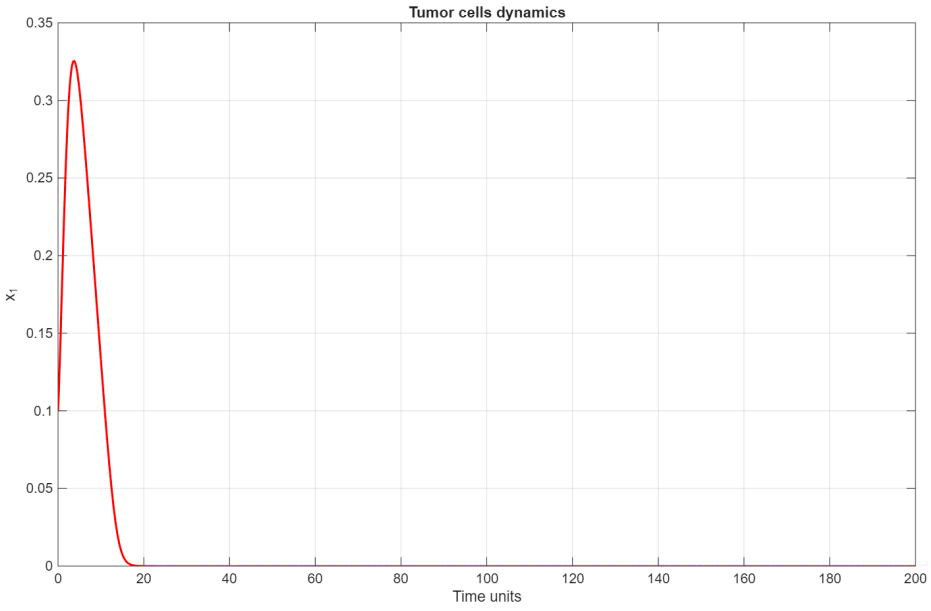


Figure 3.20: Tumor cells time evolution

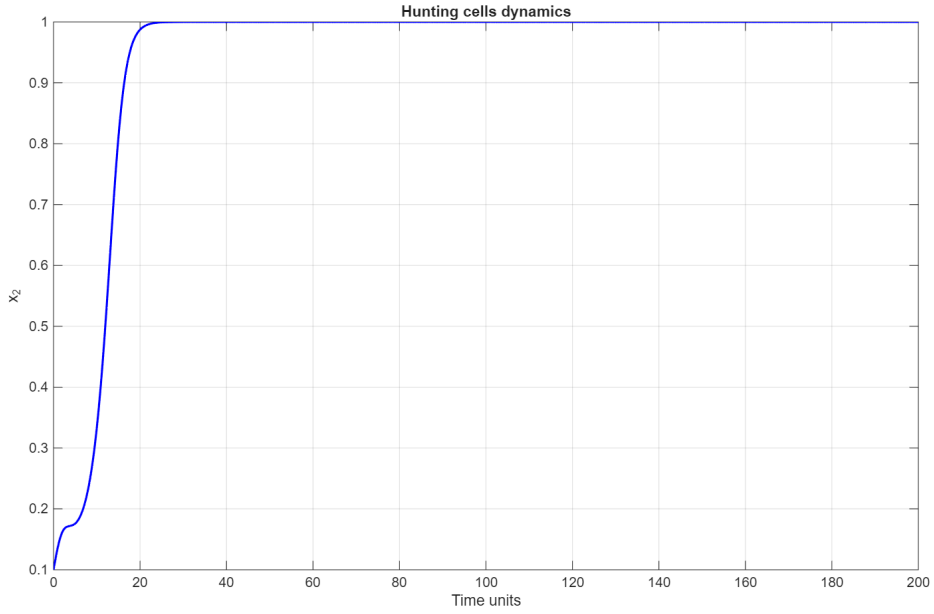


Figure 3.21: Hunting cells time evolution

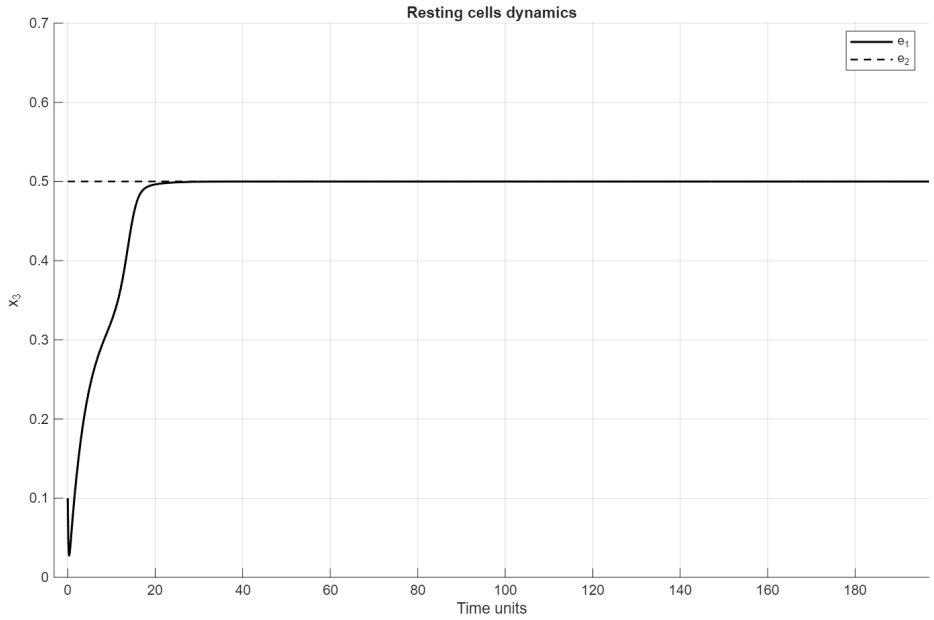


Figure 3.22: Resting cells time evolution

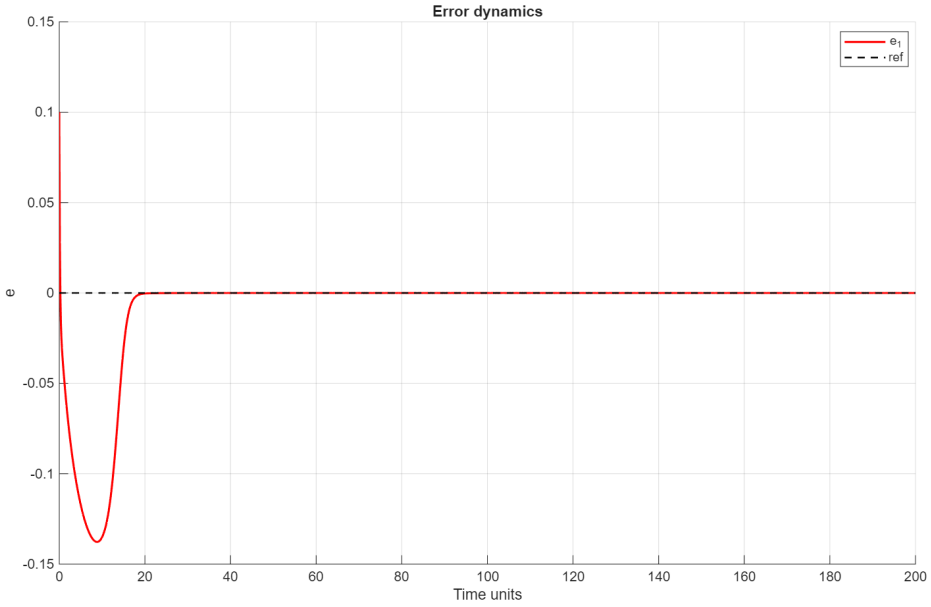


Figure 3.23: Error dynamics

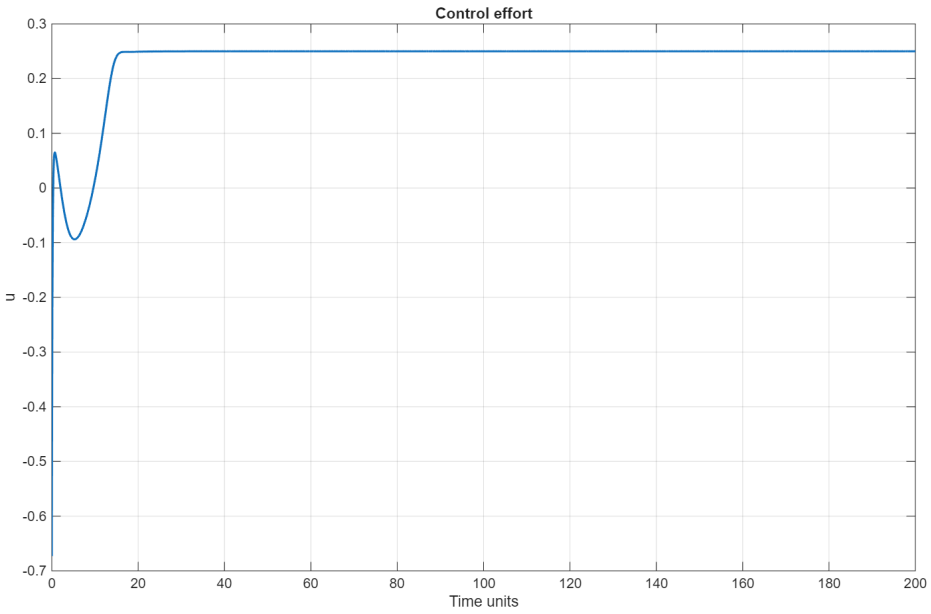


Figure 3.24: Control effort

As can be noticed, the x_3 variable successfully reaches the desired value. Tumor eradication is achieved as well, but the control input attains some negative values which are not biologically possible.

3.5.2 Impact of sigmoid saturation on steady-state error

To constrain the input to attain only positive values, we introduced a sigmoid-type saturation defined as:

$$u = \sigma(u_{raw}) = \frac{1}{1 + e^{-u_{raw}}} \tag{3.36}$$

where u_{raw} is the ideal input, while u the constrained one. By analyzing the time evolution of the x_3 variable, we can observe that now we have a steady state error which does not allow the variable to reach the desired target:

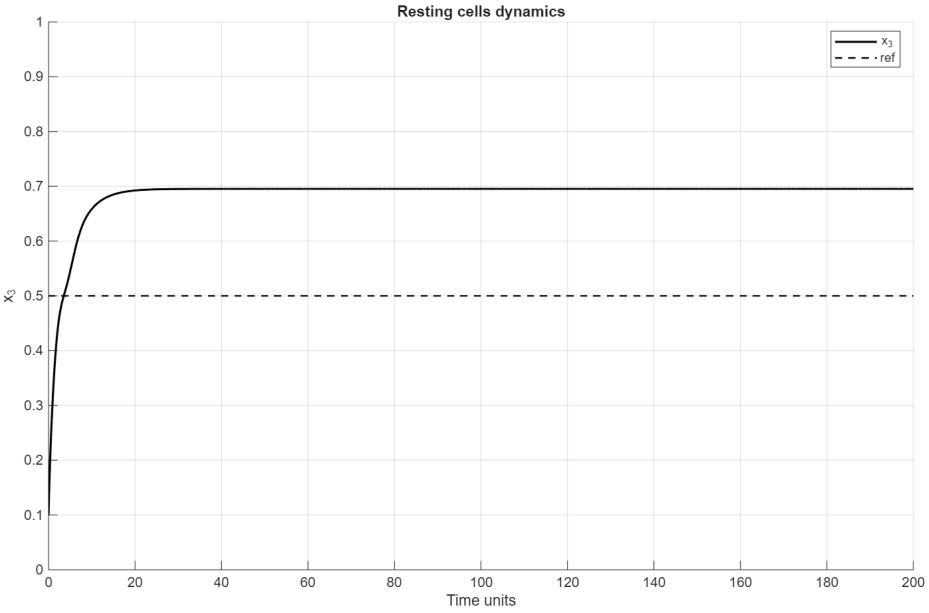


Figure 3.25: Resting cells time evolution with sigmoid saturation

3.5.3 Introduction of integral action and performance analysis

The steady-state error observed is caused by the nature of the sigmoid function employed. To maintain the target equilibrium, the required steady-state control input u_{ss} must satisfy $\dot{x}_3 = 0$. Using the system parameters and the reference $x_{3d} = 0.5$, a specific value of u_{ss} is required to balance the biological decay and interaction terms.

However, the control input u is provided through a sigmoid saturation $\sigma(v)$, where $\sigma(0) = 0.5$. This structural property implies that at zero the actuator provides a fixed dosage that likely differs from

u_{ss}. This discrepancy can be modeled as a constant input disturbance ω acting on the third state equation:

$$\omega = 0.5 - u_{ss} \tag{3.37}$$

Since ω represents a persistent offset, a purely proportional controller cannot achieve asymptotic convergence to the reference. This behaviour justifies the introduction of an integral term to the auxiliary control law. First, we define the new state ξ:

$$\xi(t) = \int_0^t e_3(\tau)d\tau \implies \dot{\xi} = x_3 - x_{3,ref} \tag{3.38}$$

The complete PI control law is then formulated as:

$$v(t) = -K_p e_3(t) - K_i \xi(t) \tag{3.39}$$

where K_p and K_i are the proportional and integral gains respectively. The new control scheme with the integral action implemented is the following:

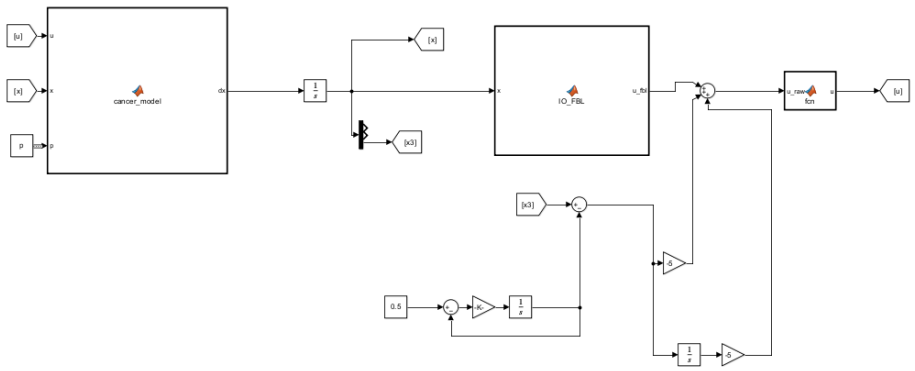


Figure 3.26: Simulink control scheme of I/O FBL controller with integral action

Simulations results are now presented, showing how the presence of the integral action makes it possible for the controller to achieve zero steady-state error:

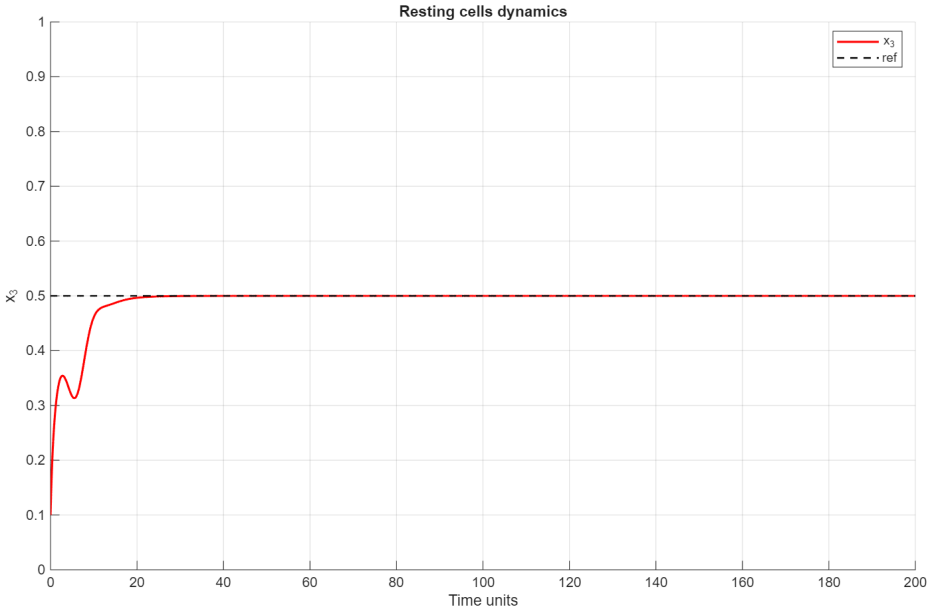


Figure 3.27: Resting cells time evolution with sigmoid saturation and integral action

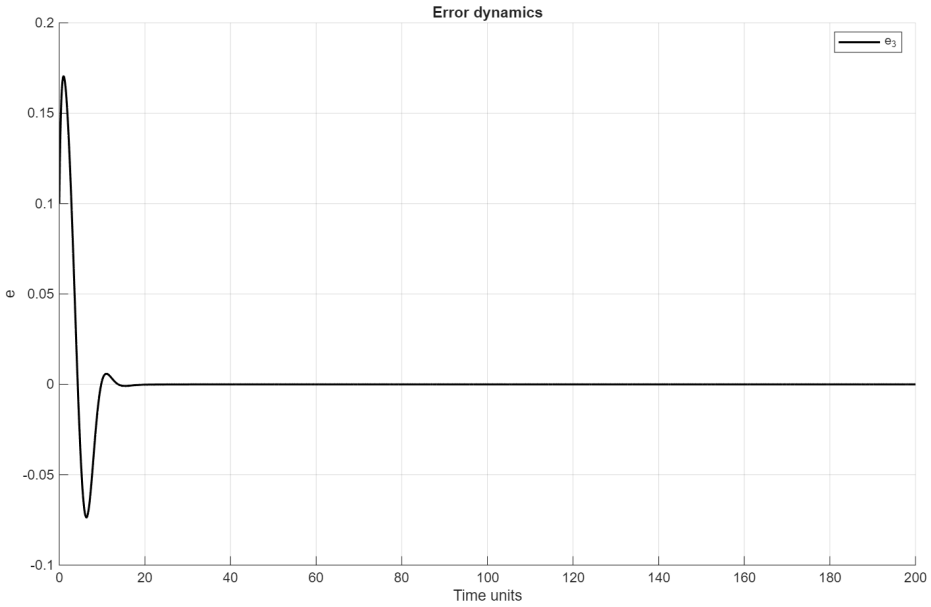


Figure 3.28: Error dynamics with sigmoid saturation and integral action

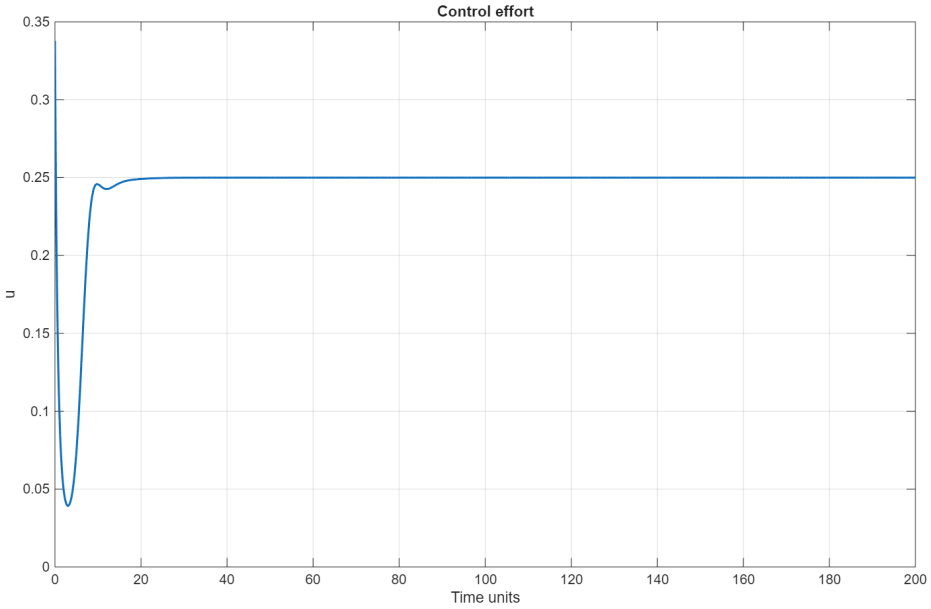


Figure 3.29: Control effort with sigmoid saturation and integral action

3.5.4 Robustness analysis of the I/O FBL controller against parametric uncertainties

The integral action was added to counteract a phenomenon which arised from an implementation choice for the control structure. Nevertheless, it proved to be quite effective even in the presence of parametric uncertainties. To clarify this mechanism, let us denote the real parameters of the system as $P = \{R_3, A_{31}, D_3\}$ and the nominal parameters used to design the control law as $\hat{P} = \{\hat{R}_3, \hat{A}_{31}, \hat{D}_3\}$.

The I/O FBL attempts to cancel the nonlinear dynamics by choosing the control input u as in (3.19). Substituting this control law into (1.3), the closed-loop dynamics take this form:

$$\dot{e}_3 = v + \Delta(x_1, x_3) \tag{3.40}$$

The term $\Delta(x_1, x_3)$ represents the **parametric mismatch residual**, defined as the difference between the real and modeled dynamics:

$$\Delta = (R_3 - \hat{R}_3)\frac{x_1 x_3}{x_1 + k_3} - (A_{31} - \hat{A}_{31})x_1 x_3 - (D_3 - \hat{D}_3)x_3 \tag{3.41}$$

Under the assumption that the system reaches a steady-state equilibrium $(x_{1,ss}, x_{3,ss})$, the residual Δ converges to a constant value Δ_{ss}. In this context, Δ_{ss} acts as a persistent input disturbance. This

demonstrates that the integral action provides the structural robustness necessary to overcome the sensitivity of pure FBL to parametric uncertainties. We consider a parametric uncertainty of -30% to the parameters of interest for the computation of the control law. The plots show the robustness of the controller:

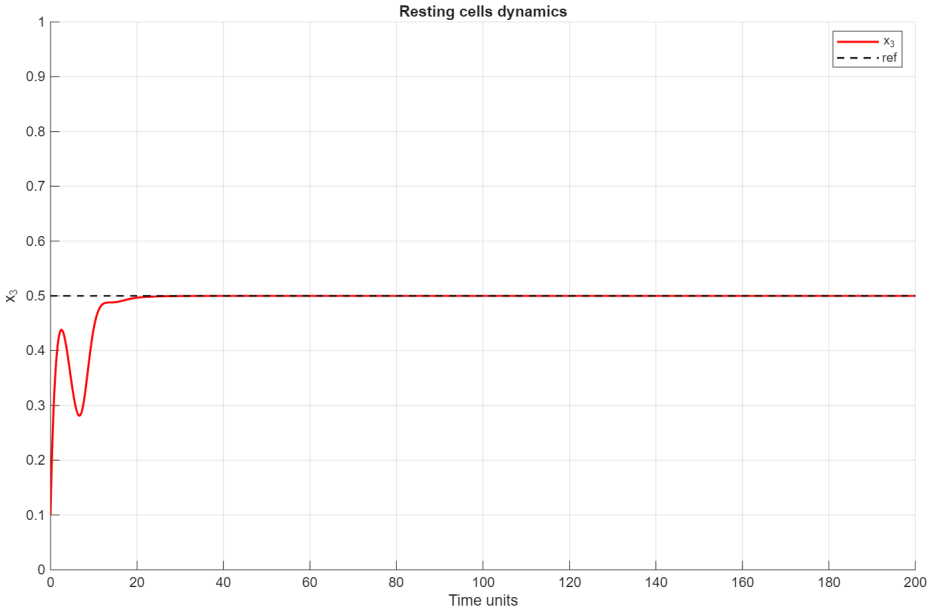


Figure 3.30: Resting cells time evolution with parametric uncertainty

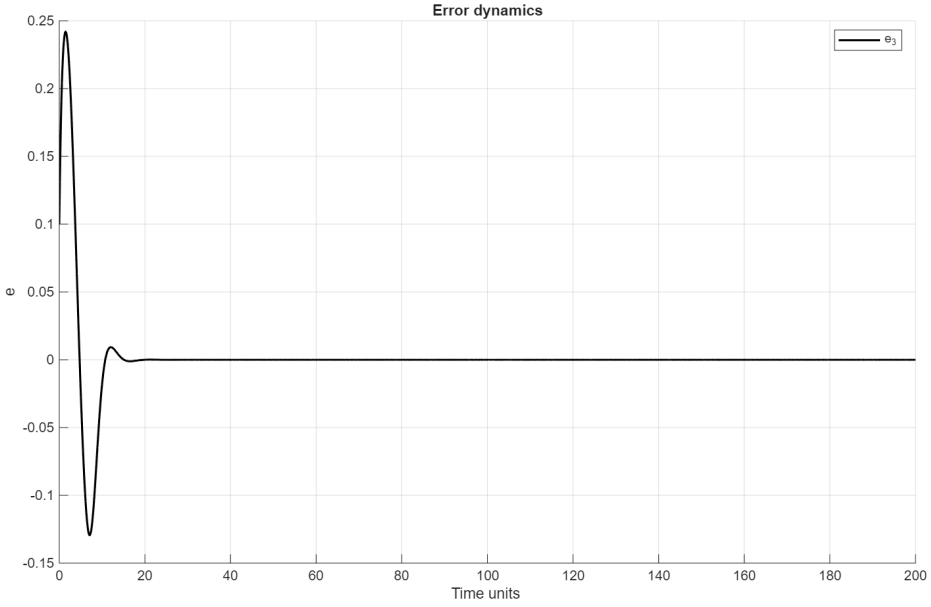


Figure 3.31: Error dynamics with with parametric uncertainty

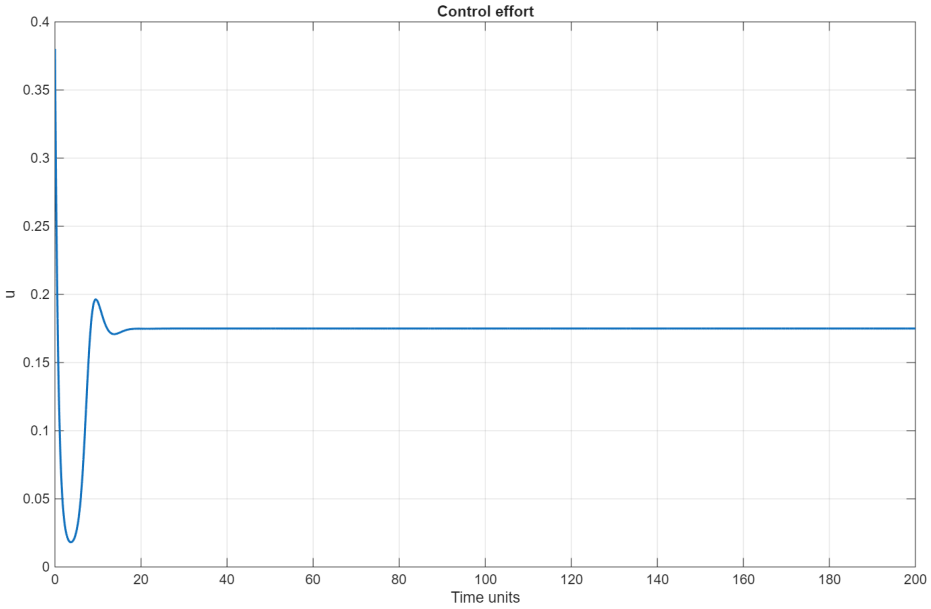


Figure 3.32: Control effort with parametric uncertainty

3.6 Sliding Mode Control design

Sliding Mode Control (SMC) is a class of variable structure control systems characterized by a control law that switches between different modes based on the system state. The design objective is to define a switching logic that makes a predefined manifold Σ, the sliding surface, globally attractive, dividing the control task into two distinct phases:

1. **Reaching phase**: The control law drives the state trajectories from any initial condition toward the manifold Σ in finite time.

2. **Sliding phase**: Once the trajectories hit the manifold, they are trapped and forced to slide along it toward the desired equilibrium point.

3.6.1 Selection of the sliding surface

The first step is to choose a scalar switching function $\sigma(x)$ such that the control specifications are satisfied when the system is on the surface $\Sigma := \{x \in \mathbb{R}^n : \sigma(x) = 0\}$. For the regulation of the resting predator cells x_3, we define:

$$\sigma(x) = \lambda(x_3 - x_{3,d}) \tag{3.42}$$

where $x_{3,d}$ is the reference and $\lambda > 0$ is a design parameter.

3.6.2 Attractivity of the sliding surface and control law compuation

To ensure the sliding surface is globally attractive, we consider the Lyapunov candidate function $V = \frac{1}{2}\sigma^2(x)$. For the surface to be attractive, we must satisfy the condition $\dot{V} = \sigma\dot{\sigma} < 0$. Expanding $\dot{\sigma}$ using the Lie derivatives:

$$\dot{\sigma} = L_f(\sigma) + L_g(\sigma)u \tag{3.43}$$

The control input u is chosen as:

$$u = \frac{1}{L_g(\sigma)}[-L_f(\sigma) - \kappa\text{sgn}(\sigma)] \tag{3.44}$$

where $\kappa > 0$ is a design parameter. Substituting Eq. (3.44) into the Lyapunov derivative yields $\dot{V} = -\kappa|\sigma|$, ensuring the attractivity of the surface.

3.6.3 Verification of the transversality condition

The design in (3.44) is only admissible if the transversality condition $L_g(\sigma) \neq 0$ is satisfied. For the chosen surface and the input vector field g we get:

$$L_g(\sigma) = \nabla\sigma \cdot g = \begin{bmatrix} 0 & 0 & \lambda \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \lambda \tag{3.45}$$

Since λ is a positive constant, the condition is always verified.

3.7 Simulation tests with signum function

In this section we present the results of the simulations carried out adopting the SMC with the *sign* function and no saturation mechanism implemented. The gain K and the parameter λ are set to 10 and 2 respectively. The control structure implemented in Simulink is the following:

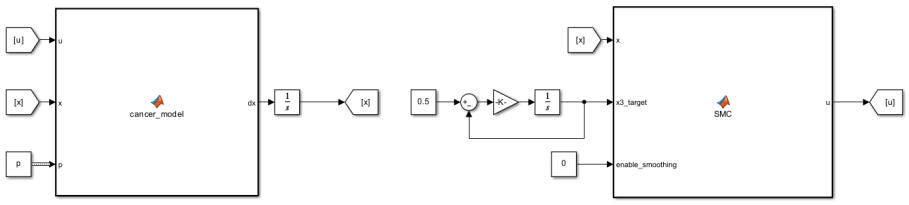


Figure 3.33: Simulink control scheme implementing the SMC controller

Hereafter, simulations showing the time evolution of the state variables, error and control input are shown:

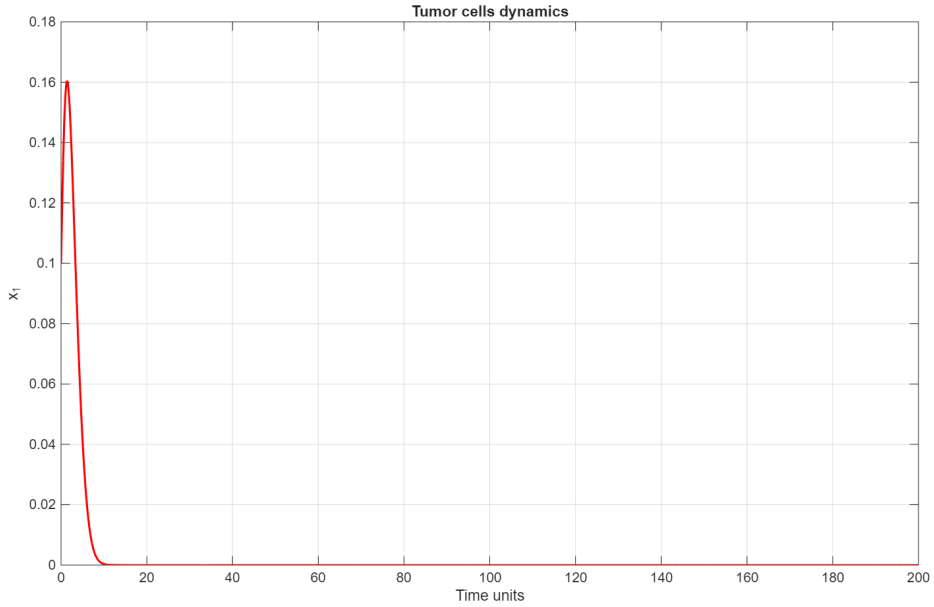


Figure 3.34: Tumor cells time evolution

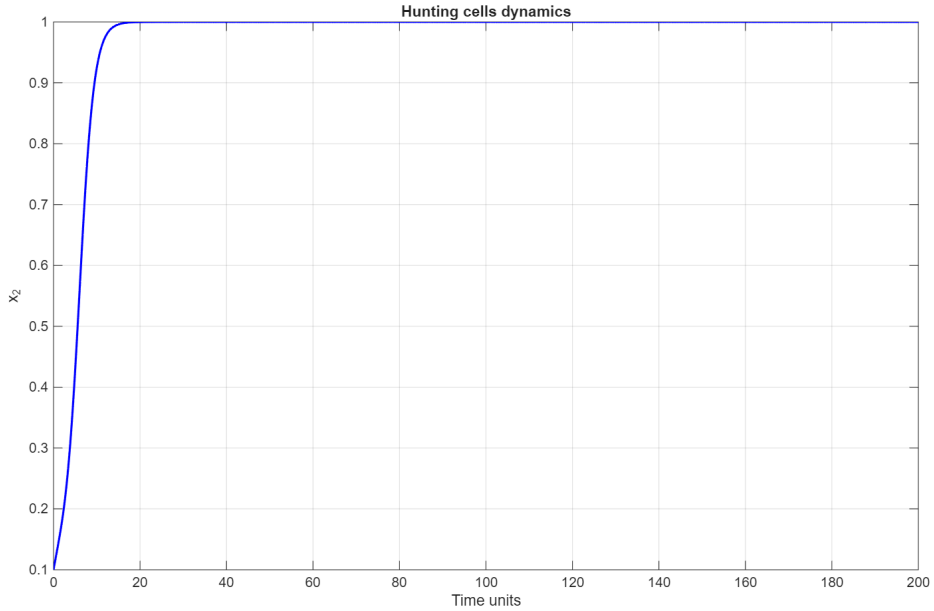


Figure 3.35: Hunting cells time evolution

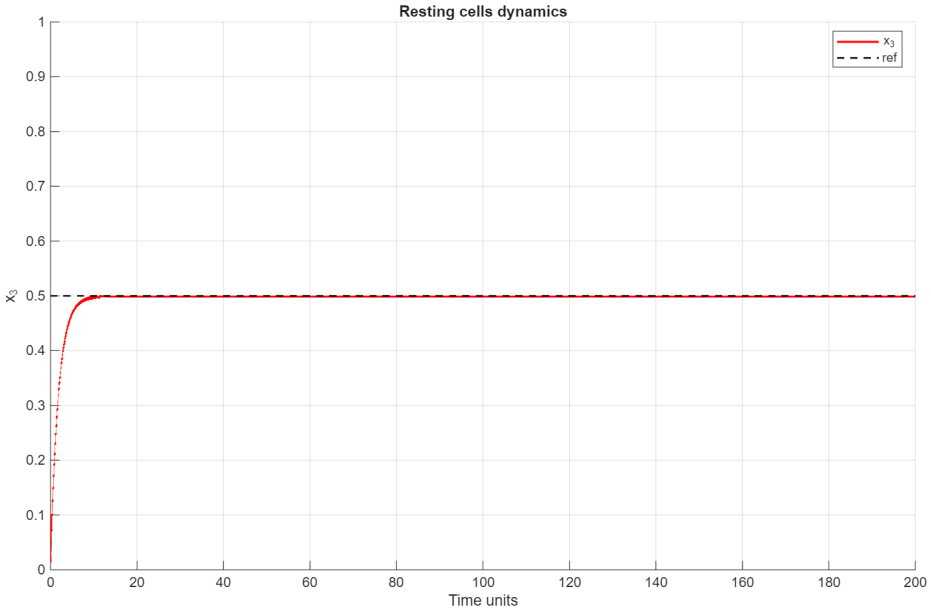


Figure 3.36: Resting cells time evolution

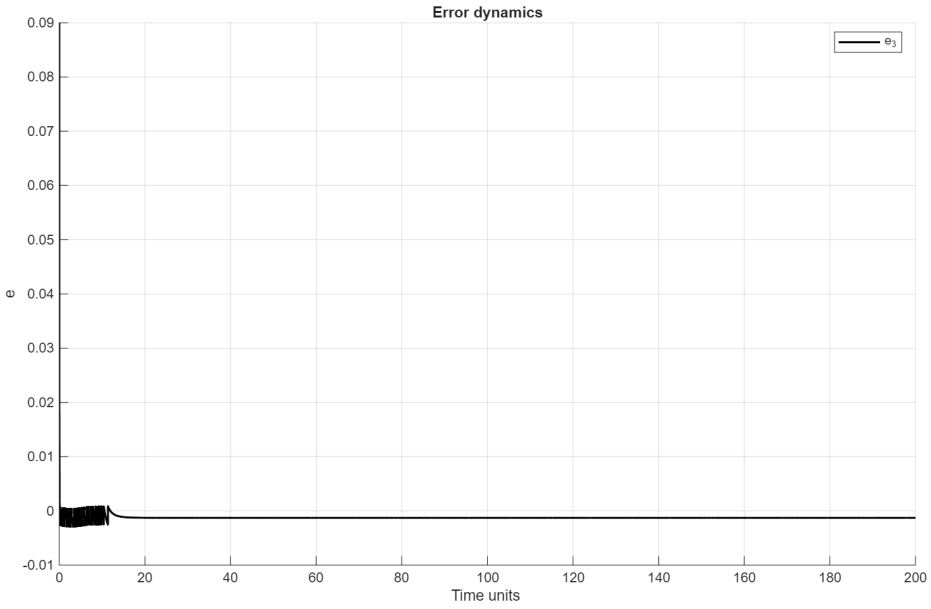


Figure 3.37: Error dynamics

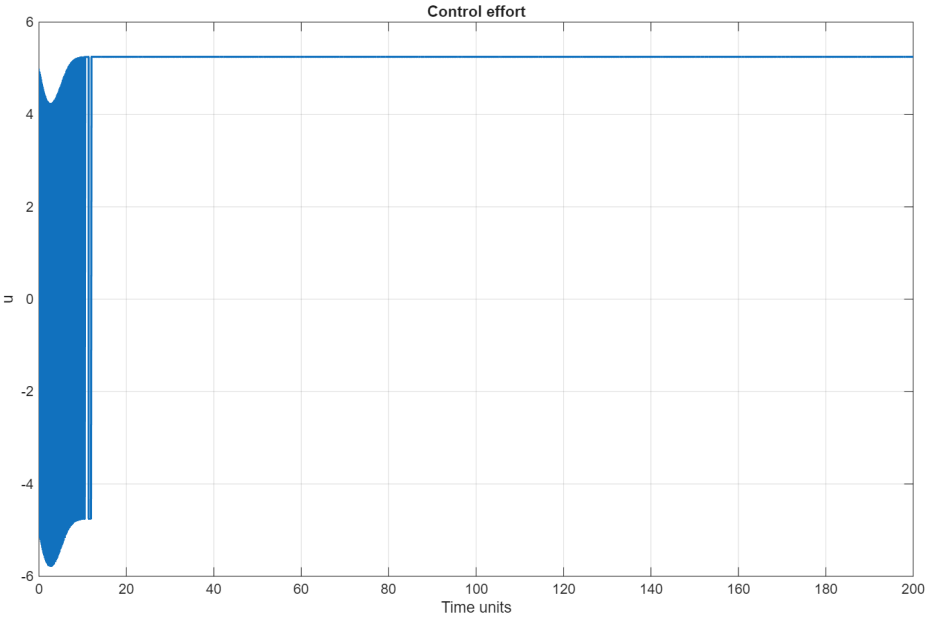


Figure 3.38: Control effort

3.7.1 Control effort analysis

A characteristic phenomenon of SMC is *chattering*, which consists of high-frequency oscillations of the control input u around the sliding manifold Σ due to the discontinuous nature of the signum function.

An atypical but significant result observed in the simulations is the complete disappearance of chattering as the tumor cell population x_1 reaches extinction.

When $x_1(t) \to 0$, the system's drift term $f_3(x_1, x_3)$ becomes:

$$f_3(0, x_3) = -D_3 x_3 \tag{3.46}$$

Once the sliding surface $\sigma = 0$ is reached, the condition $\dot{\sigma} = 0$ must hold to maintain the equilibrium. Substituting the simplified dynamics into the reachability condition, we find that the required control effort is:

$$\dot{\sigma} = \lambda(-D_3 x_{3,d} + u) = 0 \implies u_{ss} = D_3 x_{3,d} \tag{3.47}$$

At this stage, since no further perturbations act on the states of the system, the switching term $-\kappa \text{sgn}(\sigma)$ no longer needs to oscillate to keep the trajectory on the manifold. The control signal naturally settles to the constant value u_{ss}.

This behaviour is confirmed by the following simulation, where

a finite-time constant disturbance is fed as an input to the equation governing the tumor growth:

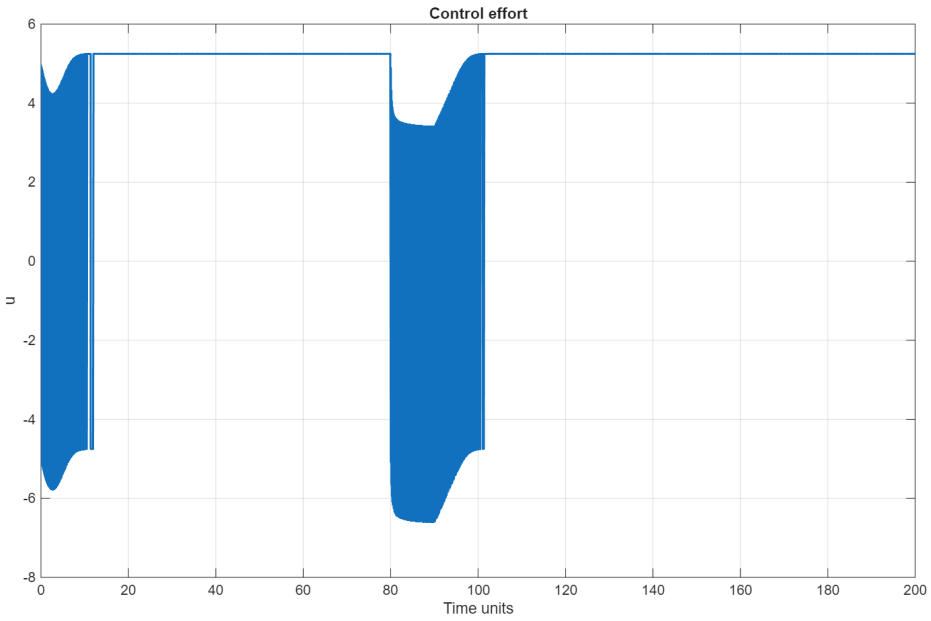


Figure 3.39: Control effort with finite-time constant input disturbance

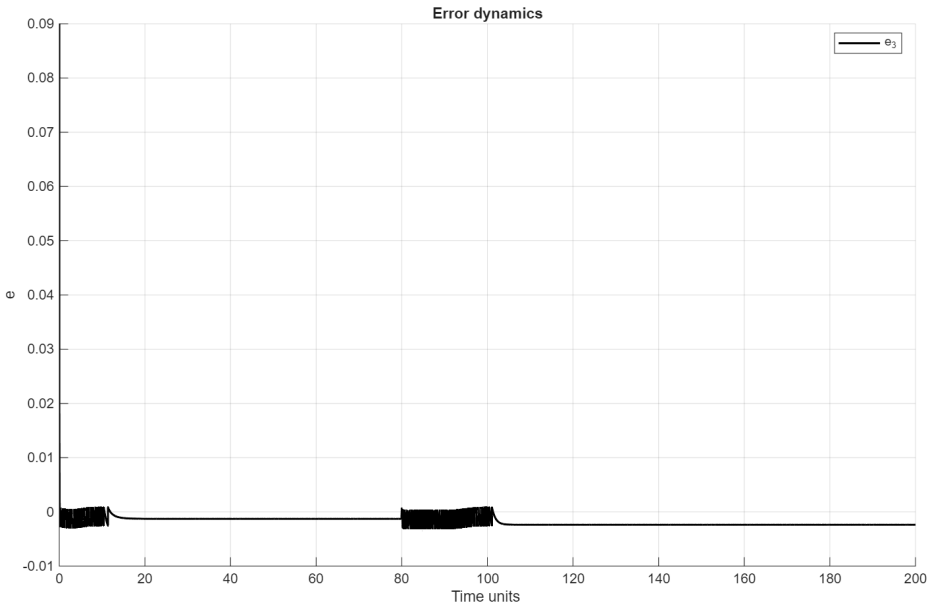


Figure 3.40: Error dynamics with finite-time constant input disturbance

As can be seen, when the disturbance enters the system at time step 80, the control law starts exhibiting the chattering again, for it to assess to a constant value when the desired value of x_3 is reached.

3.7.2 Mitigation of chattering

As we observed in the previous simulations, the control law is characterized by the chattering phenomenon. In order to mitigate the high frequency control input, the *sign* is substituted by a hyperbolic tangent. As a first attempt, a sigmoid-type of saturation was implemented as in the case of the I/O FBL controller, but simulation results showed that a steady-state error was present even when the gain was chosen to be much higher (in the simulations presented it was set to $K = 50$);

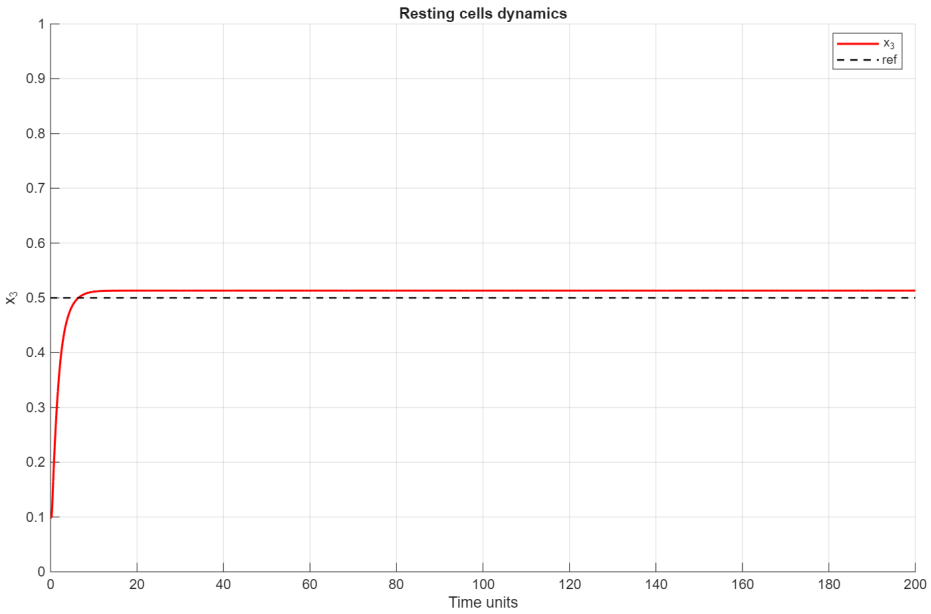


Figure 3.41: Resting cells time evolution with sigmoid function

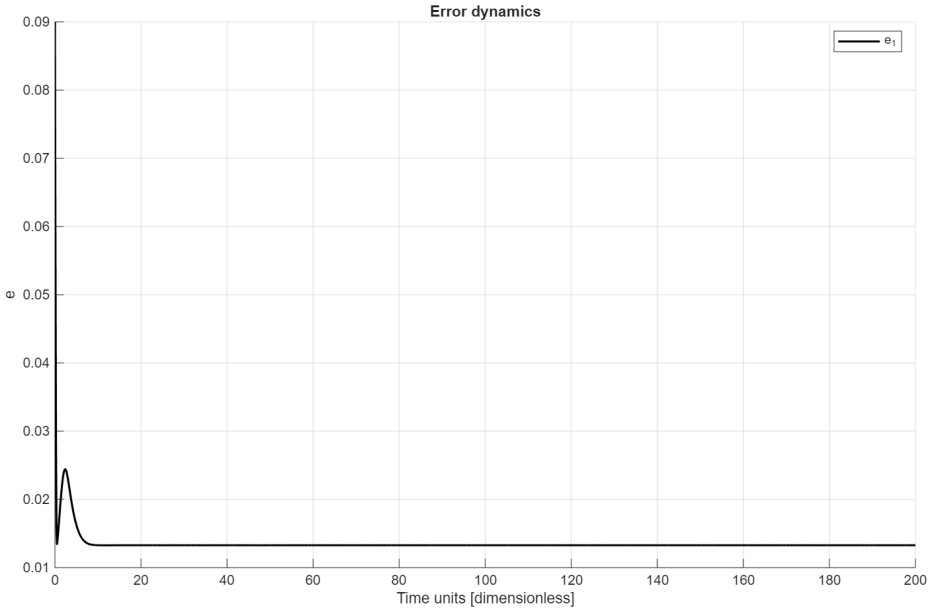


Figure 3.42: Error dynamics with sigmoid function

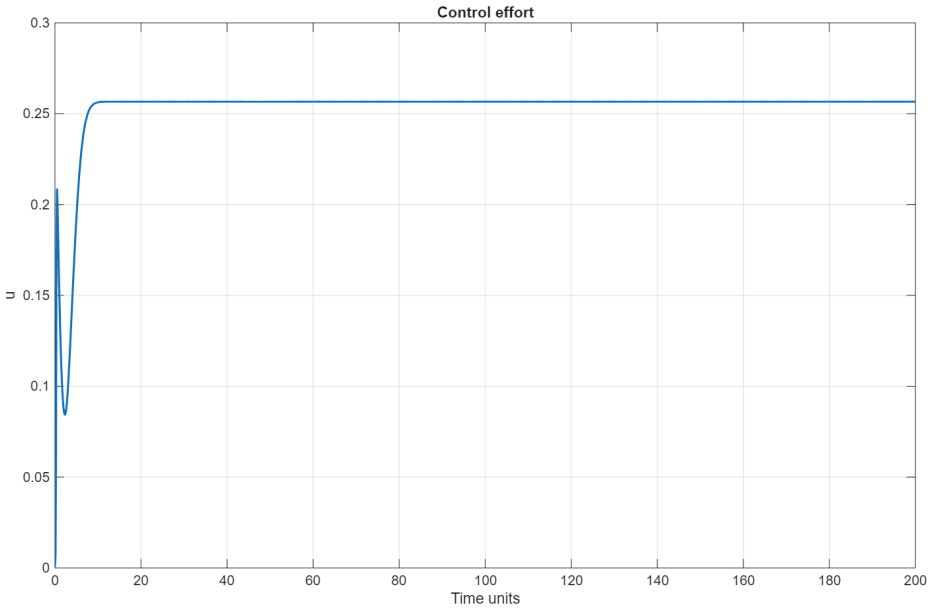


Figure 3.43: Control effort with sigmoid function

Due to the presence of this steady-state error, a saturation block was chosen in place of the sigmoid, showing performances as good as the chattering controller while keeping the value of the gain K set to 10:

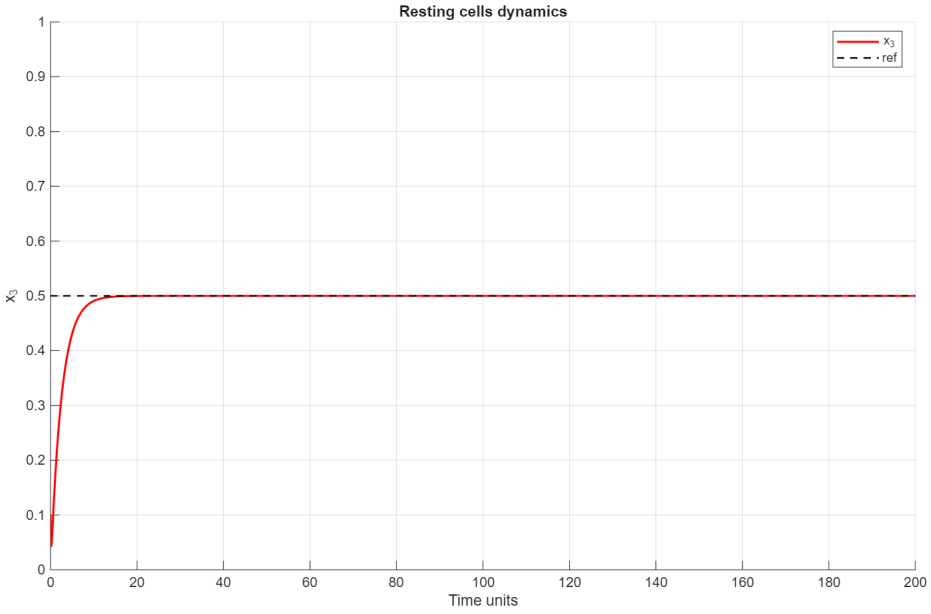


Figure 3.44: Resting cells time evolution with hyperbolic tangent and saturation

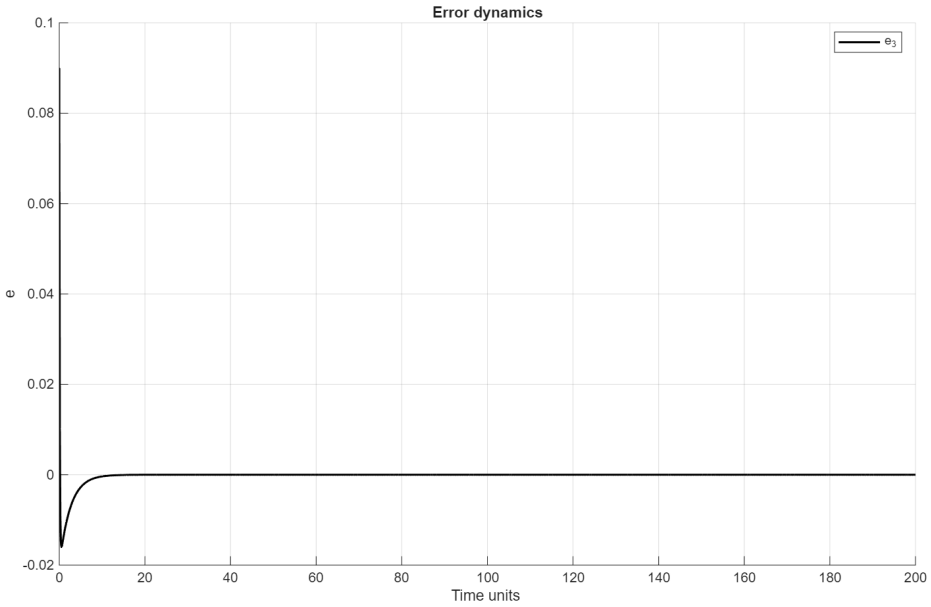


Figure 3.45: Error dynamics with hyperbolic tangent and saturation

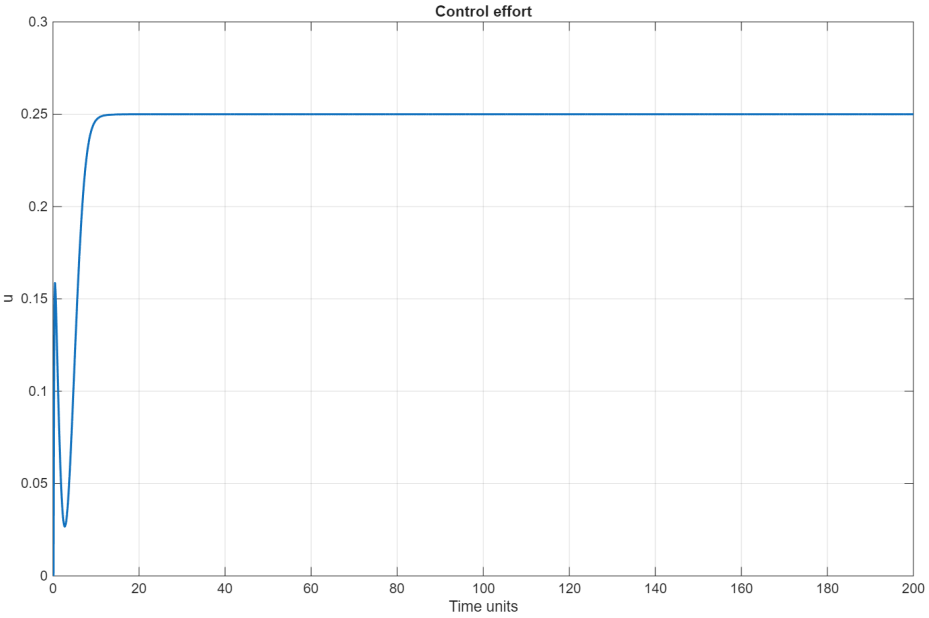


Figure 3.46: Control effort with hyperbolic tangent and saturation

3.8 Robustness analysis

In this section, we analyze the robustness of the control structure that was just derived against matched disturbances and parametric uncertainty.

3.8.1 Rejection of matched disturbances

A disturbance is defined as "matched" if it enters the system dynamics through the same channel as the control input u:

$$\dot{x}_3 = f_3(x) + g_3(x)(u + d(t)) \tag{3.48}$$

The sliding mode controller is characterized by the peculiar property of being able to completely reject matched disturbances when in chattering mode. In the following, a simulation test with a sinusoidal wave of unitary frequency and amplitude was introduced in the input channel:

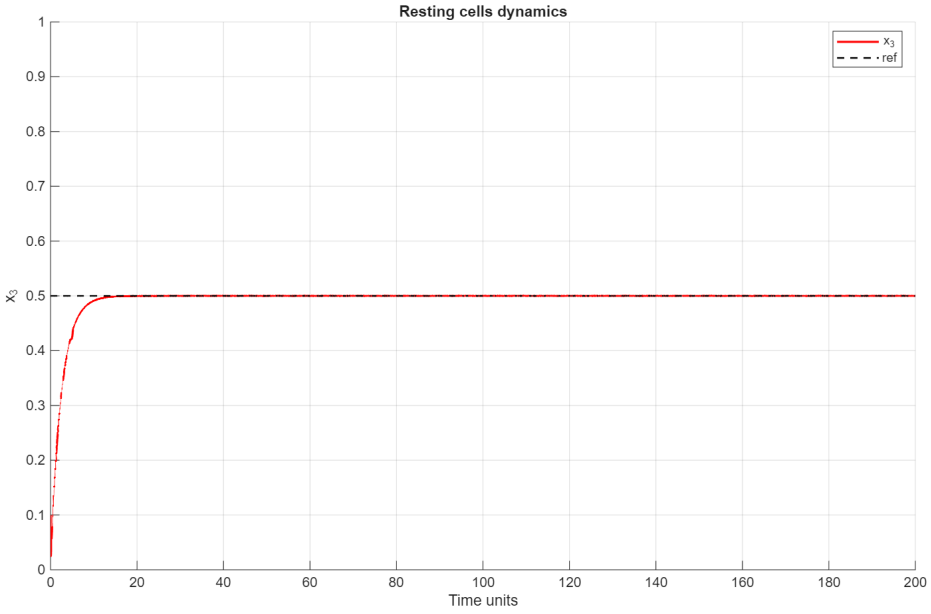


Figure 3.47: Resting cells time evolution with sinusoidal disturbance

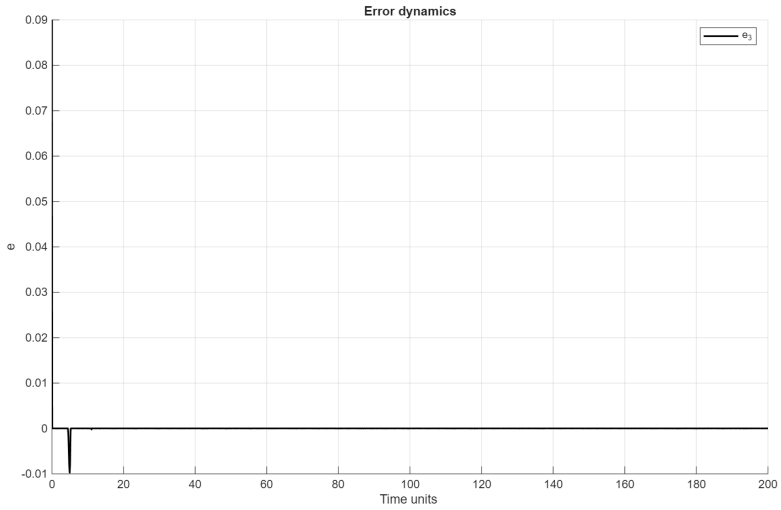


Figure 3.48: Error dynamics with sinusoidal disturbance

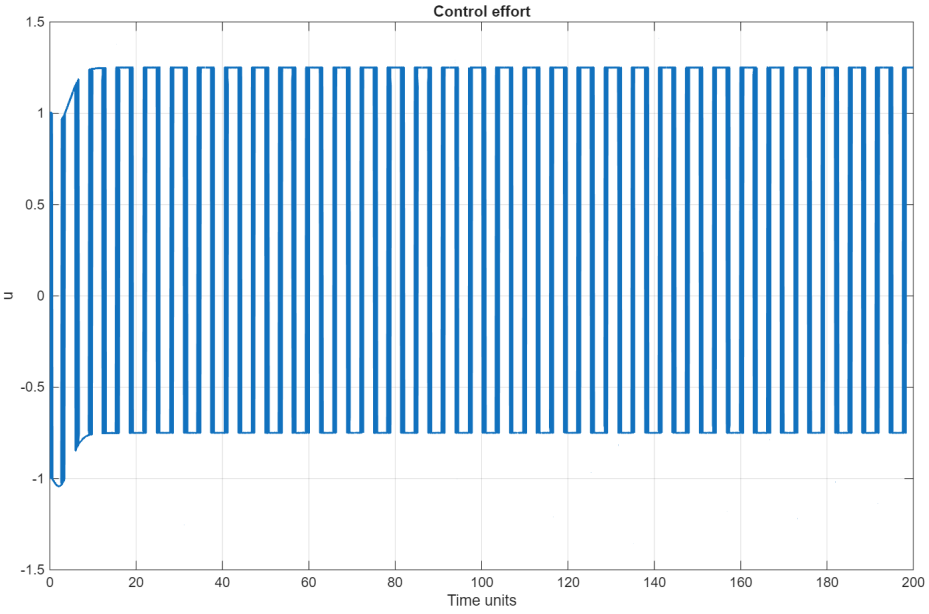


Figure 3.49: Control effort with sinusoidal disturbance

This property is lost when switching to the *tanh*. Nevertheless, the effect of matched disturbances can be mitigated by increasing the value of the gain K. In the following, a simulation with the same type of disturbance acting on the system but with the *tanh* function is presented, with K set to 50 and no saturation applied:

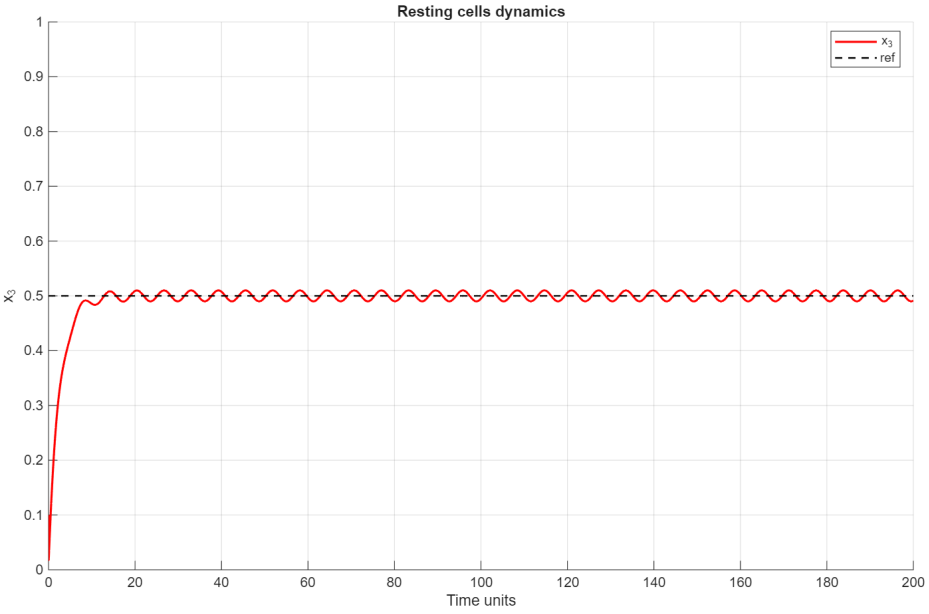


Figure 3.50: Resting cells time evolution with sinusoidal disturbance and hyperbolic tangent

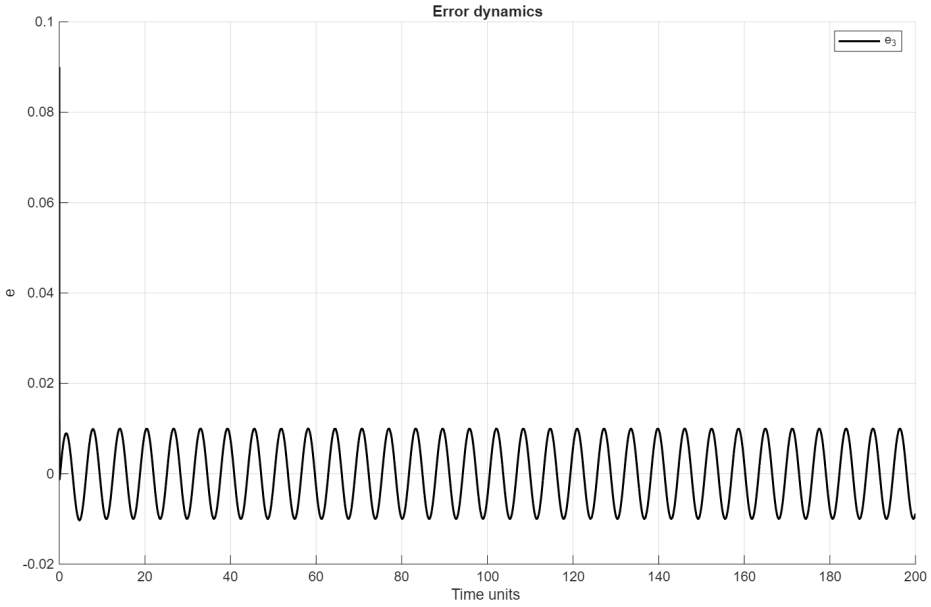


Figure 3.51: Error dynamics with sinusoidal disturbance and hyperbolic tangent

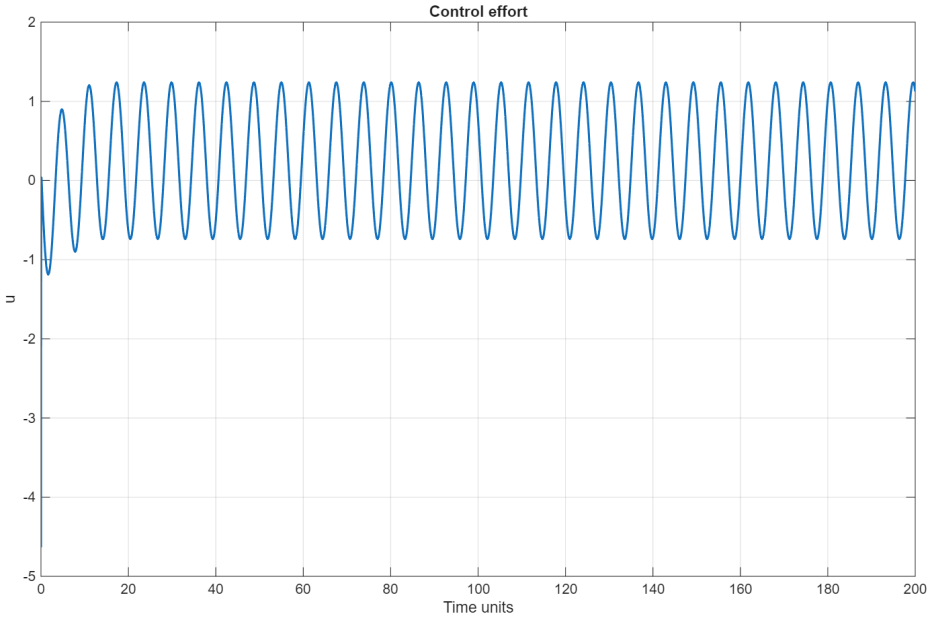


Figure 3.52: Control effort with sinusoidal disturbance and hyperbolic tangent

3.8.2 Parametric uncertainty robustness analysis

As in the case of the I/O FBL controller, a parametric uncertainty of -30% was added to those parameters that intervene in the computation of the control law, while their nominal value was used for the derivation of the control law itself. By selecting a gain K which

is sufficiently big (it was set to 25), it can be observed that the controller almost perfectly cancels the effect of the parametric uncertainty:

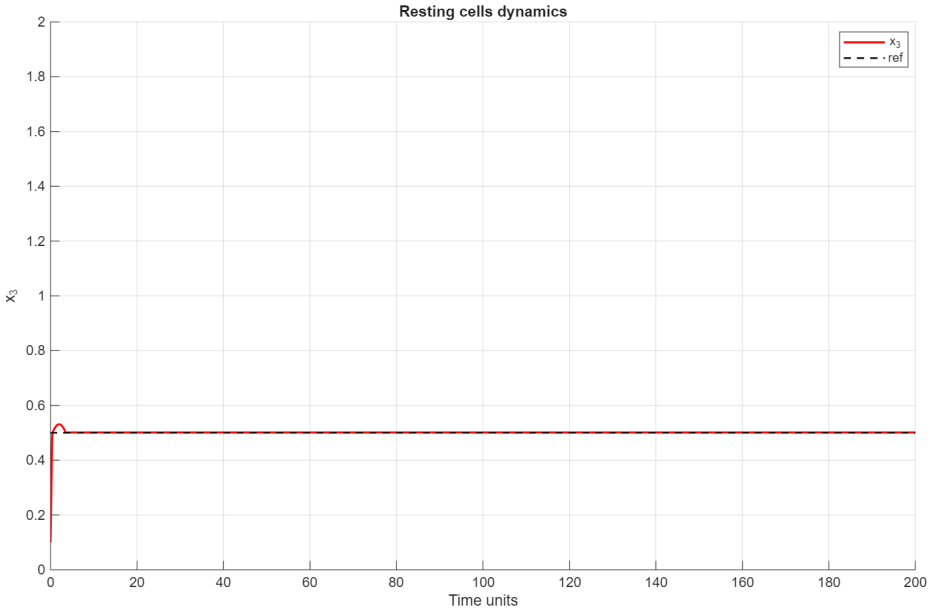


Figure 3.53: Resting cells time evolution with parametric uncertainty

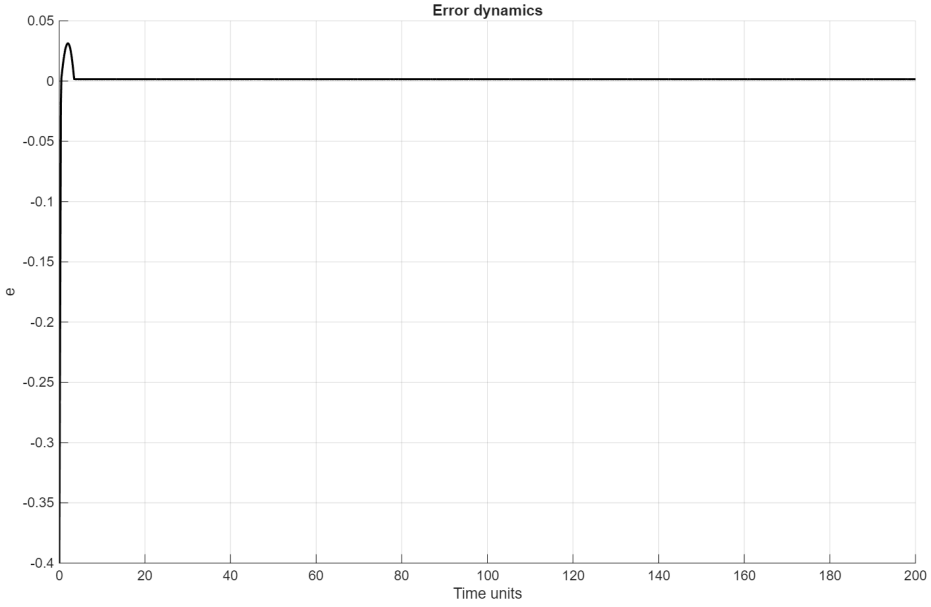


Figure 3.54: Error dynamics with parametric uncertainty

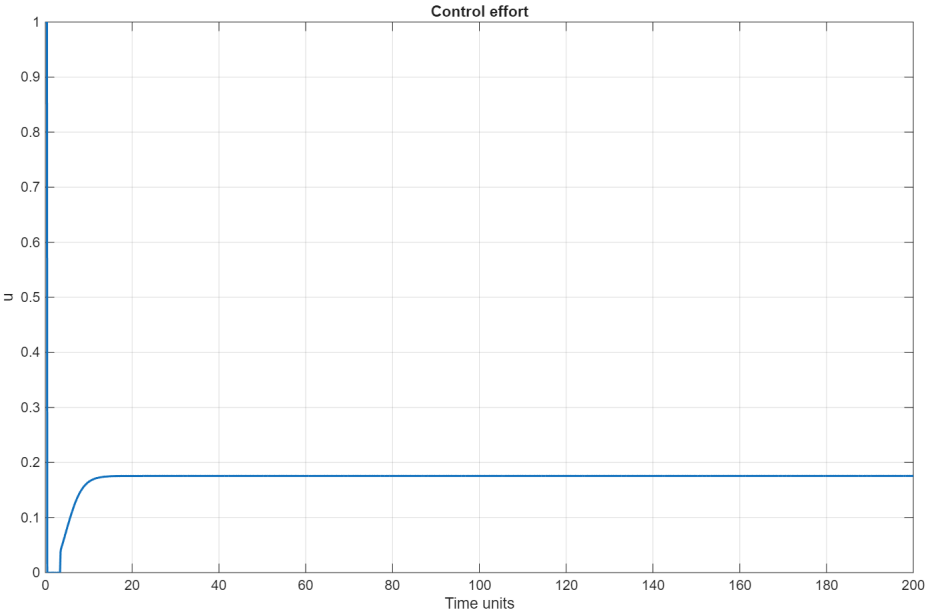


Figure 3.55: Control effort with parametric uncertainty

3.9 EKF implementation

3.9.1 Implementation-oriented design framework

Throughout the analysis presented in this paper, the focus has consistently maintained a duality between theoretical rigor and implementation feasibility. This approach justified several key design choices, such as the imposition of strict constraints on the control input, and the introduction of reference signal pre-filtering.

3.9.2 The measurement constraint

However, one critical aspect has primarily been addressed under an idealized assumption: the availability of the full state vector for feedback. The design of the nonlinear controllers presented so far assumes that x_3 is measurable in real-time. In a clinical scenario, this assumption is often unrealistic, as monitoring the immune population requires invasive procedures that cannot be performed continuously for feedback control. Conversely, the tumor mass volume x_1 is the only state variable that can be reliably monitored non-invasively via medical imaging techniques. Consequently, to bridge the gap between the proposed control logic and clinical applicability, it is necessary to transition towards an architecture where the

unmeasurable states are reconstructed starting from the available measurement $y = x_1$.

3.9.3 Extended Kalman Filter implementation

To address this challenge, we integrated an *Extended Kalman Filter* into the control loop. Its operation is based on a recursive two-step process:

1. **Prediction**: The filter estimates the future state using the nonlinear model of the system: $\dot{\hat{x}} = f(\hat{x}, u)$.

2. **Correction**: The filter linearizes the nonlinear dynamics around the current estimate at each time step via the jacobian matrix of the system. This linear approximation is used to compute the optimal Kalman Gain.

3.9.4 Configuration in Simulink environment

The estimator was implemented using the *Extended Kalman Filter* block provided by the *Control System Toolbox* in Simulink. The configuration process involved defining the following functions:

- The **state transition function**, which was defined using the discretized TDCM equations via Euler integration ($dt = 0.001s$).

- The **state jacobian function**, which was computed analytically to provide the linearized system matrix A_k at each step.

Finally, the **covariance matrices** Q and R were tuned assuming model dynamics were known with great precision and measurements are reliable.

This configuration allows the SMC to operate on the estimated state $\hat{x}_3$, effectively closing the loop using only x_1. The final control scheme implemented in Simulink is the following:

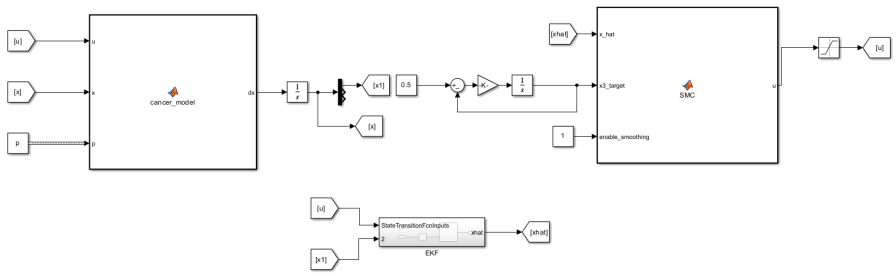


Figure 3.56: Simulink control scheme with EKF

The following results demonstrate the performance of this architecture:

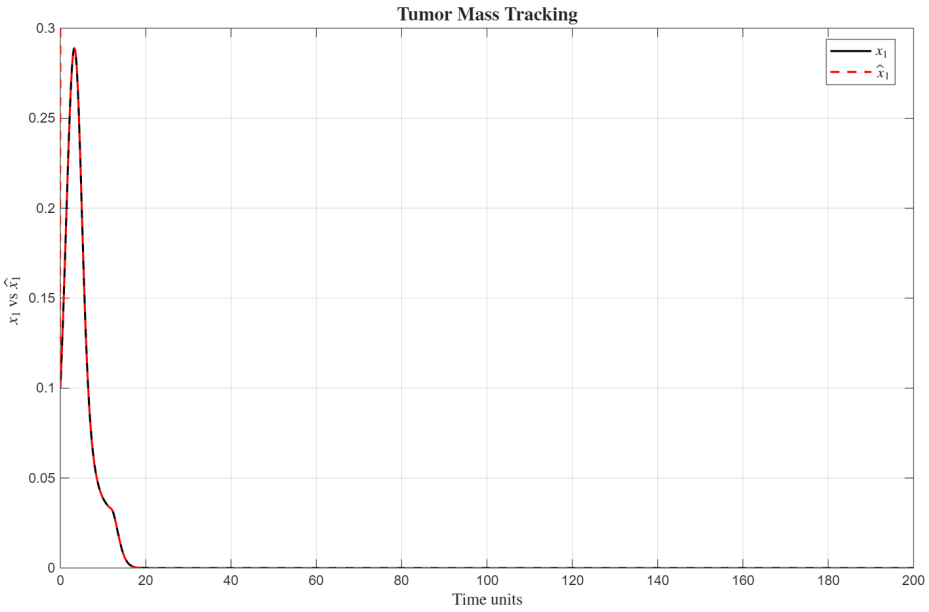


Figure 3.57: Tumor cells state estimation

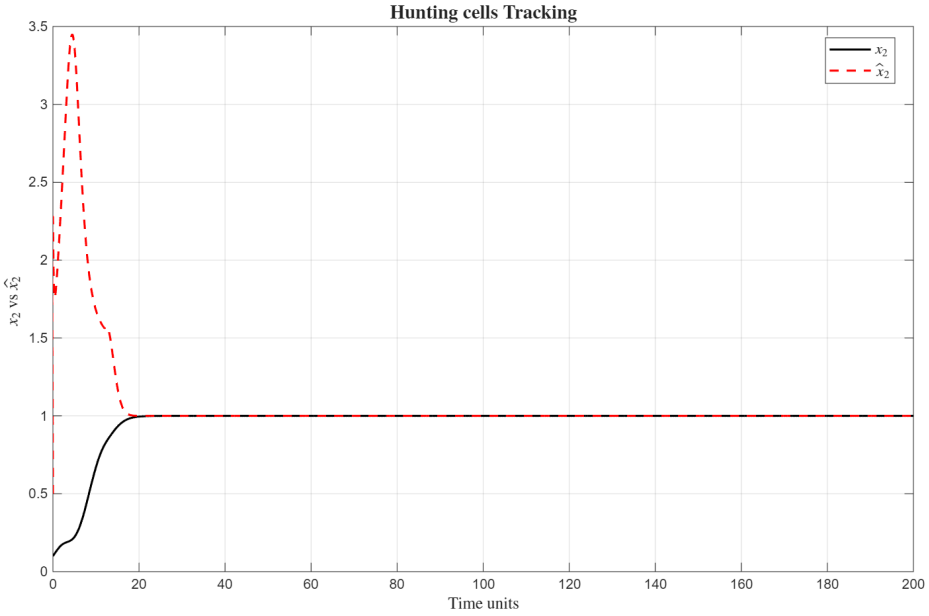


Figure 3.58: Hunting cells state estimation

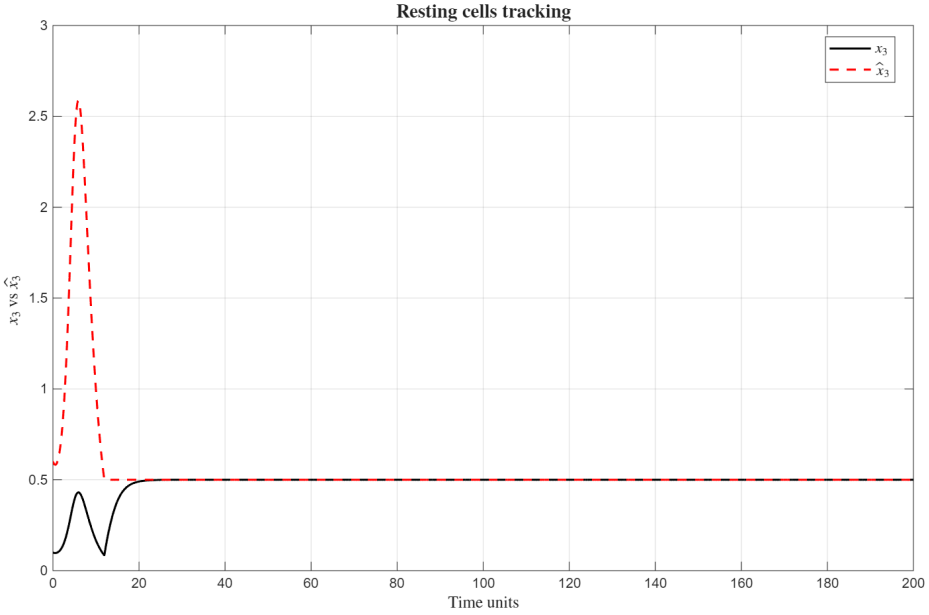


Figure 3.59: Resting cells state estimation

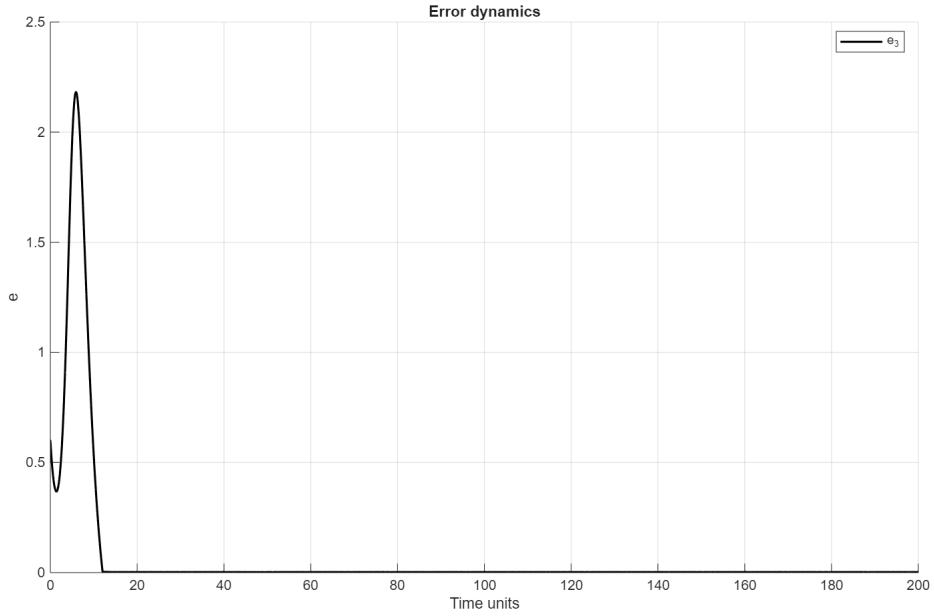


Figure 3.60: Error dynamics with EKF

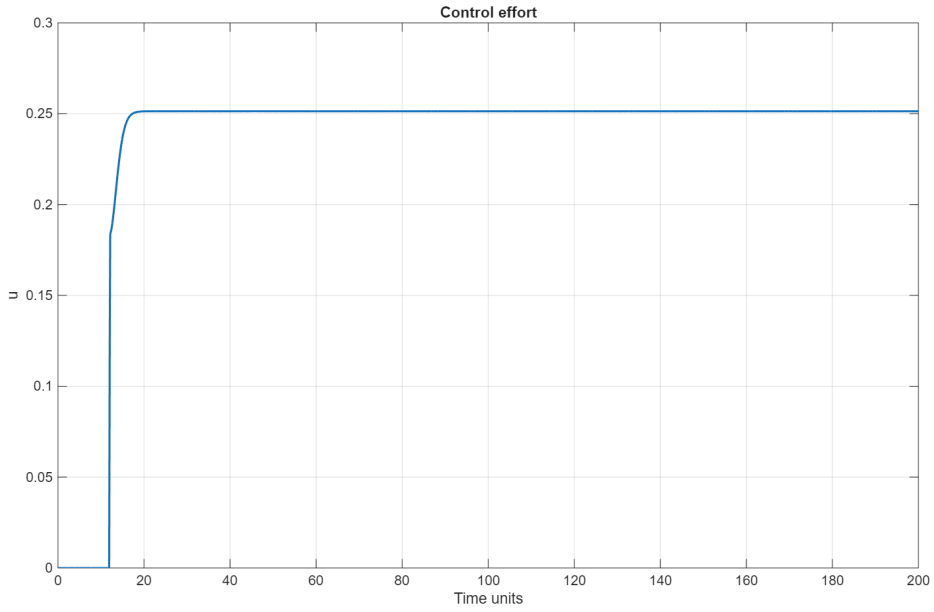


Figure 3.61: Control effort with EKF

Chapter 4

Discussion and comparative analysis

In this chapter, we provide a critical comparison between the linear and nonlinear control strategies designed in Chapter 3. The analysis focuses on three key aspects: the operational domain, compliance with clinical safety constraints and robustness against parametric uncertainties and disturbances acting on the system.

4.1 Robustness analysis

As we have already seen in the chapter regarding the synthesis of the LQR controller, the way the control structure reacts to parametric uncertainties guarantees a certain degree of robustness, but it is way outclassed by the nonlinear controllers that were implemented. As was shown previously in chapter 3, both the I/O FBL and the SMC were capable of perfectly rejecting the parametric uncertainties. For the sake of completeness, the results of two simulations carried out with the nonlinear controllers are shown, where the parameters were globally affected by a $\pm 30\%$ uncertainty:

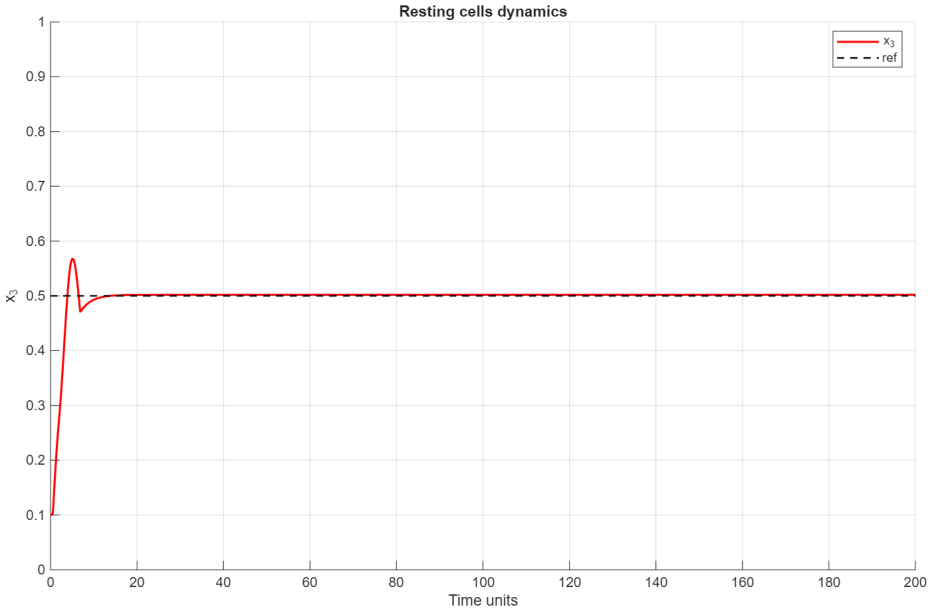


Figure 4.1: Resting cells time evolution with SMC and global parametric uncertainty

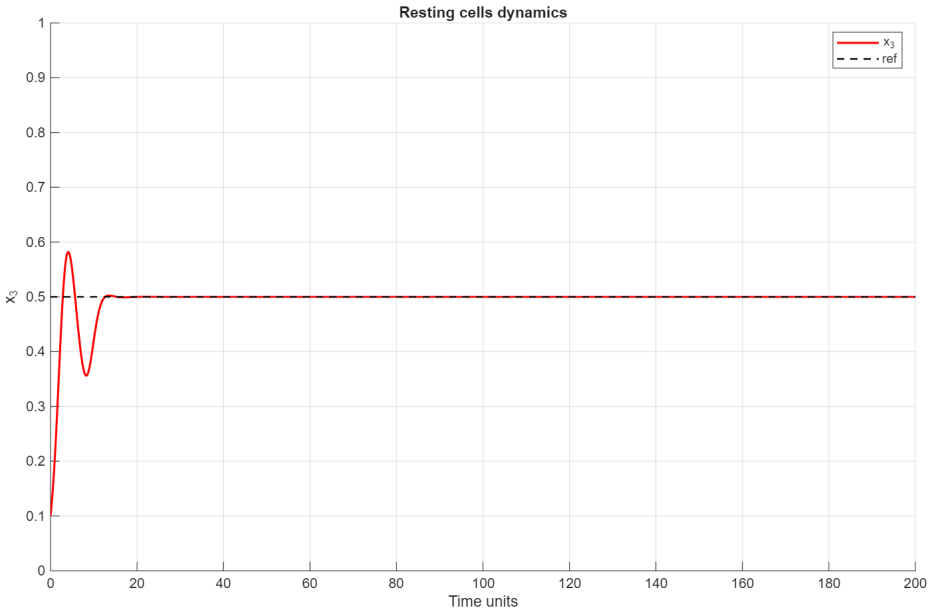


Figure 4.2: Resting cells time evolution with I/O FBL and global parametric uncertainty

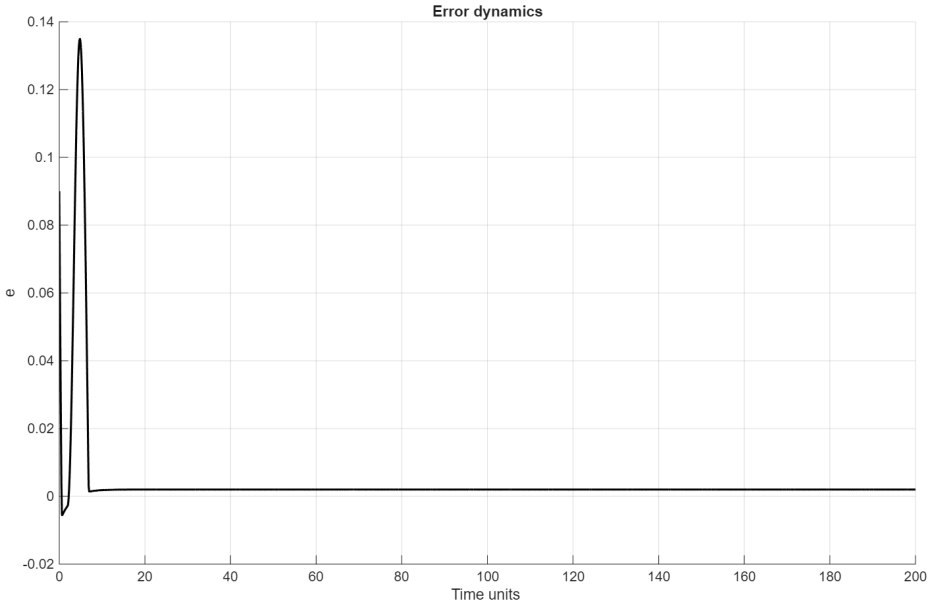


Figure 4.3: Error time evolution with SMC and global parametric uncertainty

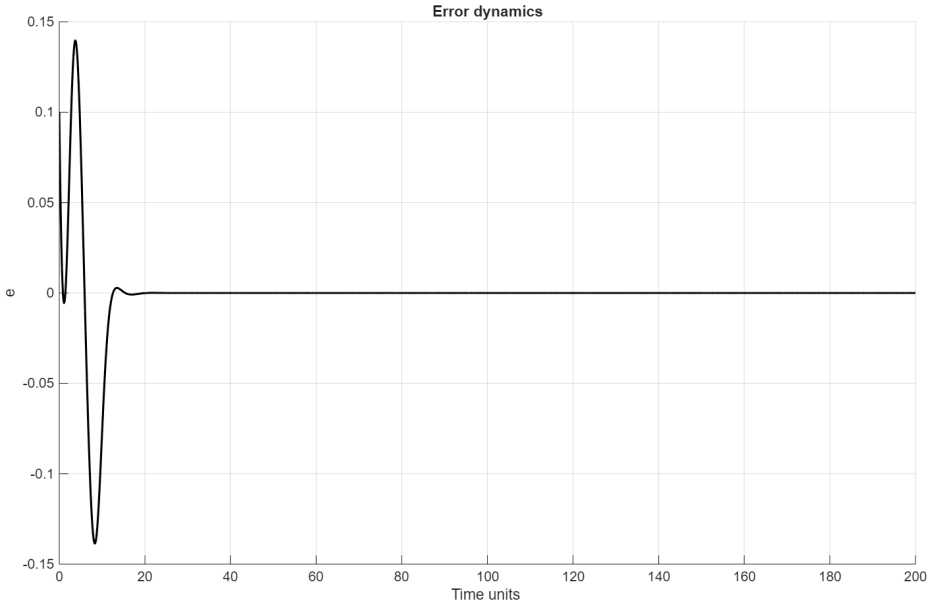


Figure 4.4: Error time evolution with I/O FBL and global parametric uncertainty

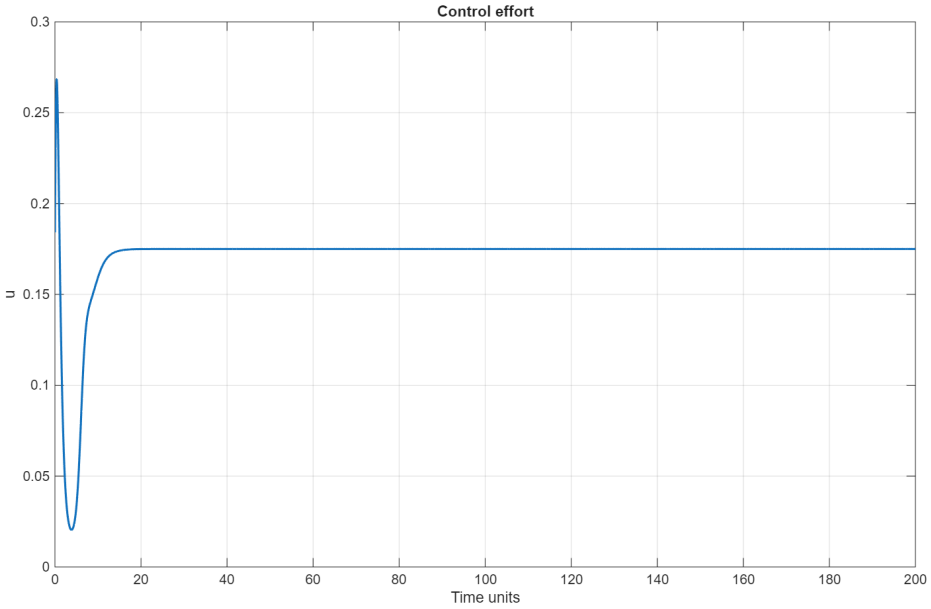


Figure 4.5: Control effort I/O FBL and global parametric uncertainty

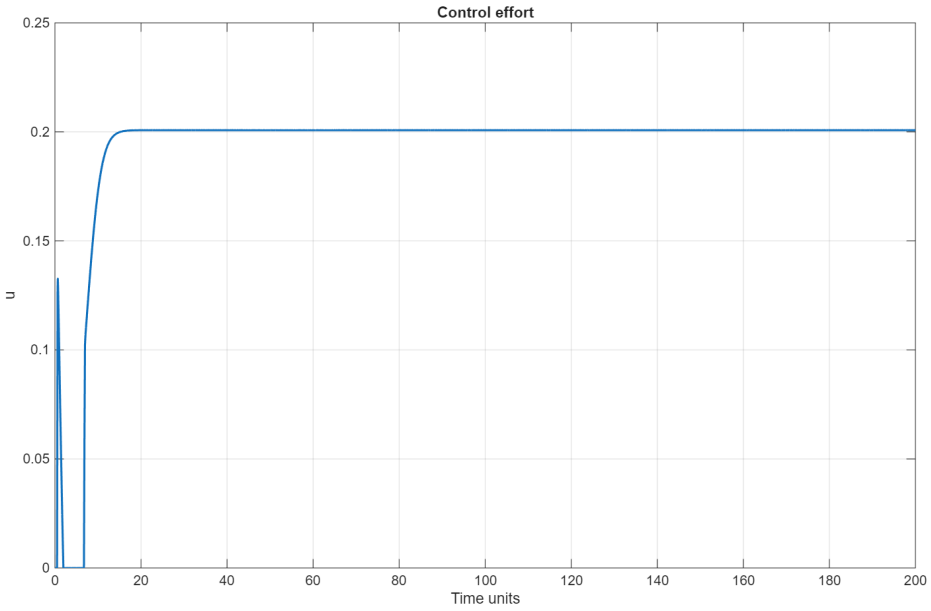


Figure 4.6: Control effort with *tanh* SMC and global parametric uncertainty under saturation

4.1.1 Disturbance rejection performance comparison

Even if at the beginning of chapter 3 we had analyzed how the LQR reacted to a finite-time disturbance applied on the first equation, it's time to see how the system responds to matched disturbances, in particular when a constant disturbance is applied:

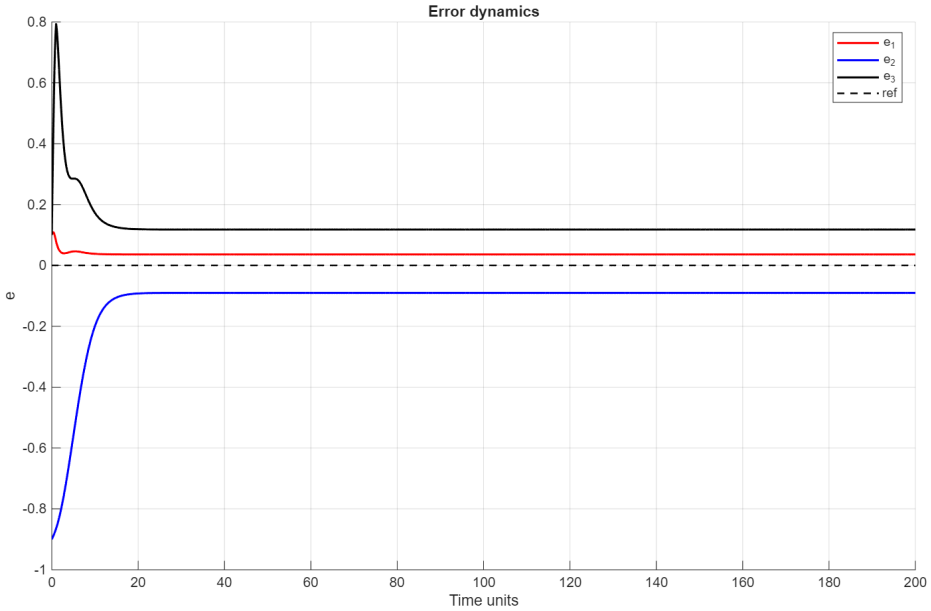


Figure 4.7: LQR Error dynamics with constant disturbance applied on the third equation

As it can be noticed, even if the error is small in terms of magnitude, the system is not capable of eliminating the steady-state error.

4.2 The role of linear control

The Linear Quadratic Regulator was synthesized based on the linearization of the system around the perturbed healthy equilibrium P_ϵ. Simulations confirm that the LQR is an extremely efficient controller when operating within a small neighborhood of the healthy state (P_H):

- **Energy Optimality:** In the scenario of small tumor recurrences, the LQR successfully suppresses the tumor growth with minimal control effort, optimizing the trade-off between drug administration and convergence speed.
- **Operational Interpretation:** Consequently, the LQR should be interpreted not as a therapeutic tool for acute phases, but as a *maintenance controller*.

4.3 Transient analysis

What has been said in the previous section on the role of the LQR controller is based on the simulation results observed when the controller starts operating with the system "far" from the equilibrium point. Suppose we start from an initial condition where the tumor is not close to the 0, for instance $x_1(0) = 0.6$, while the other two initial conditions remain unchanged. First, let's look at the x_1 and x_2 plots:

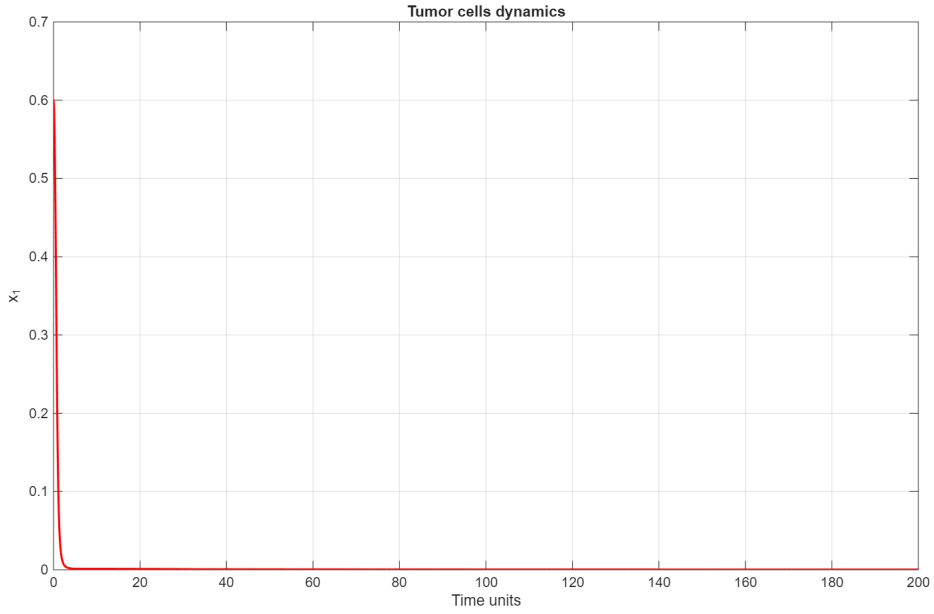


Figure 4.8: Tumor cells time evolution with $x_1(0) = 0.6$

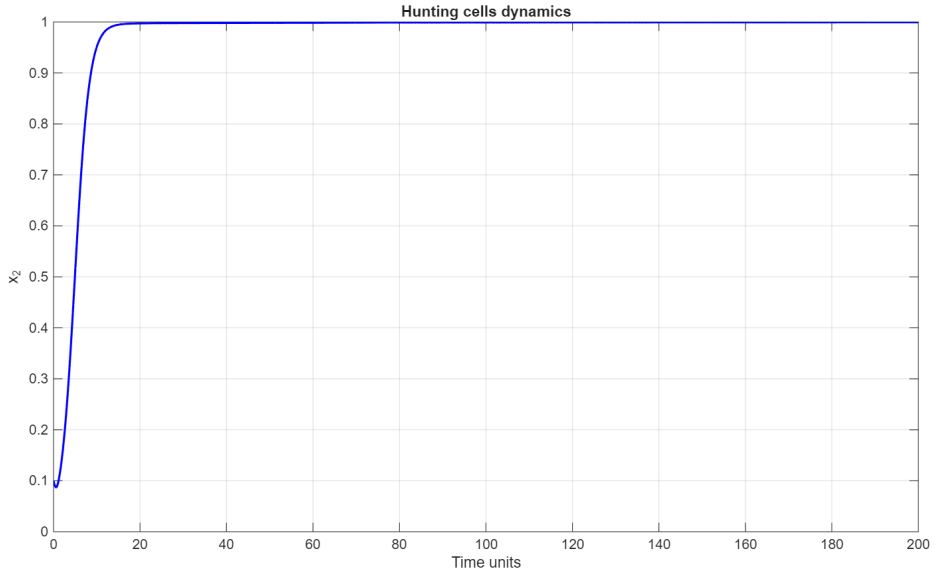


Figure 4.9: Hunting cells time evolution with $x_1(0) = 0.6$

At first glance it would look perfect, since both x_1 and x_2 converge to the desired values. Nevertheless, a deeper inspection of the system behavior reveals that asymptotic stability of the tumor population is not sufficient to guarantee clinical feasibility. To this purpose, let's analyze the x_3 plot:

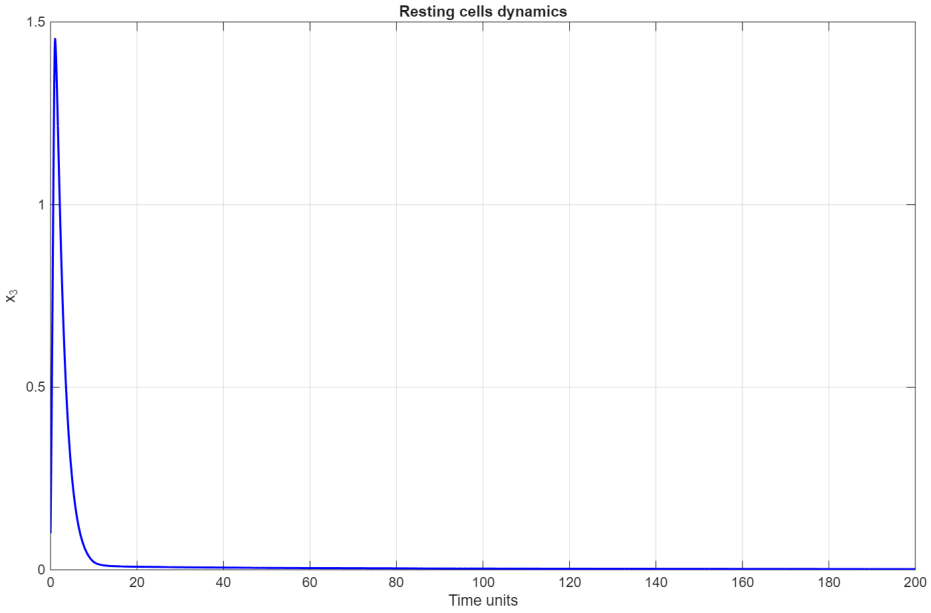


Figure 4.10: Resting cells time evolution with $x_1(0) = 0.6$

4.3.1 Physical interpretation: the "blindness" of linear control

The phenomenon of the x_3 peak is a direct consequence of the intrinsic nonlinear dynamics of the immune system, which the LQR structure ignores.

Recalling the equation (1.3), the recruitement term $\frac{R_3 x_1 x_3}{x_1 + k_3}$ acts as a **positive feedback action**, since the presence of a large tumor mass accelerates the growth of x_3. This analysis reveals the difference between the LQR controller and the nonlinear ones:

- **LQR behavior:** LQR is synthesized on the linearized model, where this nonlinear interaction is approximated by the Jacobian matrix. Consequently, the LQR is partially "blind" to the recruitment term. It perceives the growing error and applies the control input u. However, it does not actively account for the fact that the tumor itself is pushing x_3 upwards. The combination of the external input u and the strong internal

recruitment cause the variable x_3 to surge uncontrollably in an open-loop fashion during the transient phase.

- **Nonlinear controllers:** Conversely, the nonlinear control technique presented are model-based. Take for instance the I/O FBL: the control law explicitly contains the term $-L_f h(x)$, which includes the exact expression of the recruitment dynamics. The nonlinear controller *subtracts* an amount of control input exactly equal to the internal recruitment. It "sees" that the tumor is trying to accelerate x_3 and controls the external drug administration accordingly.

4.3.2 Impact of physical constraints on control performance

The analysis of the LQR behavior has prompted significant reflections regarding nonlinear control structures as well. In particular, it's interesting to analyze what happens when we evaluate their performances when x_1 is initialized far from 0, for instance $x_1 = 0.9$:

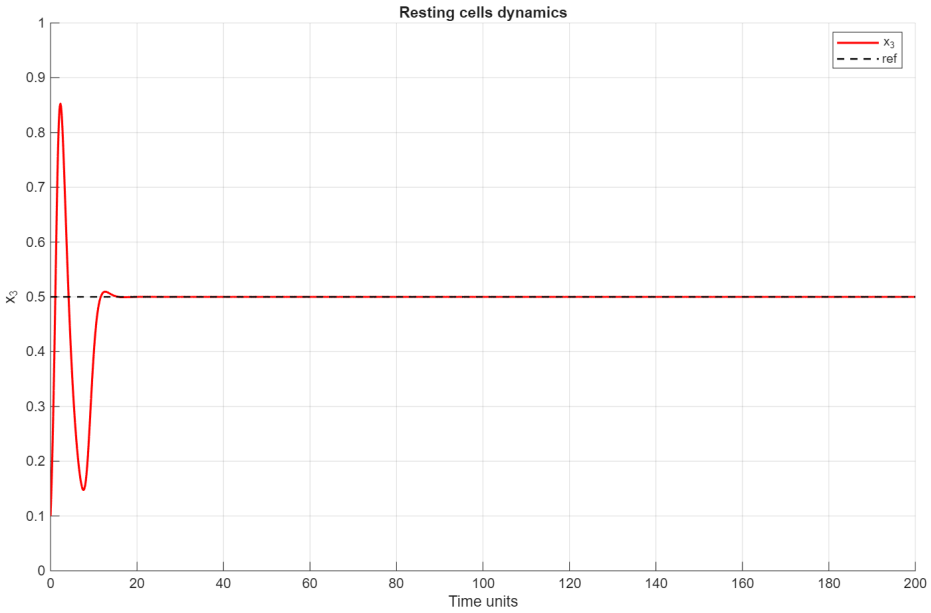


Figure 4.11: Resting cells time evolution with $x_1(0) = 0.9$

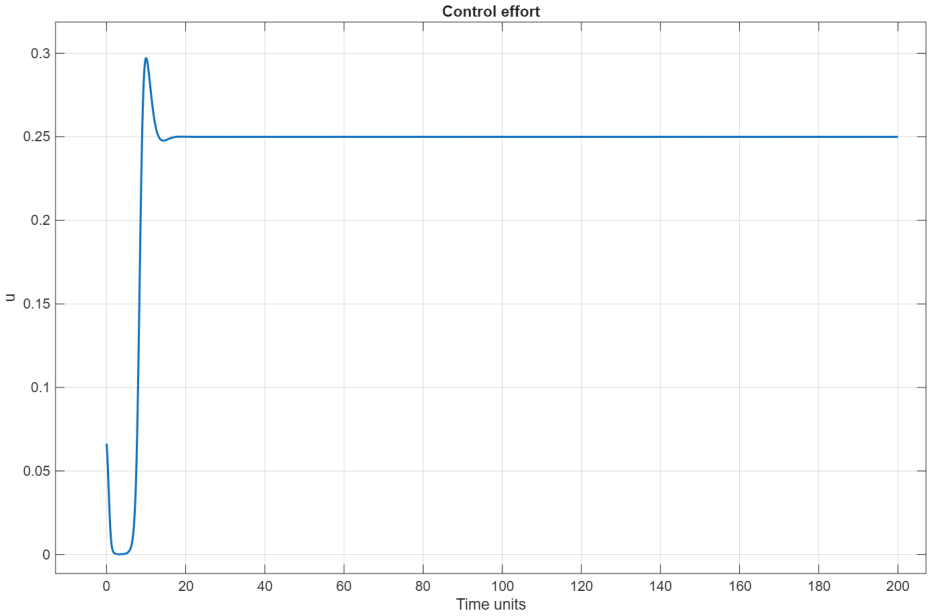


Figure 4.12: Control effort with $x_1(0) = 0.9$

What can be noticed by analyzing these graphs are two things:

1. The transient of x_3 exhibits the same initial spike that was observed with the LQR controller

2. For a small time window, the control output was close to 0.

Now, what if we start from the same initial condition, but we consider the parametric uncertainty we had accounted for before? Let's look at the graphs:

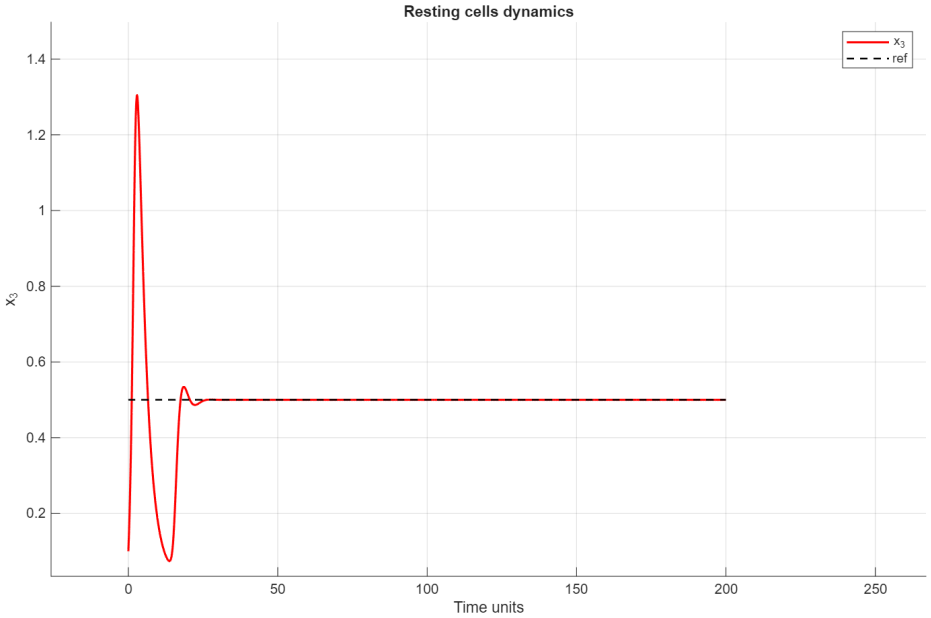


Figure 4.13: Resting cells time evolution with $x_1(0) = 0.9$ and parametric uncertainty

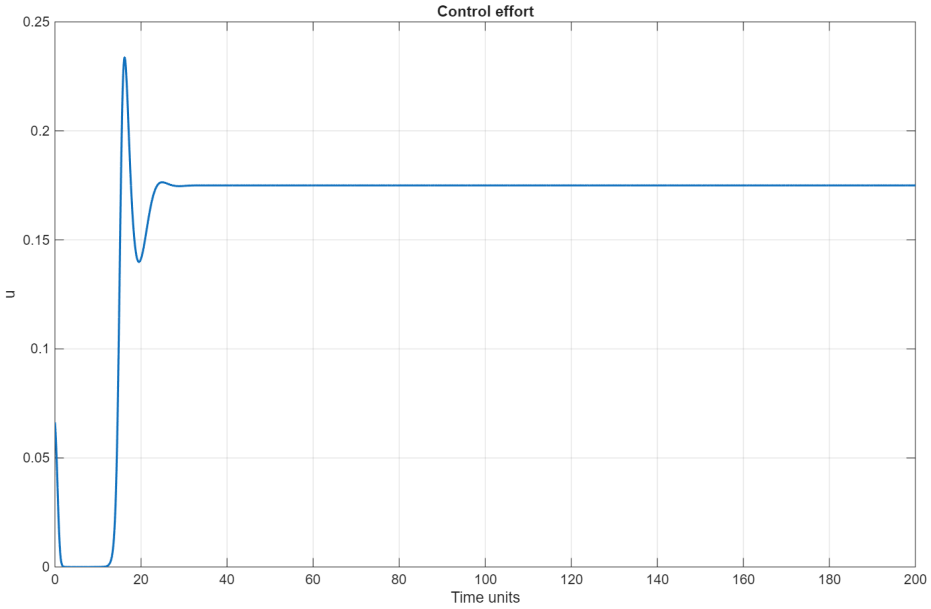


Figure 4.14: Control effort with $x_1(0) = 0.9$ and parametric uncertainty

By examining the transient response of the I/O FBL, it can be observed that the biological limit associated with the variable x_3 is violated in this configuration as well. This phenomenon is a direct consequence of the implementation constraints adopted, specifically the requirement that the control input must be non-negative.

To understand the underlying mechanism, one must consider the recruitement term again. When the tumor burden x_1 is high, the

magnitude of this term increases significantly. Consequently, when the controller attempts to cancel this nonlinear dynamic, the control law demands a negative signal. However, such an input is physically inadmissible given the physics of the drug administration.

The result is that the controller saturates at zero. In this phase, the system effectively enters an **open-loop regime**: the concentration of x_3 rises rapidly and uncontrollably, driven by the high tumor mass, and active control is resumed only when the required control effort returns to a positive value.

A similar scenario arises with the SMC controller. As pointed out in Chapter 3, the simulation that showed how the controller reacted to a sinusoidal-like disturbance was carried out without adopting saturation. Let's look at the same simulation, but with the saturation applied:

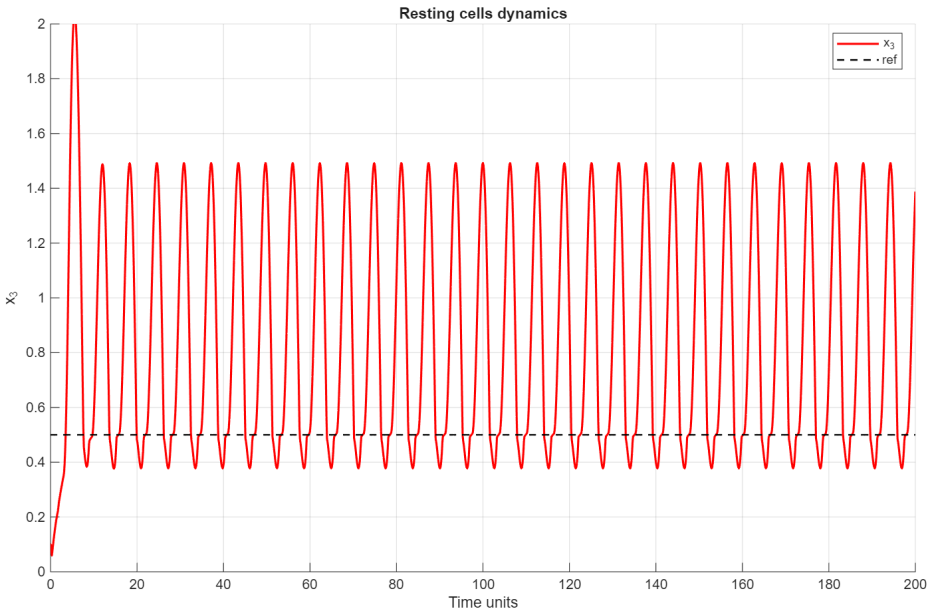


Figure 4.15: Resting cells time evolution with SMC with saturation and sinusoidal disturbance

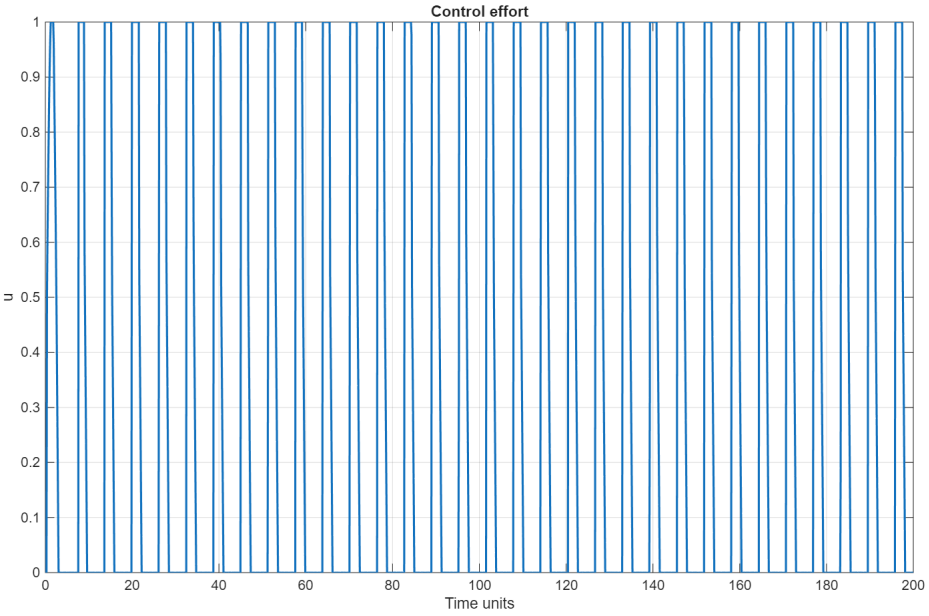


Figure 4.16: Control effort with SMC with saturation and sinusoidal disturbance

4.4 Overcoming physical constraints: the dual-phase control strategy

The analysis conducted in the previous sections leads to a fundamental conclusion: the inability of the nonlinear controllers to stabilize the system during stressful scenarios is a structural consequence of the **constraint on the control input**.

To resolve this controllability bottleneck without altering the fundamental structure of the model, we adopt the "alternating phase" strategy proposed by **Reppas, Alfonso, and Hatzikirou** [8], based on the switch between immunostimulating and immunosuppressive actions.

By integrating this logic into our controller's design, we transition from a monotherapy architecture to a dual-input one. This approach effectively keeps the system in a closed-loop regime throughout the entire transient, preventing the spikes of x_3 observed in the previous scenario. The last thing that needs to be shown is the performance of this new control structure:

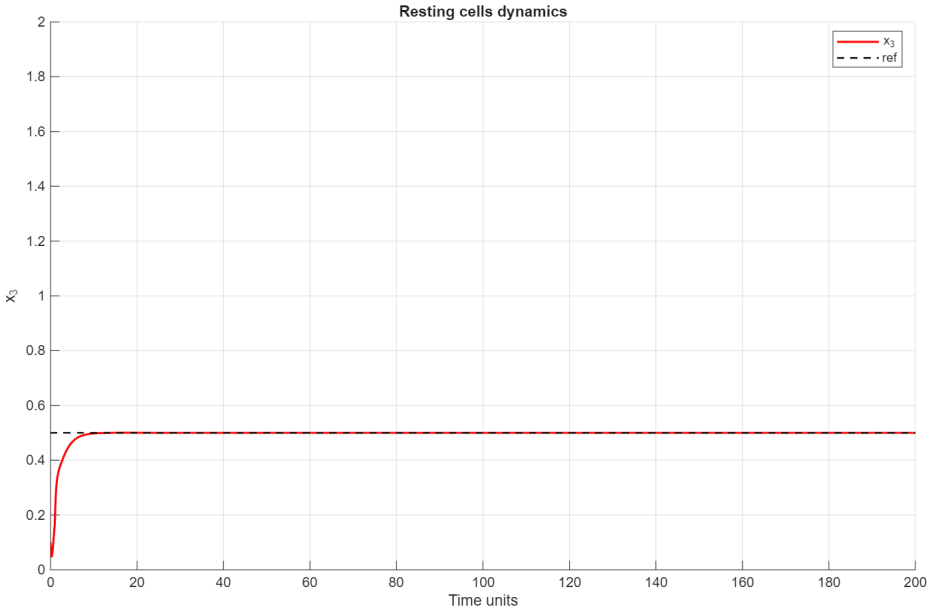


Figure 4.17: Resting cells time evolution with dual-architecture I/O FBL

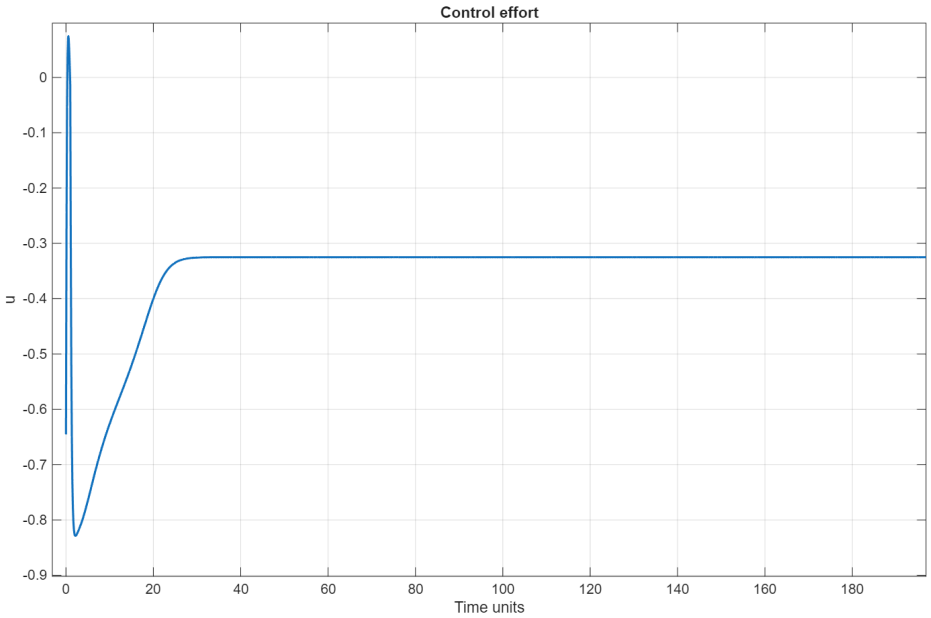


Figure 4.18: Control effort with dual-architecture I/O FBL

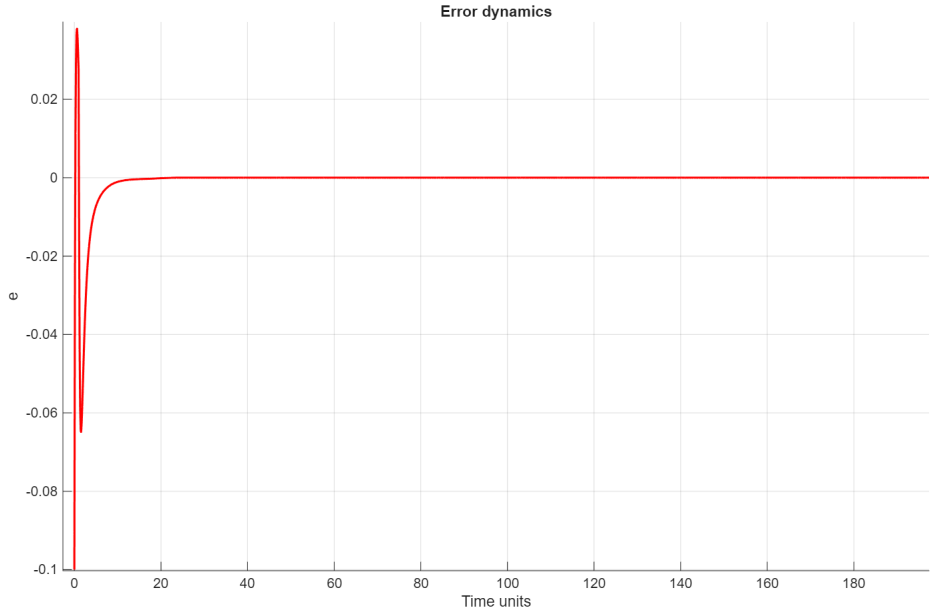


Figure 4.19: Error dynamics with dual-architecture I/O FBL

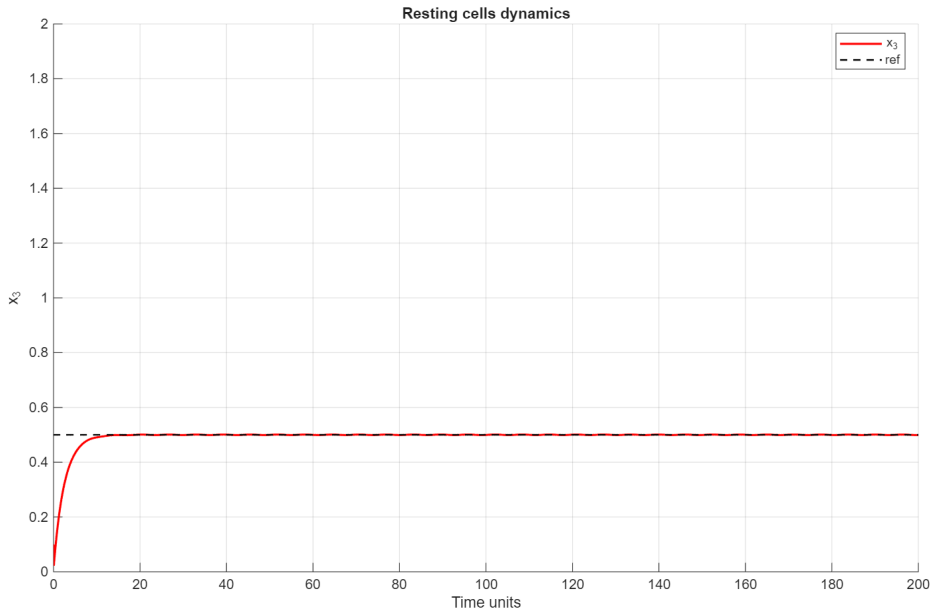


Figure 4.20: Resting cells time evolution with dual-architecture SMC

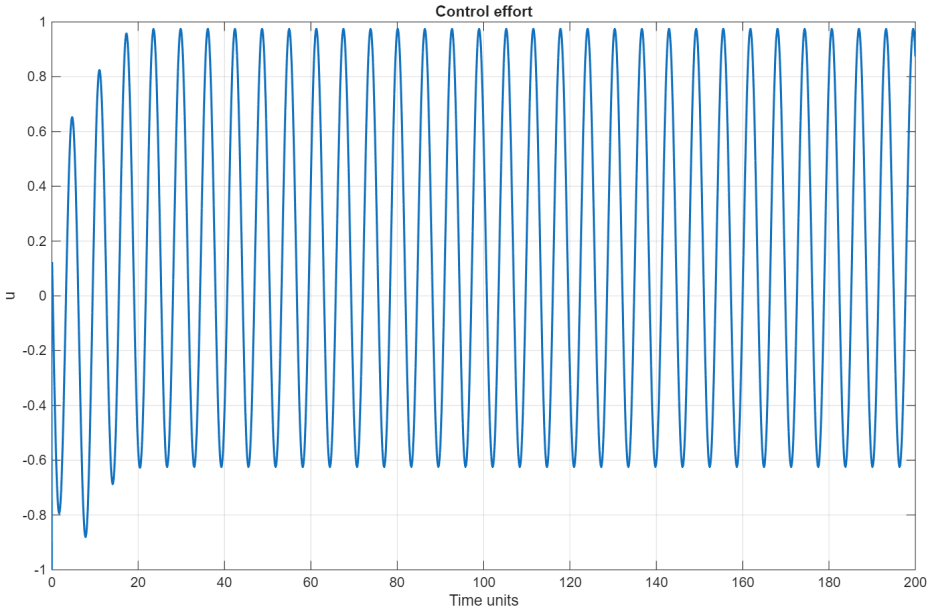


Figure 4.21: Control effort with dual-architecture SMC

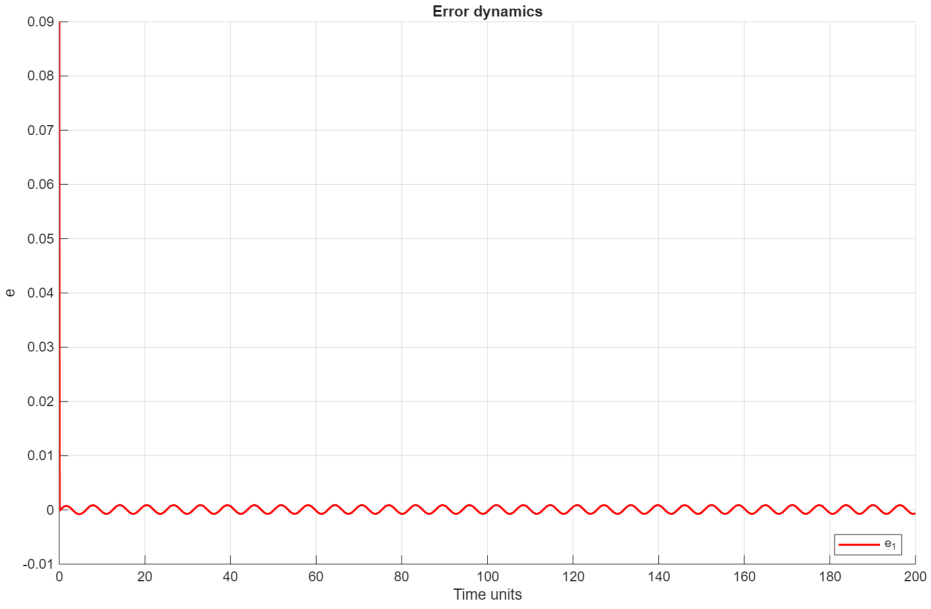


Figure 4.22: Error dynamics with dual-architecture SMC

These final simulations were carried out taking into account every possible destabilizing effect: the I/O FBL controller was stressed with a constant disturbance while the SMC had a sine wave as an input disturbance. Moreover, both started from x_1 set at 0.9, and both of them were suffering from a global parametric uncertainty of $\pm 30\%$.

4.4.1 Post-eradication protocol

Although the simulation results depict a continuous maintenance dosage to sustain the target $x_{3,ref}$ post-eradication, the clinical protocol dictates the **discontinuation of therapy** once the tumor is eliminated. This decision is mathematically justified by the stability of the tumor-free equilibrium ($x_1 = 0$): as the system lacks internal mechanisms for spontaneous resurgence, the zero-control input becomes the appropriate steady-state strategy. Consequently, the maintenance dose observed in the graphs is merely an artifact of the reference tracking and would be suspended in a realistic scenario upon confirmation of eradication.

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